

Kinora Upper Extremity — Vascular, Fascia & Biomechanics (Part 2)

Structured reference for RAG / AI-assisted physiotherapy consultation

Part 2: Vascular Supply, Fascial Anatomy & Upper Extremity Biomechanics

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Disclaimer

This document is an educational orientation resource for Kinora AI clinical consultation support. It is NOT a substitute for professional clinical judgment, direct patient examination, or licensed medical/physiotherapy care. Content reflects established vascular anatomy, fascial science, and sports biomechanics concepts but must be verified against current peer-reviewed literature, local protocols, and individual patient presentation. Always screen for vascular and neurologic red flags requiring urgent medical referral.

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1. Blood Supply — Arteries

RECORD: Vertebral Artery

ID: VES-UE-001

Name: Vertebral Artery

Latin Name: Arteria vertebralis

Type: Artery

Region: Cervical

Origin: Subclavian artery (first branch)

Termination: Basilar artery

Course: Ascends transverse foramina C6–C1, curves around atlas, enters cranium

Major Branches: Meningeal and spinal branches

Minor Branches: Muscular branches to deep cervical muscles

Structures Supplied: Brainstem, cerebellum, upper cervical spinal cord

Muscles Supplied: Deep cervical flexors and suboccipital muscles

Bones Supplied: Cervical vertebrae via periosteal branches

Joints Supplied: Atlanto-occipital and atlanto-axial joints

Ligaments Supplied: Posterior atlanto-occipital and atlanto-axial ligaments

Tendons Supplied: None directly

Skin Distribution: None — intracranial/intraforaminal

Venous Drainage: Vertebral venous plexus

Anastomoses: Circle of Willis; contralateral vertebral; muscular collaterals

Clinical Importance: Vertebrobasilar insufficiency; dissection risk with rotation trauma

Common Injuries: Arterial dissection; iatrogenic injury during cervical manipulation

Compression Sites: Transverse foramina; atlanto-axial rotation

Pulse Location: Not palpable externally

Ultrasound Appearance: Color Doppler: Vertebral Artery appears as pulsatile, anechoic/hypoechoic tubular structure with spectral broadening on stenosis; triphasic flow in proximal segments; assess for intimal flap (dissection), thrombus (non-compressible hypoechoic filling), or pseudoaneurysm.

MRI Appearance: MRA/time-resolved CE-MRA: Vertebral Artery flow void on T2; contrast-enhanced MRA delineates stenosis, occlusion, variant anatomy; TOF sensitive to slow flow; assess collateral reconstitution distal to occlusion.

CT Angiography Findings: CTA: Vertebral Artery opacifies with iodinated contrast; evaluate stenosis (>50% hemodynamically significant), dissection (double lumen, intimal flap), aneurysm, embolus, or post-traumatic extravasation in acute trauma.

Collateral Circulation: Contralateral vertebral; posterior communicating arteries

Surgical Considerations: Avoid aggressive rotation/extension with known dissection

Rehabilitation Considerations: Monitor neurologic symptoms during cervicothoracic mobility

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Vertebral Artery]

RECORD: Ascending Cervical Artery

ID: VES-UE-002

Name: Ascending Cervical Artery

Latin Name: Arteria cervicalis ascendens

Type: Artery

Region: Cervical — lateral neck

Origin: Inferior thyroid artery (thyrocervical trunk)

Termination: Anastomoses with vertebral and deep cervical arteries

Course: Ascends on scalenus anterior along cervical transverse processes

Major Branches: Spinal branches to cervical cord and meninges

Minor Branches: Muscular branches to scalene and longus colli

Structures Supplied: Cervical spinal cord segments and nerve roots

Muscles Supplied: Scalenes, longus colli/capitis, levator scapulae region

Bones Supplied: Cervical vertebral bodies and transverse processes

Joints Supplied: Facet joints via medial branches

Ligaments Supplied: Posterior longitudinal ligament region

Tendons Supplied: None

Skin Distribution: None direct

Venous Drainage: Vertebral venous plexus and thyroid venous plexus

Anastomoses: Vertebral, deep cervical, occipital muscular network

Clinical Importance: Collateral pathway in vertebral artery hypoplasia

Common Injuries: Rare direct trauma; compromised in anterior cervical surgery

Compression Sites: Scalene triangle in thoracic outlet syndrome

Pulse Location: Not palpable

Ultrasound Appearance: Color Doppler: Ascending Cervical Artery appears as pulsatile, anechoic/hypoechoic tubular structure with spectral broadening on stenosis; triphasic flow in proximal segments; assess for intimal flap (dissection), thrombus (non-compressible hypoechoic filling), or pseudoaneurysm.

MRI Appearance: MRA/time-resolved CE-MRA: Ascending Cervical Artery flow void on T2; contrast-enhanced MRA delineates stenosis, occlusion, variant anatomy; TOF sensitive to slow flow; assess collateral reconstitution distal to occlusion.

CT Angiography Findings: CTA: Ascending Cervical Artery opacifies with iodinated contrast; evaluate stenosis (>50% hemodynamically significant), dissection (double lumen, intimal flap), aneurysm, embolus, or post-traumatic extravasation in acute trauma.

Collateral Circulation: Vertebral and deep cervical anastomotic arcade

Surgical Considerations: Identify during anterior cervical approaches

Rehabilitation Considerations: Thoracic outlet rehab considers reduced collateral flow

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Ascending Cervical Artery]

RECORD: Deep Cervical Artery

ID: VES-UE-003

Name: Deep Cervical Artery

Latin Name: Arteria cervicalis profunda

Type: Artery

Region: Cervical — posterior neck

Origin: Costocervical trunk

Termination: Anastomoses with vertebral, ascending cervical, occipital arteries

Course: Ascends posterior to cervical transverse processes with cervical nerves

Major Branches: Spinal branches to posterior cervical cord

Minor Branches: Muscular branches to semispinalis, splenius, multifidus

Structures Supplied: Posterior cervical spinal cord and nerve roots

Muscles Supplied: Semispinalis, splenius, multifidus, suboccipital muscles

Bones Supplied: Cervical laminae, spinous processes, facet pillars

Joints Supplied: Facet joints posteriorly

Ligaments Supplied: Nuchal ligament and interspinous ligaments

Tendons Supplied: None

Skin Distribution: None

Venous Drainage: Vertebral venous plexus

Anastomoses: Vertebral, occipital, ascending cervical network

Clinical Importance: Posterior cervical muscle perfusion; collateral in vertebral stenosis

Common Injuries: Blunt posterior neck trauma rarely causes pseudoaneurysm

Compression Sites: Not typically compressed by facet hypertrophy

Pulse Location: Not palpable

Ultrasound Appearance: Color Doppler: Deep Cervical Artery appears as pulsatile, anechoic/hypoechoic tubular structure with spectral broadening on stenosis; triphasic flow in proximal segments; assess for intimal flap (dissection), thrombus (non-compressible hypoechoic filling), or pseudoaneurysm.

MRI Appearance: MRA/time-resolved CE-MRA: Deep Cervical Artery flow void on T2; contrast-enhanced MRA delineates stenosis, occlusion, variant anatomy; TOF sensitive to slow flow; assess collateral reconstitution distal to occlusion.

CT Angiography Findings: CTA: Deep Cervical Artery opacifies with iodinated contrast; evaluate stenosis (>50% hemodynamically significant), dissection (double lumen, intimal flap), aneurysm, embolus, or post-traumatic extravasation in acute trauma.

Collateral Circulation: Rich posterior cervical muscular anastomoses

Surgical Considerations: Posterior cervical fusion midline generally avoids major trunk

Rehabilitation Considerations: Deep cervical extensor strengthening supports segmental stability

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Deep Cervical Artery]

RECORD: Occipital Artery

ID: VES-UE-004

Name: Occipital Artery

Latin Name: Arteria occipitalis

Type: Artery

Region: Cervical — posterior scalp/suboccipital

Origin: External carotid artery

Termination: Superior nuchal line anastomoses

Course: Crosses atlas apex, pierces trapezius to scalp

Major Branches: SCM, auricular, meningeal, descending branches

Minor Branches: Muscular branches to suboccipital triangle

Structures Supplied: Posterior scalp, posterior fossa dura, SCM

Muscles Supplied: Trapezius, semispinalis capitis, suboccipital muscles

Bones Supplied: Occipital bone periosteum

Joints Supplied: Atlanto-occipital joint region

Ligaments Supplied: Nuchal ligament

Tendons Supplied: None

Skin Distribution: Posterior scalp to vertex

Venous Drainage: Occipital vein to suboccipital venous plexus

Anastomoses: Vertebral muscular, superficial temporal, posterior auricular

Clinical Importance: Suboccipital landmark; migraine trigger region

Common Injuries: Scalp laceration hemorrhage; rare dissection

Compression Sites: Trapezius tunnel at superior nuchal line

Pulse Location: Palpable behind mastoid in thin individuals

Ultrasound Appearance: Color Doppler: Occipital Artery appears as pulsatile, anechoic/hypoechoic tubular structure with spectral broadening on stenosis; triphasic flow in proximal segments; assess for intimal flap (dissection), thrombus (non-compressible hypoechoic filling), or pseudoaneurysm.

MRI Appearance: MRA/time-resolved CE-MRA: Occipital Artery flow void on T2; contrast-enhanced MRA delineates stenosis, occlusion, variant anatomy; TOF sensitive to slow flow; assess collateral reconstitution distal to occlusion.

CT Angiography Findings: CTA: Occipital Artery opacifies with iodinated contrast; evaluate stenosis (>50% hemodynamically significant), dissection (double lumen, intimal flap), aneurysm, embolus, or post-traumatic extravasation in acute trauma.

Collateral Circulation: Contralateral occipital and superficial temporal arcade

Surgical Considerations: Cautery during posterior occipital approach

Rehabilitation Considerations: Suboccipital release and upper trapezius mobility for tension headache

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Occipital Artery]

RECORD: Subclavian Artery

ID: VES-UE-005

Name: Subclavian Artery

Latin Name: Arteria subclavia

Type: Artery

Region: Shoulder girdle / thoracic inlet

Origin: Aortic arch (left) or brachiocephalic trunk (right)

Termination: Axillary artery at lateral border of 1st rib

Course: Arches over 1st rib between scalene anterior and medius

Major Branches: Vertebral, internal thoracic, thyrocervical, costocervical

Minor Branches: Dorsal scapular branch (variant)

Structures Supplied: Upper limb, brain via vertebral, anterior chest wall

Muscles Supplied: Scalenes, subclavius

Bones Supplied: Clavicle and 1st rib

Joints Supplied: Sternoclavicular and costoclavicular region

Ligaments Supplied: Costoclavicular ligament area

Tendons Supplied: Subclavius tendon region

Skin Distribution: Infraclavicular skin indirectly

Venous Drainage: Subclavian vein companion

Anastomoses: Vertebral, internal mammary, intercostal collaterals

Clinical Importance: Thoracic outlet and clavicle fracture vascular injury

Common Injuries: Compression in TOS; iatrogenic central line injury

Compression Sites: Interscalene triangle; costoclavicular space

Pulse Location: Supraclavicular fossa (avoid bilateral compression)

Ultrasound Appearance: Color Doppler: Subclavian Artery appears as pulsatile, anechoic/hypoechoic tubular structure with spectral broadening on stenosis; triphasic flow in proximal segments; assess for intimal flap (dissection), thrombus (non-compressible hypoechoic filling), or pseudoaneurysm.

MRI Appearance: MRA/time-resolved CE-MRA: Subclavian Artery flow void on T2; contrast-enhanced MRA delineates stenosis, occlusion, variant anatomy; TOF sensitive to slow flow; assess collateral reconstitution distal to occlusion.

CT Angiography Findings: CTA: Subclavian Artery opacifies with iodinated contrast; evaluate stenosis (>50% hemodynamically significant), dissection (double lumen, intimal flap), aneurysm, embolus, or post-traumatic extravasation in acute trauma.

Collateral Circulation: Contralateral subclavian and vertebral collaterals

Surgical Considerations: First rib resection and decompression planning

Rehabilitation Considerations: Avoid overhead loading early post-TOS surgery

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Subclavian Artery]

RECORD: Axillary Artery

ID: VES-UE-006

Name: Axillary Artery

Latin Name: Arteria axillaris

Type: Artery

Region: Shoulder / axilla

Origin: Subclavian at lateral 1st rib

Termination: Brachial artery at inferior teres major

Course: Three parts relative to pectoralis minor

Major Branches: Superior thoracic, thoracoacromial, lateral thoracic, subscapular, circumflex humeral

Minor Branches: Muscular branches to pectoralis and deltoid

Structures Supplied: Shoulder girdle, proximal arm, breast territory

Muscles Supplied: Pectoralis, deltoid, rotator cuff region

Bones Supplied: Proximal humerus and scapula

Joints Supplied: Glenohumeral and acromioclavicular joints

Ligaments Supplied: Glenohumeral capsular region

Tendons Supplied: Long head biceps groove region

Skin Distribution: Axillary skin

Venous Drainage: Axillary vein

Anastomoses: Scapular circumflex and rich shoulder anastomoses

Clinical Importance: Fracture/dislocation vascular compromise

Common Injuries: Anterior dislocation axillary injury

Compression Sites: Pectoralis minor; anterior humeral head dislocation

Pulse Location: Palpable in axilla (deep)

Ultrasound Appearance: Color Doppler: Axillary Artery appears as pulsatile, anechoic/hypoechoic tubular structure with spectral broadening on stenosis; triphasic flow in proximal segments; assess for intimal flap (dissection), thrombus (non-compressible hypoechoic filling), or pseudoaneurysm.

MRI Appearance: MRA/time-resolved CE-MRA: Axillary Artery flow void on T2; contrast-enhanced MRA delineates stenosis, occlusion, variant anatomy; TOF sensitive to slow flow; assess collateral reconstitution distal to occlusion.

CT Angiography Findings: CTA: Axillary Artery opacifies with iodinated contrast; evaluate stenosis (>50% hemodynamically significant), dissection (double lumen, intimal flap), aneurysm, embolus, or post-traumatic extravasation in acute trauma.

Collateral Circulation: Scapular and circumflex collaterals

Surgical Considerations: Proximal humerus ORIF vascular mapping

Rehabilitation Considerations: Progressive ROM after dislocation with neurovascular monitoring

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Axillary Artery]

RECORD: Superior Thoracic Artery

ID: VES-UE-007

Name: Superior Thoracic Artery

Latin Name: Arteria thoracica superior

Type: Artery

Region: Shoulder apex / upper chest

Origin: Axillary artery first part

Termination: Anastomoses with internal thoracic and intercostals

Course: Descends along superior thoracic wall

Major Branches: None major

Minor Branches: Periosteal twigs to upper ribs

Structures Supplied: Upper anterior chest wall

Muscles Supplied: Subclavius and upper pectoralis minor fibers

Bones Supplied: 1st and 2nd ribs; clavicle region

Joints Supplied: Sternoclavicular region

Ligaments Supplied: Costoclavicular ligament area

Tendons Supplied: None

Skin Distribution: Upper chest skin

Venous Drainage: Axillary venous plexus

Anastomoses: Internal thoracic and intercostal anastomoses

Clinical Importance: Minor but relevant in apical chest trauma

Common Injuries: Rare isolated injury

Compression Sites: Apical thoracic outlet region

Pulse Location: Not routinely palpated

Ultrasound Appearance: Color Doppler: Superior Thoracic Artery appears as pulsatile, anechoic/hypoechoic tubular structure with spectral broadening on stenosis; triphasic flow in proximal segments; assess for intimal flap (dissection), thrombus (non-compressible hypoechoic filling), or pseudoaneurysm.

MRI Appearance: MRA/time-resolved CE-MRA: Superior Thoracic Artery flow void on T2; contrast-enhanced MRA delineates stenosis, occlusion, variant anatomy; TOF sensitive to slow flow; assess collateral reconstitution distal to occlusion.

CT Angiography Findings: CTA: Superior Thoracic Artery opacifies with iodinated contrast; evaluate stenosis (>50% hemodynamically significant), dissection (double lumen, intimal flap), aneurysm, embolus, or post-traumatic extravasation in acute trauma.

Collateral Circulation: Intercostal collateral network

Surgical Considerations: Apical surgery hemostasis awareness

Rehabilitation Considerations: Upper chest breathing retraining post-thoracic surgery

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Superior Thoracic Artery]

RECORD: Thoracoacromial Artery

ID: VES-UE-008

Name: Thoracoacromial Artery

Latin Name: Arteria thoracoacromialis

Type: Artery

Region: Shoulder

Origin: Axillary artery second part

Termination: Acromion, deltoid, clavicle, pectoralis territories

Course: Pierces clavipectoral fascia with coracoclavicular ligaments

Major Branches: Acromial, deltoid, clavicular, pectoral branches

Minor Branches: Cutaneous twigs to anterior shoulder

Structures Supplied: Deltoid, pectoralis major clavicular head, AC joint

Muscles Supplied: Deltoid and pectoralis major

Bones Supplied: Distal clavicle and acromion

Joints Supplied: Acromioclavicular joint

Ligaments Supplied: Coracoacromial ligament region

Tendons Supplied: Deltoid tendon region

Skin Distribution: Anterior shoulder skin

Venous Drainage: Acromial veins

Anastomoses: Suprascapular and circumflex humeral anastomoses

Clinical Importance: AC joint and clavicle perfusion

Common Injuries: Clavicle fracture bleeding source

Compression Sites: Coracoid and clavipectoral tunnel

Pulse Location: Not palpable as discrete pulse

Ultrasound Appearance: Color Doppler: Thoracoacromial Artery appears as pulsatile, anechoic/hypoechoic tubular structure with spectral broadening on stenosis; triphasic flow in proximal segments; assess for intimal flap (dissection), thrombus (non-compressible hypoechoic filling), or pseudoaneurysm.

MRI Appearance: MRA/time-resolved CE-MRA: Thoracoacromial Artery flow void on T2; contrast-enhanced MRA delineates stenosis, occlusion, variant anatomy; TOF sensitive to slow flow; assess collateral reconstitution distal to occlusion.

CT Angiography Findings: CTA: Thoracoacromial Artery opacifies with iodinated contrast; evaluate stenosis (>50% hemodynamically significant), dissection (double lumen, intimal flap), aneurysm, embolus, or post-traumatic extravasation in acute trauma.

Collateral Circulation: Deltoid branch collaterals

Surgical Considerations: Deltoid-preserving surgical approaches

Rehabilitation Considerations: Early isometric deltoid activation post AC injury

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Thoracoacromial Artery]

RECORD: Lateral Thoracic Artery

ID: VES-UE-009

Name: Lateral Thoracic Artery

Latin Name: Arteria thoracica lateralis

Type: Artery

Region: Shoulder / lateral chest wall

Origin: Axillary artery second part

Termination: Anastomoses with intercostals and mammary branches

Course: Descends along lateral chest with long thoracic nerve

Major Branches: Mammary branches when present

Minor Branches: Serratus anterior muscular branches

Structures Supplied: Lateral chest wall and serratus anterior

Muscles Supplied: Serratus anterior and pectoralis minor

Bones Supplied: Ribs 2–6

Joints Supplied: Scapulothoracic interface

Ligaments Supplied: Serratus fascia

Tendons Supplied: None major

Skin Distribution: Lateral thorax skin

Venous Drainage: Lateral thoracic vein

Anastomoses: Intercostal and internal thoracic network

Clinical Importance: Serratus perfusion in long thoracic nerve injury

Common Injuries: Rare direct vascular injury

Compression Sites: Pectoralis minor compression in TOS variants

Pulse Location: Not palpable

Ultrasound Appearance: Color Doppler: Lateral Thoracic Artery appears as pulsatile, anechoic/hypoechoic tubular structure with spectral broadening on stenosis; triphasic flow in proximal segments; assess for intimal flap (dissection), thrombus (non-compressible hypoechoic filling), or pseudoaneurysm.

MRI Appearance: MRA/time-resolved CE-MRA: Lateral Thoracic Artery flow void on T2; contrast-enhanced MRA delineates stenosis, occlusion, variant anatomy; TOF sensitive to slow flow; assess collateral reconstitution distal to occlusion.

CT Angiography Findings: CTA: Lateral Thoracic Artery opacifies with iodinated contrast; evaluate stenosis (>50% hemodynamically significant), dissection (double lumen, intimal flap), aneurysm, embolus, or post-traumatic extravasation in acute trauma.

Collateral Circulation: Intercostal collateral network

Surgical Considerations: Breast and chest wall surgery awareness

Rehabilitation Considerations: Serratus punch rehab after nerve injury

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Lateral Thoracic Artery]

RECORD: Subscapular Artery

ID: VES-UE-010

Name: Subscapular Artery

Latin Name: Arteria subscapularis

Type: Artery

Region: Shoulder / posterior axilla

Origin: Axillary artery third part

Termination: Circumflex scapular and thoracodorsal arteries

Course: Largest branch along subscapular fossa border

Major Branches: Circumflex scapular and thoracodorsal arteries

Minor Branches: Multiple muscular branches

Structures Supplied: Scapula, latissimus dorsi, teres major region

Muscles Supplied: Subscapularis, teres major, latissimus dorsi

Bones Supplied: Scapula and proximal humerus

Joints Supplied: Glenohumeral joint capsule region

Ligaments Supplied: Glenohumeral ligaments indirectly

Tendons Supplied: Subscapularis tendon

Skin Distribution: Posterior axillary skin

Venous Drainage: Subscapular vein

Anastomoses: Rich scapular anastomotic network

Clinical Importance: Posterior shoulder perfusion; flap surgery landmark

Common Injuries: Latissimus flap harvest affects regional perfusion

Compression Sites: Quadrangular space region (circumflex scapular)

Pulse Location: Not palpable

Ultrasound Appearance: Color Doppler: Subscapular Artery appears as pulsatile, anechoic/hypoechoic tubular structure with spectral broadening on stenosis; triphasic flow in proximal segments; assess for intimal flap (dissection), thrombus (non-compressible hypoechoic filling), or pseudoaneurysm.

MRI Appearance: MRA/time-resolved CE-MRA: Subscapular Artery flow void on T2; contrast-enhanced MRA delineates stenosis, occlusion, variant anatomy; TOF sensitive to slow flow; assess collateral reconstitution distal to occlusion.

CT Angiography Findings: CTA: Subscapular Artery opacifies with iodinated contrast; evaluate stenosis (>50% hemodynamically significant), dissection (double lumen, intimal flap), aneurysm, embolus, or post-traumatic extravasation in acute trauma.

Collateral Circulation: Circumflex scapular collaterals

Surgical Considerations: Posterior shoulder surgical approaches

Rehabilitation Considerations: Rowing and pulling rehab respects posterior supply

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Subscapular Artery]

RECORD: Anterior Circumflex Humeral Artery

ID: VES-UE-011

Name: Anterior Circumflex Humeral Artery

Latin Name: Arteria circumflexa humeri anterior

Type: Artery

Region: Shoulder / anterior humerus

Origin: Axillary artery third part

Termination: Anastomoses with posterior circumflex humeral

Course: Wraps anterior surgical neck with ascending branch to head

Major Branches: Ascending branch to humeral head

Minor Branches: Articular branches to head and biceps groove

Structures Supplied: Humeral head and biceps groove region

Muscles Supplied: Coracobrachialis and short head biceps region

Bones Supplied: Proximal humerus and greater tuberosity

Joints Supplied: Glenohumeral joint head perfusion

Ligaments Supplied: Rotator interval region

Tendons Supplied: Biceps long head tendon sheath

Skin Distribution: Anterior shoulder skin

Venous Drainage: Axillary venous plexus

Anastomoses: Posterior circumflex humeral arcade

Clinical Importance: Critical head perfusion — AVN risk in surgical neck fracture

Common Injuries: Proximal humerus fracture with head ischemia

Compression Sites: Surgical neck fracture; anterior dislocation

Pulse Location: Not palpable

Ultrasound Appearance: Color Doppler: Anterior Circumflex Humeral Artery appears as pulsatile, anechoic/hypoechoic tubular structure with spectral broadening on stenosis; triphasic flow in proximal segments; assess for intimal flap (dissection), thrombus (non-compressible hypoechoic filling), or pseudoaneurysm.

MRI Appearance: MRA/time-resolved CE-MRA: Anterior Circumflex Humeral Artery flow void on T2; contrast-enhanced MRA delineates stenosis, occlusion, variant anatomy; TOF sensitive to slow flow; assess collateral reconstitution distal to occlusion.

CT Angiography Findings: CTA: Anterior Circumflex Humeral Artery opacifies with iodinated contrast; evaluate stenosis (>50% hemodynamically significant), dissection (double lumen, intimal flap), aneurysm, embolus, or post-traumatic extravasation in acute trauma.

Collateral Circulation: Posterior circumflex collaterals

Surgical Considerations: ORIF planning for humeral head blood supply

Rehabilitation Considerations: Protected ROM after proximal humerus fracture

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Anterior Circumflex Humeral Artery]

RECORD: Posterior Circumflex Humeral Artery

ID: VES-UE-012

Name: Posterior Circumflex Humeral Artery

Latin Name: Arteria circumflexa humeri posterior

Type: Artery

Region: Shoulder / posterior humerus

Origin: Axillary artery third part

Termination: Anastomoses with anterior circumflex humeral

Course: Passes quadrangular space with axillary nerve

Major Branches: Descending branch along posterior humerus

Minor Branches: Muscular branches to deltoid and teres minor

Structures Supplied: Posterior humerus, deltoid, teres minor

Muscles Supplied: Deltoid, teres minor, infraspinatus region

Bones Supplied: Proximal humerus posterior surface

Joints Supplied: Glenohumeral joint posterior capsule

Ligaments Supplied: Posterior glenohumeral ligaments

Tendons Supplied: Infraspinatus and teres minor tendons

Skin Distribution: Posterior deltoid skin

Venous Drainage: Axillary vein

Anastomoses: Anterior circumflex humeral network

Clinical Importance: Quadrangular space syndrome vascular component

Common Injuries: Axillary nerve palsy with space-occupying lesion

Compression Sites: Quadrangular space fibrous bands or cysts

Pulse Location: Not palpable

Ultrasound Appearance: Color Doppler: Posterior Circumflex Humeral Artery appears as pulsatile, anechoic/hypoechoic tubular structure with spectral broadening on stenosis; triphasic flow in proximal segments; assess for intimal flap (dissection), thrombus (non-compressible hypoechoic filling), or pseudoaneurysm.

MRI Appearance: MRA/time-resolved CE-MRA: Posterior Circumflex Humeral Artery flow void on T2; contrast-enhanced MRA delineates stenosis, occlusion, variant anatomy; TOF sensitive to slow flow; assess collateral reconstitution distal to occlusion.

CT Angiography Findings: CTA: Posterior Circumflex Humeral Artery opacifies with iodinated contrast; evaluate stenosis (>50% hemodynamically significant), dissection (double lumen, intimal flap), aneurysm, embolus, or post-traumatic extravasation in acute trauma.

Collateral Circulation: Anterior circumflex collaterals

Surgical Considerations: Posterior shoulder approach awareness

Rehabilitation Considerations: External rotation strengthening after space decompression

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Posterior Circumflex Humeral Artery]

RECORD: Brachial Artery

ID: VES-UE-013

Name: Brachial Artery

Latin Name: Arteria brachialis

Type: Artery

Region: Arm

Origin: Axillary at inferior teres major

Termination: Bifurcates into radial and ulnar at cubital fossa

Course: Medial arm in medial bicipital groove with median nerve

Major Branches: Profunda brachii and nutrient artery

Minor Branches: Muscular branches to flexor compartment

Structures Supplied: Elbow, forearm, and hand via terminal branches

Muscles Supplied: Biceps, brachialis, coracobrachialis, flexor compartment

Bones Supplied: Humerus

Joints Supplied: Elbow joint

Ligaments Supplied: Medial and lateral collateral regions

Tendons Supplied: Distal biceps insertion region

Skin Distribution: Medial arm skin

Venous Drainage: Paired brachial veins (venae comitantes)

Anastomoses: Profunda brachii and recurrent collaterals

Clinical Importance: Blood pressure site; laceration risk in trauma

Common Injuries: Supracondylar fracture vascular injury in children

Compression Sites: Medial epicondyle and lacertus fibrosus region

Pulse Location: Palpable throughout medial arm to antecubital fossa

Ultrasound Appearance: Color Doppler: Brachial Artery appears as pulsatile, anechoic/hypoechoic tubular structure with spectral broadening on stenosis; triphasic flow in proximal segments; assess for intimal flap (dissection), thrombus (non-compressible hypoechoic filling), or pseudoaneurysm.

MRI Appearance: MRA/time-resolved CE-MRA: Brachial Artery flow void on T2; contrast-enhanced MRA delineates stenosis, occlusion, variant anatomy; TOF sensitive to slow flow; assess collateral reconstitution distal to occlusion.

CT Angiography Findings: CTA: Brachial Artery opacifies with iodinated contrast; evaluate stenosis (>50% hemodynamically significant), dissection (double lumen, intimal flap), aneurysm, embolus, or post-traumatic extravasation in acute trauma.

Collateral Circulation: Recurrent radial and ulnar collaterals around elbow

Surgical Considerations: Tourniquet time limits in arm surgery

Rehabilitation Considerations: Gradual elbow ROM after supracondylar fracture with monitoring

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Brachial Artery]

RECORD: Profunda Brachii Artery

ID: VES-UE-014

Name: Profunda Brachii Artery

Latin Name: Arteria profunda brachii

Type: Artery

Region: Arm / posterior elbow

Origin: Brachial artery

Termination: Anastomoses with radial and middle collateral arteries

Course: Travels with radial nerve in spiral groove

Major Branches: Radial collateral and middle collateral arteries

Minor Branches: Muscular branches to triceps

Structures Supplied: Posterior arm and elbow extension mechanism

Muscles Supplied: Triceps brachii all heads

Bones Supplied: Posterior humerus

Joints Supplied: Elbow olecranon region

Ligaments Supplied: Olecranon bursa region

Tendons Supplied: Triceps tendon

Skin Distribution: Posterior arm skin

Venous Drainage: Deep brachial veins

Anastomoses: Elbow collateral anastomotic arcade

Clinical Importance: Elbow perfusion when brachial injured

Common Injuries: Humeral shaft fracture with radial nerve

Compression Sites: Spiral groove fracture site

Pulse Location: Not palpable

Ultrasound Appearance: Color Doppler: Profunda Brachii Artery appears as pulsatile, anechoic/hypoechoic tubular structure with spectral broadening on stenosis; triphasic flow in proximal segments; assess for intimal flap (dissection), thrombus (non-compressible hypoechoic filling), or pseudoaneurysm.

MRI Appearance: MRA/time-resolved CE-MRA: Profunda Brachii Artery flow void on T2; contrast-enhanced MRA delineates stenosis, occlusion, variant anatomy; TOF sensitive to slow flow; assess collateral reconstitution distal to occlusion.

CT Angiography Findings: CTA: Profunda Brachii Artery opacifies with iodinated contrast; evaluate stenosis (>50% hemodynamically significant), dissection (double lumen, intimal flap), aneurysm, embolus, or post-traumatic extravasation in acute trauma.

Collateral Circulation: Middle and posterior collateral collaterals

Surgical Considerations: Posterior approach to humerus

Rehabilitation Considerations: Triceps progressive loading after humeral shaft fracture

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Profunda Brachii Artery]

RECORD: Superior Ulnar Collateral Artery

ID: VES-UE-015

Name: Superior Ulnar Collateral Artery

Latin Name: Arteria collateralis ulnaris superior

Type: Artery

Region: Elbow — medial

Origin: Brachial artery

Termination: Anastomoses with posterior ulnar recurrent

Course: Descends with ulnar nerve posterior to medial epicondyle

Major Branches: None major

Minor Branches: Periarticular branches to medial elbow

Structures Supplied: Medial elbow and ulnar nerve region

Muscles Supplied: Flexor carpi ulnaris origin region

Bones Supplied: Medial epicondyle and olecranon

Joints Supplied: Ulnohumeral joint medial aspect

Ligaments Supplied: UCL complex region

Tendons Supplied: Common flexor origin

Skin Distribution: Medial forearm skin

Venous Drainage: Ulnar venous plexus

Anastomoses: Inferior ulnar collateral and recurrent ulnar network

Clinical Importance: UCL reconstruction considerations; cubital tunnel region; thrower elbow medial perfusion

Common Injuries: UCL sprain and cubital tunnel syndrome

Compression Sites: Cubital tunnel retinaculum

Pulse Location: Not palpable

Ultrasound Appearance: Color Doppler: Superior Ulnar Collateral Artery appears as pulsatile, anechoic/hypoechoic tubular structure with spectral broadening on stenosis; triphasic flow in proximal segments; assess for intimal flap (dissection), thrombus (non-compressible hypoechoic filling), or pseudoaneurysm.

MRI Appearance: MRA/time-resolved CE-MRA: Superior Ulnar Collateral Artery flow void on T2; contrast-enhanced MRA delineates stenosis, occlusion, variant anatomy; TOF sensitive to slow flow; assess collateral reconstitution distal to occlusion.

CT Angiography Findings: CTA: Superior Ulnar Collateral Artery opacifies with iodinated contrast; evaluate stenosis (>50% hemodynamically significant), dissection (double lumen, intimal flap), aneurysm, embolus, or post-traumatic extravasation in acute trauma.

Collateral Circulation: Recurrent ulnar collaterals

Surgical Considerations: UCL surgery graft harvest planning

Rehabilitation Considerations: Throwing progression monitors medial elbow symptoms

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Superior Ulnar Collateral Artery]

RECORD: Inferior Ulnar Collateral Artery

ID: VES-UE-016

Name: Inferior Ulnar Collateral Artery

Latin Name: Arteria collateralis ulnaris inferior

Type: Artery

Region: Elbow — medial anterior

Origin: Brachial artery

Termination: Anastomoses with anterior ulnar recurrent

Course: Crosses anterior to medial epicondyle

Major Branches: None major

Minor Branches: Periarticular twigs

Structures Supplied: Anterior medial elbow

Muscles Supplied: Pronator teres and flexor origin region

Bones Supplied: Medial epicondyle

Joints Supplied: Ulnohumeral joint

Ligaments Supplied: UCL anterior band region

Tendons Supplied: Flexor-pronator mass

Skin Distribution: Medial antecubital skin

Venous Drainage: Median and basilic tributaries

Anastomoses: Superior ulnar collateral network

Clinical Importance: Elbow flexion stability perfusion

Common Injuries: Valgus extension overload in throwers

Compression Sites: Medial epicondyle compression

Pulse Location: Not palpable

Ultrasound Appearance: Color Doppler: Inferior Ulnar Collateral Artery appears as pulsatile, anechoic/hypoechoic tubular structure with spectral broadening on stenosis; triphasic flow in proximal segments; assess for intimal flap (dissection), thrombus (non-compressible hypoechoic filling), or pseudoaneurysm.

MRI Appearance: MRA/time-resolved CE-MRA: Inferior Ulnar Collateral Artery flow void on T2; contrast-enhanced MRA delineates stenosis, occlusion, variant anatomy; TOF sensitive to slow flow; assess collateral reconstitution distal to occlusion.

CT Angiography Findings: CTA: Inferior Ulnar Collateral Artery opacifies with iodinated contrast; evaluate stenosis (>50% hemodynamically significant), dissection (double lumen, intimal flap), aneurysm, embolus, or post-traumatic extravasation in acute trauma.

Collateral Circulation: Superior ulnar collateral collaterals

Surgical Considerations: Medial elbow surgical approach

Rehabilitation Considerations: Eccentric wrist flexor training for medial tendinopathy

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Inferior Ulnar Collateral Artery]

RECORD: Radial Recurrent Artery

ID: VES-UE-017

Name: Radial Recurrent Artery

Latin Name: Arteria recurrens radialis

Type: Artery

Region: Elbow — lateral

Origin: Radial artery

Termination: Anastomoses with radial and middle collateral arteries

Course: Ascends lateral elbow with radial nerve

Major Branches: None major

Minor Branches: Muscular branches to brachioradialis and ECRB

Structures Supplied: Lateral elbow and mobile wad

Muscles Supplied: Brachioradialis and extensor carpi radialis brevis

Bones Supplied: Lateral epicondyle

Joints Supplied: Radiocapitellar joint

Ligaments Supplied: Lateral collateral ligament region

Tendons Supplied: Common extensor origin

Skin Distribution: Lateral forearm skin

Venous Drainage: Radial veins

Anastomoses: Elbow collateral arcade

Clinical Importance: Lateral elbow perfusion; collateral in brachial injury

Common Injuries: Lateral epicondylalgia context

Compression Sites: Radial tunnel region

Pulse Location: Not palpable

Ultrasound Appearance: Color Doppler: Radial Recurrent Artery appears as pulsatile, anechoic/hypoechoic tubular structure with spectral broadening on stenosis; triphasic flow in proximal segments; assess for intimal flap (dissection),

thrombus (non-compressible hypoechoic filling), or pseudoaneurysm.

MRI Appearance: MRA/time-resolved CE-MRA: Radial Recurrent Artery flow void on T2; contrast-enhanced MRA delineates stenosis, occlusion, variant anatomy; TOF sensitive to slow flow; assess collateral reconstitution distal to occlusion.

CT Angiography Findings: CTA: Radial Recurrent Artery opacifies with iodinated contrast; evaluate stenosis (>50% hemodynamically significant), dissection (double lumen, intimal flap), aneurysm, embolus, or post-traumatic extravasation in acute trauma.

Collateral Circulation: Radial collateral collaterals

Surgical Considerations: Lateral epicondyle release hemostasis

Rehabilitation Considerations: Eccentric wrist extension progression for lateral elbow pain

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Radial Recurrent Artery]

RECORD: Ulnar Recurrent Artery

ID: VES-UE-018

Name: Ulnar Recurrent Artery

Latin Name: Arteria recurrens ulnaris

Type: Artery

Region: Elbow — medial

Origin: Ulnar artery

Termination: Anastomoses with inferior and superior ulnar collateral

Course: Anterior and posterior branches around medial elbow

Major Branches: None major

Minor Branches: Periarticular branches

Structures Supplied: Medial elbow stability zone

Muscles Supplied: Flexor carpi ulnaris and flexor mass

Bones Supplied: Medial elbow osseous landmarks

Joints Supplied: Ulnohumeral joint

Ligaments Supplied: Ulnar collateral ligament

Tendons Supplied: Flexor tendons medially

Skin Distribution: Medial forearm skin

Venous Drainage: Ulnar veins

Anastomoses: Ulnar collateral network

Clinical Importance: Medial elbow collateral circulation

Common Injuries: UCL injury in throwers

Compression Sites: Cubital tunnel compression

Pulse Location: Not palpable

Ultrasound Appearance: Color Doppler: Ulnar Recurrent Artery appears as pulsatile, anechoic/hypoechoic tubular structure with spectral broadening on stenosis; triphasic flow in proximal segments; assess for intimal flap (dissection), thrombus (non-compressible hypoechoic filling), or pseudoaneurysm.

MRI Appearance: MRA/time-resolved CE-MRA: Ulnar Recurrent Artery flow void on T2; contrast-enhanced MRA delineates stenosis, occlusion, variant anatomy; TOF sensitive to slow flow; assess collateral reconstitution distal to occlusion.

CT Angiography Findings: CTA: Ulnar Recurrent Artery opacifies with iodinated contrast; evaluate stenosis (>50% hemodynamically significant), dissection (double lumen, intimal flap), aneurysm, embolus, or post-traumatic extravasation in acute trauma.

Collateral Circulation: Collateral ulnar network

Surgical Considerations: Medial elbow surgery

Rehabilitation Considerations: Medial isometrics progressing to functional valgus control

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Ulnar Recurrent Artery]

RECORD: Radial Artery

ID: VES-UE-019

Name: Radial Artery

Latin Name: Arteria radialis

Type: Artery

Region: Forearm and wrist

Origin: Brachial bifurcation

Termination: Deep palmar arch (dominant variant) with superficial contribution

Course: Lateral forearm with flexor carpi radialis tendon boundary

Major Branches: Princeps pollicis and radialis indicis arteries

Minor Branches: Carpal and muscular branches

Structures Supplied: Lateral forearm, thumb, radial hand

Muscles Supplied: Brachioradialis, extensors, thenar muscles

Bones Supplied: Radius, scaphoid, trapezium

Joints Supplied: Wrist and carpometacarpal joints

Ligaments Supplied: Volar radiocarpal ligaments

Tendons Supplied: Brachioradialis and thumb tendons

Skin Distribution: Thenar and radial hand skin

Venous Drainage: Radial veins

Anastomoses: Ulnar artery via palmar arches

Clinical Importance: Allen test patency; catheterization access

Common Injuries: Laceration at wrist; iatrogenic radial access injury

Compression Sites: Anatomical snuffbox; scaphoid groove

Pulse Location: Palpable at wrist radial to FCR tendon

Ultrasound Appearance: Color Doppler: Radial Artery appears as pulsatile, anechoic/hypoechoic tubular structure with spectral broadening on stenosis; triphasic flow in proximal segments; assess for intimal flap (dissection), thrombus (non-compressible hypoechoic filling), or pseudoaneurysm.

MRI Appearance: MRA/time-resolved CE-MRA: Radial Artery flow void on T2; contrast-enhanced MRA delineates stenosis, occlusion, variant anatomy; TOF sensitive to slow flow; assess collateral reconstitution distal to occlusion.

CT Angiography Findings: CTA: Radial Artery opacifies with iodinated contrast; evaluate stenosis (>50% hemodynamically significant), dissection (double lumen, intimal flap), aneurysm, embolus, or post-traumatic extravasation in acute trauma.

Collateral Circulation: Ulnar artery palmar arch collaterals

Surgical Considerations: Flap harvest and radial artery graft

Rehabilitation Considerations: Grip strengthening after harvest with ulnar dominance check

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Radial Artery]

RECORD: Ulnar Artery

ID: VES-UE-020

Name: Ulnar Artery

Latin Name: Arteria ulnaris

Type: Artery

Region: Forearm and wrist

Origin: Brachial bifurcation

Termination: Superficial palmar arch (usually) with deep arch contribution

Course: Medial forearm under flexor digitorum superficialis

Major Branches: Common palmar digital arteries

Minor Branches: Muscular and carpal branches

Structures Supplied: Medial forearm, hypothenar, ulnar digits

Muscles Supplied: Flexor carpi ulnaris, flexor digitorum, hypothenar muscles

Bones Supplied: Ulna, pisiform, hamate

Joints Supplied: Wrist and elbow via collaterals

Ligaments Supplied: UCL and TFCC region

Tendons Supplied: Flexor tendons

Skin Distribution: Hypothenar and ulnar hand skin

Venous Drainage: Ulnar veins

Anastomoses: Radial artery palmar arches

Clinical Importance: Guyon's canal compression in cyclists

Common Injuries: Handlebar palsy with vascular and neural components

Compression Sites: Guyon's canal; hook of hamate fracture

Pulse Location: Palpable at wrist medial to FCU tendon

Ultrasound Appearance: Color Doppler: Ulnar Artery appears as pulsatile, anechoic/hypoechoic tubular structure with spectral broadening on stenosis; triphasic flow in proximal segments; assess for intimal flap (dissection), thrombus (non-compressible hypoechoic filling), or pseudoaneurysm.

MRI Appearance: MRA/time-resolved CE-MRA: Ulnar Artery flow void on T2; contrast-enhanced MRA delineates stenosis, occlusion, variant anatomy; TOF sensitive to slow flow; assess collateral reconstitution distal to occlusion.

CT Angiography Findings: CTA: Ulnar Artery opacifies with iodinated contrast; evaluate stenosis (>50% hemodynamically significant), dissection (double lumen, intimal flap), aneurysm, embolus, or post-traumatic extravasation in acute trauma.

Collateral Circulation: Radial collaterals via palmar arches

Surgical Considerations: Ulnar artery transposition in trauma

Rehabilitation Considerations: Progressive grip after Guyon's canal release

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Ulnar Artery]

RECORD: Common Interosseous Artery

ID: VES-UE-021

Name: Common Interosseous Artery

Latin Name: Arteria interossea communis

Type: Artery

Region: Forearm / elbow region

Origin: Ulnar artery

Termination: Bifurcates into anterior and posterior interosseous arteries

Course: Short trunk before passing deep in forearm

Major Branches: Anterior and posterior interosseous branches

Minor Branches: None on trunk

Structures Supplied: Deep forearm compartment

Muscles Supplied: Deep flexors and extensors

Bones Supplied: Radius and ulna

Joints Supplied: Proximal radioulnar joint

Ligaments Supplied: Interosseous membrane

Tendons Supplied: Deep flexor and extensor tendons

Skin Distribution: Deep forearm region

Venous Drainage: Interosseous veins

Anastomoses: Recurrent interosseous at elbow

Clinical Importance: Forearm compartment perfusion

Common Injuries: Compartment syndrome vascular assessment

Compression Sites: Interosseous membrane

Pulse Location: Not palpable

Ultrasound Appearance: Color Doppler: Common Interosseous Artery appears as pulsatile, anechoic/hypoechoic tubular structure with spectral broadening on stenosis; triphasic flow in proximal segments; assess for intimal flap (dissection), thrombus (non-compressible hypoechoic filling), or pseudoaneurysm.

MRI Appearance: MRA/time-resolved CE-MRA: Common Interosseous Artery flow void on T2; contrast-enhanced MRA delineates stenosis, occlusion, variant anatomy; TOF sensitive to slow flow; assess collateral reconstitution distal to occlusion.

CT Angiography Findings: CTA: Common Interosseous Artery opacifies with iodinated contrast; evaluate stenosis (>50% hemodynamically significant), dissection (double lumen, intimal flap), aneurysm, embolus, or post-traumatic extravasation in acute trauma.

Collateral Circulation: Elbow recurrent interosseous collaterals

Surgical Considerations: Forearm fasciotomy with vascular preservation

Rehabilitation Considerations: Compartment rehab only after medical clearance

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Common Interosseous Artery]

RECORD: Anterior Interosseous Artery

ID: VES-UE-022

Name: Anterior Interosseous Artery

Latin Name: Arteria interossea anterior

Type: Artery

Region: Forearm — deep anterior

Origin: Common interosseous artery

Termination: Palmar carpal network and muscular branches

Course: Descends on anterior interosseous membrane with AIN

Major Branches: Muscular branches to deep flexors

Minor Branches: Palmar carpal branches

Structures Supplied: Deep flexor compartment and pronator quadratus

Muscles Supplied: FDP, FPL, pronator quadratus

Bones Supplied: Radius and ulna shafts

Joints Supplied: Distal radioulnar joint

Ligaments Supplied: Interosseous membrane anterior surface

Tendons Supplied: FDP and FPL tendons

Skin Distribution: None specific cutaneous

Venous Drainage: Anterior interosseous veins

Anastomoses: Posterior interosseous anastomoses

Clinical Importance: Forearm perfusion; AIN syndrome context

Common Injuries: Forearm fracture vascular compromise

Compression Sites: Interosseous membrane

Pulse Location: Not palpable

Ultrasound Appearance: Color Doppler: Anterior Interosseous Artery appears as pulsatile, anechoic/hypoechoic tubular structure with spectral broadening on stenosis; triphasic flow in proximal segments; assess for intimal flap (dissection), thrombus (non-compressible hypoechoic filling), or pseudoaneurysm.

MRI Appearance: MRA/time-resolved CE-MRA: Anterior Interosseous Artery flow void on T2; contrast-enhanced MRA delineates stenosis, occlusion, variant anatomy; TOF sensitive to slow flow; assess collateral reconstitution distal to occlusion.

CT Angiography Findings: CTA: Anterior Interosseous Artery opacifies with iodinated contrast; evaluate stenosis (>50% hemodynamically significant), dissection (double lumen, intimal flap), aneurysm, embolus, or post-traumatic extravasation in acute trauma.

Collateral Circulation: Posterior interosseous collaterals

Surgical Considerations: Forearm fracture ORIF

Rehabilitation Considerations: Intrinsic hand rehab after AIN recovery

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Anterior Interosseous Artery]

RECORD: Posterior Interosseous Artery

ID: VES-UE-023

Name: Posterior Interosseous Artery

Latin Name: Arteria interossea posterior

Type: Artery

Region: Forearm / dorsum hand

Origin: Common interosseous artery

Termination: Dorsal carpal arch

Course: Passes with PIN through or above supinator

Major Branches: Recurrent interosseous branch to elbow

Minor Branches: Muscular branches to extensor compartment

Structures Supplied: Extensor compartment and dorsum hand

Muscles Supplied: Extensor muscles and supinator

Bones Supplied: Dorsal radius and ulna

Joints Supplied: Distal radioulnar joint

Ligaments Supplied: Interosseous membrane posterior surface

Tendons Supplied: Extensor tendons

Skin Distribution: Dorsum hand skin

Venous Drainage: Posterior interosseous veins

Anastomoses: Anterior interosseous network

Clinical Importance: PIN syndrome and dorsal hand perfusion

Common Injuries: Monteggia fracture vascular injury

Compression Sites: Supinator arch (PIN compression)

Pulse Location: Not palpable

Ultrasound Appearance: Color Doppler: Posterior Interosseous Artery appears as pulsatile, anechoic/hypoechoic tubular structure with spectral broadening on stenosis; triphasic flow in proximal segments; assess for intimal flap (dissection),

thrombus (non-compressible hypoechoic filling), or pseudoaneurysm.

MRI Appearance: MRA/time-resolved CE-MRA: Posterior Interosseous Artery flow void on T2; contrast-enhanced MRA delineates stenosis, occlusion, variant anatomy; TOF sensitive to slow flow; assess collateral reconstitution distal to occlusion.

CT Angiography Findings: CTA: Posterior Interosseous Artery opacifies with iodinated contrast; evaluate stenosis (>50% hemodynamically significant), dissection (double lumen, intimal flap), aneurysm, embolus, or post-traumatic extravasation in acute trauma.

Collateral Circulation: Anterior interosseous collaterals

Surgical Considerations: Posterior forearm approach

Rehabilitation Considerations: Supination strengthening after PIN neuropraxia

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Posterior Interosseous Artery]

RECORD: Superficial Palmar Arch

ID: VES-UE-024

Name: Superficial Palmar Arch

Latin Name: Arcus palmaris superficialis

Type: Artery

Region: Hand — palm

Origin: Ulnar artery (usually dominant contributor)

Termination: Common palmar digital arteries to fingers

Course: Curves across palm superficial to flexor tendons

Major Branches: Common digital arteries to fingers II–V

Minor Branches: Thenar and hypothenar contributions

Structures Supplied: Palmar skin, fingers, lumbricals and interossei

Muscles Supplied: Thenar, hypothenar, lumbrical muscles

Bones Supplied: Metacarpals and phalanges

Joints Supplied: MCP and PIP joints

Ligaments Supplied: Palmar plate region

Tendons Supplied: Flexor tendons

Skin Distribution: Palmar hand skin

Venous Drainage: Superficial palmar venous plexus

Anastomoses: Deep palmar arch via perforators

Clinical Importance: Allen test; palmar laceration hemorrhage

Common Injuries: Glass and hand trauma laceration

Compression Sites: Thenar and hypothenar fascial tunnels

Pulse Location: Not separately palpable as arch pulse

Ultrasound Appearance: Color Doppler: Superficial Palmar Arch appears as pulsatile, anechoic/hypoechoic tubular structure with spectral broadening on stenosis; triphasic flow in proximal segments; assess for intimal flap (dissection), thrombus (non-compressible hypoechoic filling), or pseudoaneurysm.

MRI Appearance: MRA/time-resolved CE-MRA: Superficial Palmar Arch flow void on T2; contrast-enhanced MRA delineates stenosis, occlusion, variant anatomy; TOF sensitive to slow flow; assess collateral reconstitution distal to occlusion.

CT Angiography Findings: CTA: Superficial Palmar Arch opacifies with iodinated contrast; evaluate stenosis (>50% hemodynamically significant), dissection (double lumen, intimal flap), aneurysm, embolus, or post-traumatic extravasation in acute trauma.

Collateral Circulation: Deep arch collaterals

Surgical Considerations: Palmar wound repair with vascular exam

Rehabilitation Considerations: Progressive grip after palmar repair

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Superficial Palmar Arch]

RECORD: Deep Palmar Arch

ID: VES-UE-025

Name: Deep Palmar Arch

Latin Name: Arcus palmaris profundus

Type: Artery

Region: Hand — deep palm

Origin: Radial artery (usually dominant contributor)

Termination: Palmar metacarpal arteries

Course: Deep to flexor tendons on metacarpal bases

Major Branches: Princeps pollicis and radialis indicis

Minor Branches: Perforators to superficial arch

Structures Supplied: Thumb, index, deep hand structures

Muscles Supplied: Interossei and adductor pollicis

Bones Supplied: Metacarpals

Joints Supplied: CMC and MCP joints

Ligaments Supplied: Deep transverse metacarpal ligaments

Tendons Supplied: Flexor tendons deep surface

Skin Distribution: Thumb and index palmar skin

Venous Drainage: Deep palmar veins

Anastomoses: Superficial arch anastomoses

Clinical Importance: Thumb perfusion critical in crush injury

Common Injuries: Metacarpal fracture vascular injury

Compression Sites: Adductor pollicis space

Pulse Location: Not separately palpable

Ultrasound Appearance: Color Doppler: Deep Palmar Arch appears as pulsatile, anechoic/hypoechoic tubular structure with spectral broadening on stenosis; triphasic flow in proximal segments; assess for intimal flap (dissection), thrombus (non-compressible hypoechoic filling), or pseudoaneurysm.

MRI Appearance: MRA/time-resolved CE-MRA: Deep Palmar Arch flow void on T2; contrast-enhanced MRA delineates stenosis, occlusion, variant anatomy; TOF sensitive to slow flow; assess collateral reconstitution distal to occlusion.

CT Angiography Findings: CTA: Deep Palmar Arch opacifies with iodinated contrast; evaluate stenosis (>50% hemodynamically significant), dissection (double lumen, intimal flap), aneurysm, embolus, or post-traumatic extravasation in acute trauma.

Collateral Circulation: Superficial arch collaterals

Surgical Considerations: Hand replantation microsurgery

Rehabilitation Considerations: Pinch and grip rehab after deep hand injury

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Deep Palmar Arch]

RECORD: Dorsal Carpal Arch

ID: VES-UE-026

Name: Dorsal Carpal Arch

Latin Name: Rete carpal dorsale

Type: Artery

Region: Wrist / dorsum hand

Origin: Posterior interosseous and dorsal carpal branches

Termination: Dorsal metacarpal arteries

Course: Arches across dorsum of carpus beneath extensor retinaculum

Major Branches: Dorsal metacarpal arteries

Minor Branches: Dorsal digital branches

Structures Supplied: Dorsum hand and digits dorsally

Muscles Supplied: Extensor compartment muscles

Bones Supplied: Carpals and metacarpals

Joints Supplied: Wrist extension mechanics

Ligaments Supplied: Dorsal wrist ligaments

Tendons Supplied: Extensor tendons

Skin Distribution: Dorsum hand skin

Venous Drainage: Dorsal venous network

Anastomoses: Palmar arches via perforating branches

Clinical Importance: Dorsal hand flap perfusion

Common Injuries: Extensor tendon zone injuries

Compression Sites: Extensor retinaculum compartments

Pulse Location: Not palpable

Ultrasound Appearance: Color Doppler: Dorsal Carpal Arch appears as pulsatile, anechoic/hypoechoic tubular structure with spectral broadening on stenosis; triphasic flow in proximal segments; assess for intimal flap (dissection), thrombus (non-compressible hypoechoic filling), or pseudoaneurysm.

MRI Appearance: MRA/time-resolved CE-MRA: Dorsal Carpal Arch flow void on T2; contrast-enhanced MRA delineates stenosis, occlusion, variant anatomy; TOF sensitive to slow flow; assess collateral reconstitution distal to occlusion.

CT Angiography Findings: CTA: Dorsal Carpal Arch opacifies with iodinated contrast; evaluate stenosis (>50% hemodynamically significant), dissection (double lumen, intimal flap), aneurysm, embolus, or post-traumatic extravasation in acute trauma.

Collateral Circulation: Palmar arch collaterals

Surgical Considerations: Dorsal wrist surgery

Rehabilitation Considerations: Edema control and tendon gliding after dorsal injury

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Dorsal Carpal Arch]

RECORD: Common Digital Arteries

ID: VES-UE-027

Name: Common Digital Arteries

Latin Name: Arteriae digitales palmares communes

Type: Artery

Region: Fingers — palm

Origin: Superficial or deep palmar arch

Termination: Proper digital arteries at web spaces

Course: Run with digital nerves in fibrous sheaths toward web spaces

Major Branches: Proper digital artery pairs at each web space

Minor Branches: Dorsal perforators proximally

Structures Supplied: Fingers II–V palmar pulp and joints

Muscles Supplied: Lumbricals, interossei, flexor digitorum

Bones Supplied: Phalanges of digits II–V

Joints Supplied: MCP, PIP, and DIP joints

Ligaments Supplied: Collateral ligaments and volar plates

Tendons Supplied: Flexor digitorum tendons

Skin Distribution: Palmar digit skin

Venous Drainage: Palmar digital veins

Anastomoses: Contralateral digit collaterals via arches

Clinical Importance: Finger perfusion in crush and replantation

Common Injuries: Flexor tendon zone vascular injury

Compression Sites: Fibrous flexor sheath pulleys

Pulse Location: Not individually palpable

Ultrasound Appearance: Color Doppler: Common Digital Arteries appears as pulsatile, anechoic/hypoechoic tubular structure with spectral broadening on stenosis; triphasic flow in proximal segments; assess for intimal flap (dissection), thrombus (non-compressible hypoechoic filling), or pseudoaneurysm.

MRI Appearance: MRA/time-resolved CE-MRA: Common Digital Arteries flow void on T2; contrast-enhanced MRA delineates stenosis, occlusion, variant anatomy; TOF sensitive to slow flow; assess collateral reconstitution distal to occlusion.

CT Angiography Findings: CTA: Common Digital Arteries opacifies with iodinated contrast; evaluate stenosis (>50% hemodynamically significant), dissection (double lumen, intimal flap), aneurysm, embolus, or post-traumatic extravasation in acute trauma.

Collateral Circulation: Palmar arch collaterals

Surgical Considerations: Digit replantation microvascular repair

Rehabilitation Considerations: Tendon gliding within vascularized sheath post repair

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Common Digital Arteries]

RECORD: Proper Digital Arteries

ID: VES-UE-028

Name: Proper Digital Arteries

Latin Name: Arteriae digitales palmares propriae

Type: Artery

Region: Fingers — pulp

Origin: Common digital arteries

Termination: Terminal branches to finger pulp and nail bed

Course: Run lateral to flexor sheath with proper digital nerves

Major Branches: Terminal branches to finger pulp

Minor Branches: Dorsal branches to nail bed

Structures Supplied: Finger pulp, nail bed, DIP region

Muscles Supplied: Flexor digitorum profundus and superficialis

Bones Supplied: Phalanges

Joints Supplied: DIP, PIP, and MCP joints

Ligaments Supplied: Collateral ligaments

Tendons Supplied: FDP and FDS tendons

Skin Distribution: Finger pulp and nail bed

Venous Drainage: Digital veins

Anastomoses: Adjacent digit collaterals via arches

Clinical Importance: Fingertip amputation replantation viability

Common Injuries: Crush injury, frostbite, felon

Compression Sites: Pulp and nail bed compression

Pulse Location: Not palpable individually

Ultrasound Appearance: Color Doppler: Proper Digital Arteries appears as pulsatile, anechoic/hypoechoic tubular structure with spectral broadening on stenosis; triphasic flow in proximal segments; assess for intimal flap (dissection), thrombus (non-compressible hypoechoic filling), or pseudoaneurysm.

MRI Appearance: MRA/time-resolved CE-MRA: Proper Digital Arteries flow void on T2; contrast-enhanced MRA delineates stenosis, occlusion, variant anatomy; TOF sensitive to slow flow; assess collateral reconstitution distal to occlusion.

CT Angiography Findings: CTA: Proper Digital Arteries opacifies with iodinated contrast; evaluate stenosis (>50% hemodynamically significant), dissection (double lumen, intimal flap), aneurysm, embolus, or post-traumatic extravasation in acute trauma.

Collateral Circulation: Palmar arch collaterals

Surgical Considerations: Microvascular finger repair

Rehabilitation Considerations: Desensitization and ROM after fingertip injury

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Proper Digital Arteries]

2. Veins

RECORD: Cephalic Vein

ID: VES-UE-V01

Name: Cephalic Vein

Latin Name: Vena cephalica

Type: Vein

Region: Arm/shoulder — lateral

Origin: Dorsal venous network of hand

Termination: Axillary vein (or deltopectoral groove)

Course: Ascends lateral arm in superficial fascia with deltopectoral groove

Major Branches: Median cubital tributaries; dorsal metacarpal tributaries

Minor Branches: Cutaneous perforators

Structures Supplied: Lateral arm, shoulder, lateral thorax

Muscles Supplied: Deltoid and biceps superficial region

Bones Supplied: Clavicle lateral third

Joints Supplied: Shoulder region

Ligaments Supplied: Deltopectoral fascia

Tendons Supplied: None major

Skin Distribution: Lateral arm and forearm skin

Venous Drainage: Axillary vein

Anastomoses: Basilic and median cubital anastomoses

Clinical Importance: Primary IV access site; flap harvest conduit

Common Injuries: Laceration, thrombophlebitis after cannulation

Compression Sites: Clavipectoral groove; deltopectoral groove

Pulse Location: Not a pulse — compressible vein

Ultrasound Appearance: Compression ultrasound: Cephalic Vein compressible when patent; loss of compressibility suggests thrombosis; augment distal segments to assess proximal obstruction; color flow absent in acute thrombosis; chronic thrombus may show recanalization.

MRI Appearance: MRV/CE-MRV: Cephalic Vein flow void on T2; contrast delineates thrombosis, compression, or variant drainage in upper extremity venous mapping.

CT Angiography Findings: CTV: Cephalic Vein opacifies with contrast; evaluate DVT, central venous stenosis, catheter thrombosis, or thoracic outlet venous compression.

Collateral Circulation: Basilic and deep venous collaterals

Surgical Considerations: Preserve during shoulder surgery when possible

Rehabilitation Considerations: Avoid aggressive shoulder flexion with fresh cephalic harvest site

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Cephalic Vein]

RECORD: Basilic Vein

ID: VES-UE-V02

Name: Basilic Vein

Latin Name: Vena basilica

Type: Vein

Region: Arm — medial

Origin: Ulnar dorsal venous network

Termination: Axillary or brachial vein

Course: Ascends medial arm with medial cutaneous nerve of forearm

Major Branches: Median cubital vein; muscular tributaries

Minor Branches: Cutaneous branches

Structures Supplied: Medial arm and forearm

Muscles Supplied: Flexor compartment superficial region

Bones Supplied: Medial epicondyle region

Joints Supplied: Elbow region

Ligaments Supplied: Medial intermuscular septum region

Tendons Supplied: None major

Skin Distribution: Medial arm skin

Venous Drainage: Brachial/axillary vein

Anastomoses: Cephalic via median cubital; deep brachial veins

Clinical Importance: Preferred AV fistula conduit; IV access

Common Injuries: Thrombophlebitis; post-cannulation scarring

Compression Sites: Medial bicipital groove at elbow

Pulse Location: Visible subcutaneous medial arm

Ultrasound Appearance: Compression ultrasound: Basilic Vein compressible when patent; loss of compressibility suggests thrombosis; augment distal segments to assess proximal obstruction; color flow absent in acute thrombosis; chronic thrombus may show recanalization.

MRI Appearance: MRV/CE-MRV: Basilic Vein flow void on T2; contrast delineates thrombosis, compression, or variant drainage in upper extremity venous mapping.

CT Angiography Findings: CTV: Basilic Vein opacifies with contrast; evaluate DVT, central venous stenosis, catheter thrombosis, or thoracic outlet venous compression.

Collateral Circulation: Cephalic and deep venous collaterals

Surgical Considerations: Protect during medial elbow surgery

Rehabilitation Considerations: Elbow extension ROM after medial cannulation sites heal

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Basilic Vein]

RECORD: Median Cubital Vein

ID: VES-UE-V03

Name: Median Cubital Vein

Latin Name: Vena mediana cubiti

Type: Vein

Region: Elbow — antecubital

Origin: Cephalic and basilic veins

Termination: Drains to brachial or axillary via connecting veins

Course: Oblique superficial vein across antecubital fossa

Major Branches: None

Minor Branches: Cutaneous twigs

Structures Supplied: Antecubital fossa skin

Muscles Supplied: Biceps/brachialis superficial region

Bones Supplied: Distal humerus region

Joints Supplied: Elbow joint anterior

Ligaments Supplied: Bicipital aponeurosis region

Tendons Supplied: Biceps tendon region

Skin Distribution: Antecubital skin

Venous Drainage: Brachial vein

Anastomoses: Cephalic-basilic crossover — critical collateral

Clinical Importance: Most common phlebotomy site

Common Injuries: Hematoma, nerve injury (median/lateral antebrachial cutaneous)

Compression Sites: Bicipital aponeurosis compression with elbow flexion

Pulse Location: Visible in antecubital fossa with tourniquet

Ultrasound Appearance: Compression ultrasound: Median Cubital Vein compressible when patent; loss of compressibility suggests thrombosis; augment distal segments to assess proximal obstruction; color flow absent in acute thrombosis; chronic thrombus may show recanalization.

MRI Appearance: MRV/CE-MRV: Median Cubital Vein flow void on T2; contrast delineates thrombosis, compression, or variant drainage in upper extremity venous mapping.

CT Angiography Findings: CTV: Median Cubital Vein opacifies with contrast; evaluate DVT, central venous stenosis, catheter thrombosis, or thoracic outlet venous compression.

Collateral Circulation: Cephalic and basilic collaterals

Surgical Considerations: Avoid repeated venipuncture same site in athletes

Rehabilitation Considerations: Full elbow ROM once antecubital hematoma resolved

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Median Cubital Vein]

RECORD: Deep Veins (Paired Venae Comitantes)

ID: VES-UE-V04

Name: Deep Veins (Paired Venae Comitantes)

Latin Name: Venae comitantes

Type: Vein

Region: Arm/forearm — deep

Origin: Palmar/deep forearm venous plexus

Termination: Axillary vein via brachial veins

Course: Accompany brachial, radial, and ulnar arteries deep to fascia

Major Branches: Multiple unnamed tributaries from muscles

Minor Branches: Periosteal and articular drainage

Structures Supplied: Deep muscle compartments

Muscles Supplied: All deep flexor/extensor muscles

Bones Supplied: Humerus, radius, ulna

Joints Supplied: Elbow and wrist joints

Ligaments Supplied: Capsular plexus

Tendons Supplied: Deep tendons

Skin Distribution: Minimal cutaneous

Venous Drainage: Axillary vein

Anastomoses: Superficial cephalic/basilic via perforators

Clinical Importance: Primary drainage for muscle compartments; DVT less common UE

Common Injuries: Paget-Schroetter effort thrombosis; post-traumatic thrombosis

Compression Sites: Thoracic outlet; hyperabduction

Pulse Location: Not superficially visible

Ultrasound Appearance: Compression ultrasound: Deep Veins (Paired Venae Comitantes) compressible when patent; loss of compressibility suggests thrombosis; augment distal segments to assess proximal obstruction; color flow absent in acute thrombosis; chronic thrombus may show recanalization.

MRI Appearance: MRV/CE-MRV: Deep Veins (Paired Venae Comitantes) flow void on T2; contrast delineates thrombosis, compression, or variant drainage in upper extremity venous mapping.

CT Angiography Findings: CTV: Deep Veins (Paired Venae Comitantes) opacifies with contrast; evaluate DVT, central venous stenosis, catheter thrombosis, or thoracic outlet venous compression.

Collateral Circulation: Superficial venous collaterals

Surgical Considerations: Fasciotomy does not require deep vein ligation routinely

Rehabilitation Considerations: Gradual loading after UE DVT per hematology clearance

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Deep Veins (Paired Venae Comitantes)]

RECORD: Digital Veins

ID: VES-UE-V05

Name: Digital Veins

Latin Name: Venae digitales

Type: Vein

Region: Fingers/hand

Origin: Palmar and dorsal digital venous plexus

Termination: Deep palmar venous arch and dorsal network

Course: Parallel digital nerves along fingers

Major Branches: Connecting branches at web spaces

Minor Branches: Pulp venous plexus

Structures Supplied: Finger pulp and nail bed

Muscles Supplied: Intrinsic hand muscles

Bones Supplied: Phalanges

Joints Supplied: MCP/PIP/DIP joints

Ligaments Supplied: Collateral ligament plexus

Tendons Supplied: Flexor/extensor tendons

Skin Distribution: Finger skin dorsal and palmar

Venous Drainage: Deep palmar and dorsal arch veins

Anastomoses: Interdigital anastomoses

Clinical Importance: Replantation venous outflow critical for survival

Common Injuries: Crush injury thrombosis; felon spread

Compression Sites: Pulp and tight rings

Pulse Location: Visible on dorsum with tourniquet

Ultrasound Appearance: Compression ultrasound: Digital Veins compressible when patent; loss of compressibility suggests thrombosis; augment distal segments to assess proximal obstruction; color flow absent in acute thrombosis; chronic thrombus may show recanalization.

MRI Appearance: MRV/CE-MRV: Digital Veins flow void on T2; contrast delineates thrombosis, compression, or variant drainage in upper extremity venous mapping.

CT Angiography Findings: CTV: Digital Veins opacifies with contrast; evaluate DVT, central venous stenosis, catheter thrombosis, or thoracic outlet venous compression.

Collateral Circulation: Contralateral digital and arch collaterals

Surgical Considerations: Microvascular replantation venous repair

Rehabilitation Considerations: Edema management and ROM after finger replantation

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Digital Veins]

3. Fascia

RECORD: Investing Layer (Cervical Fascia)

ID: FAS-UE-001

Name: Investing Layer (Cervical Fascia)

Region: Neck

Layer: Superficial layer of deep cervical fascia

Thickness: 1–2 mm variable

Continuity: Encircles neck like collar; splits to invest SCM and trapezius

Attachments: Mandible, hyoid, nuchal ligament, spinous processes, clavicle

Compartments: Superficial neck compartment

Structures Enclosed: Platysma, external jugular vein

Muscles Covered: SCM, trapezius

Nerves Passing Through: Spinal accessory, cervical plexus branches

Vessels Passing Through: External jugular vein, transverse cervical vessels

Biomechanical Role: Maintains neck shape; transmits force from SCM/trapezius

Force Transmission: Links shoulder girdle elevation to cervical stabilizers

Mobility: Moderate — slides over deeper layers with swallowing/rotation

Sliding Characteristics: Platysma and SCM glide over pretracheal fascia

Clinical Importance: Infection spread (deep neck space); surgical landmark

Palpation: Palpable as tight band with SCM/trapezius tension

Ultrasound Appearance: High-resolution US: Investing Layer (Cervical Fascia) hyperechoic fibrillar layer; assess thickness, adhesion, and inter-layer sliding.

MRI Appearance: MRI: Investing Layer (Cervical Fascia) low T1/T2 signal; STIR hyperintensity with inflammation; defines compartment boundaries.

Fibrosis: Thickening in chronic neck pain and post-radiation fibrosis

Restrictions: Reduced rotation with global neck guarding

Common Injuries: Blunt neck trauma contusion

Manual Therapy: Myofascial release to SCM/trapezius; gentle skin rolling

Stretching: Upper trapezius and SCM stretching with chin tuck

Mobility Exercises: Cervical rotation and side flexion AROM

Rehabilitation: Post-whiplash gradual mobility with red flag screening

Evidence: Limited RCT evidence; widely used clinically for neck tension

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Investing Layer (Cervical Fascia)]

RECORD: Pretracheal Fascia

ID: FAS-UE-002

Name: Pretracheal Fascia

Region: Neck

Layer: Middle layer of deep cervical fascia

Thickness: Thin

Continuity: Invests thyroid, trachea, esophagus; attaches to hyoid and fibrous pericardium

Attachments: Hyoid bone, manubrium, visceral fascia

Compartments: Visceral compartment of neck

Structures Enclosed: Thyroid, trachea, esophagus

Muscles Covered: Infrahyoid strap muscles

Nerves Passing Through: Recurrent laryngeal nerves

Vessels Passing Through: Inferior thyroid arteries

Biomechanical Role: Visceral support during swallowing and neck extension

Force Transmission: Transmits hyoid stabilizer forces to thorax

Mobility: Low — moves with swallowing

Sliding Characteristics: Slides relative to prevertebral fascia during neck motion

Clinical Importance: Deep neck infection route; thyroid surgery landmark

Palpation: Not directly palpable; inferred with swallowing dysfunction

Ultrasound Appearance: High-resolution US: Pretracheal Fascia hyperechoic fibrillar layer; assess thickness, adhesion, and inter-layer sliding.

MRI Appearance: MRI: Pretracheal Fascia low T1/T2 signal; STIR hyperintensity with inflammation; defines compartment boundaries.

Fibrosis: Post-surgical adhesions after anterior neck surgery

Restrictions: Dysphagia-related neck guarding

Common Injuries: Penetrating trauma infection risk

Manual Therapy: Gentle throat/neck mobility post thyroidectomy when cleared

Stretching: Chin tuck and swallowing retraining

Mobility Exercises: Cervical AROM with swallowing coordination

Rehabilitation: Speech-language pathology integration post neck surgery

Evidence: Clinical consensus for post-surgical neck mobility

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Pretracheal Fascia]

RECORD: Prevertebral Fascia

ID: FAS-UE-003

Name: Prevertebral Fascia

Region: Neck

Layer: Deep layer anterior to vertebral column

Thickness: Thick posteriorly

Continuity: Covers anterior vertebral bodies and longus colli/capitis

Attachments: Base of skull to anterior longitudinal ligament continuity

Compartments: Prevertebral space

Structures Enclosed: Cervical spine, longus muscles, sympathetic trunk region

Muscles Covered: Longus colli, longus capitis, scalenes deep surface

Nerves Passing Through: Cervical nerve roots (nearby)

Vessels Passing Through: Vertebral and ascending cervical arteries

Biomechanical Role: Deep neck stability; barrier limiting infection spread to mediastinum

Force Transmission: Longitudinal force transmission during cervical flexion

Mobility: Moderate with flexion/extension

Sliding Characteristics: Longus colli glides during cervical flexion

Clinical Importance: Retropharyngeal abscess space boundary

Palpation: Deep anterior neck tightness with flexion loss

Ultrasound Appearance: High-resolution US: Prevertebral Fascia hyperechoic fibrillar layer; assess thickness, adhesion, and inter-layer sliding.

MRI Appearance: MRI: Prevertebral Fascia low T1/T2 signal; STIR hyperintensity with inflammation; defines compartment boundaries.

Fibrosis: Chronic postural flexion fibrosis in desk workers

Restrictions: Flexion restriction with deep neck pain

Common Injuries: Whiplash-associated deep flexor inhibition

Manual Therapy: Indirect release via deep neck flexor training

Stretching: Deep neck flexor activation before stretching

Mobility Exercises: Craniocervical flexion progression (CCF)

Rehabilitation: Progressive deep neck flexor retraining protocol

Evidence: Jull et al. deep neck flexor training evidence for neck pain

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Prevertebral Fascia]

RECORD: Carotid Sheath

ID: FAS-UE-004

Name: Carotid Sheath

Region: Neck

Layer: Condensation of deep cervical fascia

Thickness: Dense tubular sheath

Continuity: Invests common carotid, internal jugular vein, vagus nerve

Attachments: Skull base to root of neck

Compartments: Carotid sheath compartment

Structures Enclosed: Carotid artery, IJV, vagus nerve

Muscles Covered: None directly — adjacent infrahyoid muscles

Nerves Passing Through: Vagus, glossopharyngeal nearby

Vessels Passing Through: Common/internal carotid, internal jugular vein

Biomechanical Role: Protects neurovascular bundle; limits spread of infection

Force Transmission: Stabilizes carotid during neck rotation

Mobility: Low mobility

Sliding Characteristics: Minimal sliding — tethered to cervical structures

Clinical Importance: Carotid sinus hypersensitivity; central line placement

Palpation: Carotid pulse palpable lateral to trachea

Ultrasound Appearance: High-resolution US: Carotid Sheath hyperechoic fibrillar layer; assess thickness, adhesion, and inter-layer sliding.

MRI Appearance: MRI: Carotid Sheath low T1/T2 signal; STIR hyperintensity with inflammation; defines compartment boundaries.

Fibrosis: Post-radiation fibrosis encasing sheath

Restrictions: Autonomic symptoms with carotid tightness

Common Injuries: Blunt carotid injury (BCVI) in trauma

Manual Therapy: Avoid aggressive carotid triangle massage in vascular disease

Stretching: Gentle cervical rotation only when medically cleared

Mobility Exercises: Rotation AROM with monitoring for dizziness

Rehabilitation: BCVI screening protocols in high-energy trauma rehab

Evidence: AHA guidelines for blunt cerebrovascular injury screening

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Carotid Sheath]

RECORD: Clavipectoral Fascia

ID: FAS-UE-005

Name: Clavipectoral Fascia

Region: Shoulder

Layer: Deep fascia in deltopectoral groove

Thickness: Moderate

Continuity: Continuous with axillary sheath; pierced by cephalic vein and thoracoacromial vessels

Attachments: Clavicle, coracoid, axillary fascia

Compartments: Deltopectoral interval

Structures Enclosed: Cephalic vein, thoracoacromial trunk

Muscles Covered: Pectoralis major, deltoid deep surface

Nerves Passing Through: Lateral pectoral nerve

Vessels Passing Through: Cephalic vein, thoracoacromial artery

Biomechanical Role: Suspends axillary contents; resists inferior shoulder girdle displacement

Force Transmission: Transfers pectoralis force toward humerus

Mobility: Moderate with shoulder flexion/abduction

Sliding Characteristics: Deltoid glides over pectoralis in flexion

Clinical Importance: Deltopectoral approach landmark; TOS region

Palpation: Palpable groove between deltoid and pec major

Ultrasound Appearance: High-resolution US: Clavipectoral Fascia hyperechoic fibrillar layer; assess thickness, adhesion, and inter-layer sliding.

MRI Appearance: MRI: Clavipectoral Fascia low T1/T2 signal; STIR hyperintensity with inflammation; defines compartment boundaries.

Fibrosis: Post-surgical scarring after shoulder approach

Restrictions: Limited flexion with interval adhesions

Common Injuries: Pectoralis strain with fascial tearing

Manual Therapy: Cross-friction and soft tissue mobilization to interval

Stretching: Pec minor/deltoid doorway stretch

Mobility Exercises: Shoulder flexion and horizontal abduction AROM

Rehabilitation: Post-arthroscopy deltopectoral interval mobilization

Evidence: Clinical soft tissue mobilization literature

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Clavipectoral Fascia]

RECORD: Deltoid Fascia

ID: FAS-UE-006

Name: Deltoid Fascia

Region: Shoulder

Layer: Deep fascia investing deltoid

Thickness: Thin over muscle belly

Continuity: Continuous with arm fascia and pectoral fascia

Attachments: Acromion, clavicle, spine of scapula, deltoid tuberosity

Compartments: Shoulder superficial compartment

Structures Enclosed: Deltoid muscle

Muscles Covered: Deltoid

Nerves Passing Through: Axillary nerve terminal branches

Vessels Passing Through: Thoracoacromial deltoid branch

Biomechanical Role: Envelops deltoid force transmission to humerus

Force Transmission: Multi-directional shoulder force distribution

Mobility: High with shoulder ROM

Sliding Characteristics: Slides over humerus during abduction

Clinical Importance: Deltoid strain and shoulder impingement context

Palpation: Palpable over deltoid with contraction

Ultrasound Appearance: High-resolution US: Deltoid Fascia hyperechoic fibrillar layer; assess thickness, adhesion, and inter-layer sliding.

MRI Appearance: MRI: Deltoid Fascia low T1/T2 signal; STIR hyperintensity with inflammation; defines compartment boundaries.

Fibrosis: Post-injection fibrosis nodules

Restrictions: Painful arc with fascial restriction

Common Injuries: Deltoid contusion in contact sports

Manual Therapy: Myofascial release over deltoid fibers

Stretching: Cross-body and overhead deltoid stretch

Mobility Exercises: Scaption and abduction isometrics

Rehabilitation: Return to overhead sport after deltoid strain

Evidence: EMG-guided deltoid rehab protocols

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Deltoid Fascia]

RECORD: Supraspinous Fascia

ID: FAS-UE-007

Name: Supraspinous Fascia

Region: Shoulder/scapula

Layer: Thick fascia over supraspinatus fossa

Thickness: Thick

Continuity: Attached to scapular spine margins

Attachments: Scapular spine, acromion continuity

Compartments: Supraspinatus compartment

Structures Enclosed: Supraspinatus muscle and bursa region

Muscles Covered: Supraspinatus, trapezius inferior to spine

Nerves Passing Through: Suprascapular nerve (supraspinatus fossa)

Vessels Passing Through: Suprascapular artery

Biomechanical Role: Rotator cuff force transmission to scapula

Force Transmission: Links scapular elevation to humeral abduction

Mobility: Moderate with scapular motion

Sliding Characteristics: Supraspinatus tendon glides beneath acromion

Clinical Importance: Rotator cuff impingement and supraspinatus pathology

Palpation: Palpable tenderness over supraspinatus fossa

Ultrasound Appearance: High-resolution US: Supraspinous Fascia hyperechoic fibrillar layer; assess thickness, adhesion, and inter-layer sliding.

MRI Appearance: MRI: Supraspinous Fascia low T1/T2 signal; STIR hyperintensity with inflammation; defines compartment boundaries.

Fibrosis: Calcific tendinopathy associated thickening

Restrictions: Painful abduction arc

Common Injuries: Supraspinatus tear and tendinopathy

Manual Therapy: Posterior shoulder soft tissue mobilization

Stretching: Sleeper stretch for posterior capsule/fascia

Mobility Exercises: Scapular stabilization with external rotation

Rehabilitation: Rotator cuff progressive loading (Neer/Jobe protocols)

Evidence: Strong evidence for rotator cuff exercise therapy

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Supraspinous Fascia]

RECORD: Infraspinous Fascia

ID: FAS-UE-008

Name: Infraspinous Fascia

Region: Shoulder/scapula

Layer: Thick fascia over infraspinatus fossa

Thickness: Thick

Continuity: Attached to scapular spine and axillary border

Attachments: Scapular spine, teres major/minor border

Compartments: Infraspinatus compartment

Structures Enclosed: Infraspinatus, teres minor

Muscles Covered: Infraspinatus, teres minor

Nerves Passing Through: Suprascapular nerve (after spinoglenoid notch)

Vessels Passing Through: Suprascapular and circumflex scapular vessels

Biomechanical Role: Posterior rotator cuff stabilization force

Force Transmission: External rotation force transmission

Mobility: Moderate

Sliding Characteristics: Infraspinatus glides during ER

Clinical Importance: Posterior impingement and ER weakness

Palpation: Tenderness over infraspinous fossa

Ultrasound Appearance: High-resolution US: Infraspinous Fascia hyperechoic fibrillar layer; assess thickness, adhesion, and inter-layer sliding.

MRI Appearance: MRI: Infraspinous Fascia low T1/T2 signal; STIR hyperintensity with inflammation; defines compartment boundaries.

Fibrosis: Post-surgical posterior shoulder adhesions

Restrictions: ER loss with posterior shoulder pain

Common Injuries: Infraspinatus strain in throwers

Manual Therapy: Posterior glenohumeral mobilization

Stretching: Sleeper stretch; cross-body stretch

Mobility Exercises: Side-lying external rotation

Rehabilitation: Thrower's interval program posterior cuff focus

Evidence: Jobe/Kuhn thrower's ten program evidence

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Infraspinous Fascia]

RECORD: Brachial Fascia

ID: FAS-UE-009

Name: Brachial Fascia

Region: Arm

Layer: Deep fascia sleeve of arm

Thickness: Moderate

Continuity: Continuous with antebrachial fascia and axillary sheath

Attachments: Medial/lateral epicondyles, olecranon, axillary fascia

Compartments: Arm anterior and posterior compartments

Structures Enclosed: Brachial artery/veins, median/ulnar/radial nerves in places

Muscles Covered: Biceps, brachialis, triceps, coracobrachialis

Nerves Passing Through: Median, ulnar, musculocutaneous nerves

Vessels Passing Through: Brachial artery and venae comitantes

Biomechanical Role: Compartment boundary limiting muscle bulge

Force Transmission: Longitudinal tension transfer during elbow flexion/extension

Mobility: Moderate with elbow motion

Sliding Characteristics: Muscles glide within fascial sleeve

Clinical Importance: Compartment syndrome boundary; lymphatic flow

Palpation: Palpable tightness with flexion contracture

Ultrasound Appearance: High-resolution US: Brachial Fascia hyperechoic fibrillar layer; assess thickness, adhesion, and inter-layer sliding.

MRI Appearance: MRI: Brachial Fascia low T1/T2 signal; STIR hyperintensity with inflammation; defines compartment boundaries.

Fibrosis: Post-fracture arm fascial thickening

Restrictions: Elbow flexion contracture

Common Injuries: Compartment syndrome (rare arm)

Manual Therapy: Fascial stretching with elbow AROM

Stretching: Biceps/triceps stretching

Mobility Exercises: Elbow flexion/extension AROM

Rehabilitation: Post-humerus fracture mobility progression

Evidence: Compartment syndrome clinical monitoring protocols

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Brachial Fascia]

RECORD: Medial Intermuscular Septum

ID: FAS-UE-010

Name: Medial Intermuscular Septum

Region: Arm

Layer: Transverse septum arm

Thickness: Thin to moderate

Continuity: From medial humeral lip to brachial fascia

Attachments: Medial epicondyle region, medial humeral intermuscular ridge

Compartments: Medial (flexor) arm compartment boundary

Structures Enclosed: Flexor compartment vessels and nerves

Muscles Covered: Flexor compartment muscles

Nerves Passing Through: Ulnar nerve (posterior to septum at elbow)

Vessels Passing Through: Superior ulnar collateral artery region

Biomechanical Role: Separates flexor and extensor compartments

Force Transmission: Directs flexor force along humerus

Mobility: Low independent mobility

Sliding Characteristics: Minimal sliding

Clinical Importance: Cubital tunnel and flexor compartment syndrome

Palpation: Medial arm tenderness with nerve entrapment

Ultrasound Appearance: High-resolution US: Medial Intermuscular Septum hyperechoic fibrillar layer; assess thickness, adhesion, and inter-layer sliding.

MRI Appearance: MRI: Medial Intermuscular Septum low T1/T2 signal; STIR hyperintensity with inflammation; defines compartment boundaries.

Fibrosis: Post-traumatic septum thickening

Restrictions: Elbow flexion loss with medial arm pain

Common Injuries: Flexor compartment strain

Manual Therapy: Neural mobilization for ulnar nerve

Stretching: Flexor stretching with nerve glides

Mobility Exercises: Nerve gliding exercises

Rehabilitation: Cubital tunnel conservative management

Evidence: Novak cubital tunnel treatment guidelines

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Medial Intermuscular Septum]

RECORD: Lateral Intermuscular Septum

ID: FAS-UE-011

Name: Lateral Intermuscular Septum

Region: Arm

Layer: Transverse septum arm

Thickness: Thin to moderate

Continuity: From lateral humeral lip to brachial fascia

Attachments: Lateral epicondyle, deltoid tuberosity region

Compartments: Posterior/extensor boundary with anterior compartment

Structures Enclosed: Radial nerve in spiral groove region

Muscles Covered: Triceps, brachialis, mobile wad origins

Nerves Passing Through: Radial nerve

Vessels Passing Through: Profunda brachii artery

Biomechanical Role: Separates anterior and posterior compartments

Force Transmission: Extensor/triceps force guidance

Mobility: Low

Sliding Characteristics: Minimal

Clinical Importance: Radial nerve palsy with humerus fracture

Palpation: Lateral arm tenderness post fracture

Ultrasound Appearance: High-resolution US: Lateral Intermuscular Septum hyperechoic fibrillar layer; assess thickness, adhesion, and inter-layer sliding.

MRI Appearance: MRI: Lateral Intermuscular Septum low T1/T2 signal; STIR hyperintensity with inflammation; defines compartment boundaries.

Fibrosis: Humeral shaft fracture hematoma

Restrictions: Wrist drop with nerve injury

Common Injuries: Triceps/brachialis strain

Manual Therapy: Radial nerve gliding

Stretching: Wrist extensor stretching

Mobility Exercises: Radial nerve tensioner/tolerance

Rehabilitation: Humeral shaft fracture nerve recovery rehab

Evidence: Shao radial nerve palsy rehab protocols

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Lateral Intermuscular Septum]

RECORD: Antebrachial Fascia

ID: FAS-UE-012

Name: Antebrachial Fascia

Region: Forearm

Layer: Deep fascia of forearm

Thickness: Moderate

Continuity: Continuous with brachial fascia and retinacula

Attachments: Olecranon, lateral epicondyle, styloid processes, palmar/dorsal hand fascia

Compartments: Flexor and extensor compartments (mobile wad)

Structures Enclosed: Forearm muscles, neurovascular bundles

Muscles Covered: All forearm muscles

Nerves Passing Through: Median, ulnar, radial nerves

Vessels Passing Through: Radial/ulnar arteries, interosseous vessels

Biomechanical Role: Compartment envelope; retinacular anchor

Force Transmission: Grip and wrist force transmission to elbow

Mobility: High with wrist/pronation-supination

Sliding Characteristics: Tendons glide under retinacula

Clinical Importance: Compartment syndrome; De Quervain; intersection syndrome

Palpation: Palpable tightness with tenosynovitis

Ultrasound Appearance: High-resolution US: Antebrachial Fascia hyperechoic fibrillar layer; assess thickness, adhesion, and inter-layer sliding.

MRI Appearance: MRI: Antebrachial Fascia low T1/T2 signal; STIR hyperintensity with inflammation; defines compartment boundaries.

Fibrosis: Chronic tendinopathy thickening

Restrictions: Wrist/forearm pain with gripping

Common Injuries: Forearm compartment syndrome

Manual Therapy: Transverse friction massage to tendons

Stretching: Wrist flexor/extensor stretching

Mobility Exercises: Tendon gliding; nerve glides

Rehabilitation: Carpal tunnel and tendinopathy protocols

Evidence: Strong evidence for tendon gliding in CTS

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Antebrachial Fascia]

RECORD: Extensor Retinaculum

ID: FAS-UE-013

Name: Extensor Retinaculum

Region: Wrist — dorsal

Layer: Fibrous band over wrist extensors

Thickness: Thick transverse band

Continuity: Spans wrist dorsum from radius to ulna/styloid

Attachments: Radius/ulna styloids, hand fascia

Compartments: Six extensor compartments

Structures Enclosed: Extensor tendons and synovial sheaths

Muscles Covered: Extensor compartment muscles

Nerves Passing Through: Sensory branches radial/ulnar

Vessels Passing Through: Dorsal carpal branch radial artery

Biomechanical Role: Prevents bowstringing of extensors

Force Transmission: Wrist extension force concentration

Mobility: Low — tendons glide beneath

Sliding Characteristics: Tendon excursion 2–4 cm with wrist motion

Clinical Importance: De Quervain (1st compartment); extensor tenosynovitis

Palpation: Tenderness over compartments with resisted extension

Ultrasound Appearance: High-resolution US: Extensor Retinaculum hyperechoic fibrillar layer; assess thickness, adhesion, and inter-layer sliding.

MRI Appearance: MRI: Extensor Retinaculum low T1/T2 signal; STIR hyperintensity with inflammation; defines compartment boundaries.

Fibrosis: Stenosing tenosynovitis thickening

Restrictions: Painful wrist extension/grip

Common Injuries: ECU subsheath tear in athletes

Manual Therapy: Retinacular soft tissue mobilization

Stretching: Wrist extensor stretches; Tyler twist for lateral epicondyle

Mobility Exercises: Eccentric wrist extension

Rehabilitation: De Quervain splinting and loading

Evidence: Cochrane review supports splinting and injection for De Quervain

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Extensor Retinaculum]

RECORD: Flexor Retinaculum

ID: FAS-UE-014

Name: Flexor Retinaculum

Region: Wrist — volar

Layer: Transverse carpal ligament

Thickness: Thick (~2–4 mm)

Continuity: Carpal arch to pisiform/hook hamate

Attachments: Scaphoid/triquetrum pillars, pisiform, hook hamate

Compartments: Carpal tunnel

Structures Enclosed: Median nerve, flexor tendons FDS/FPL

Muscles Covered: Thenar/hypothenar origins

Nerves Passing Through: Median nerve, ulnar nerve (Guyon's canal separate)

Vessels Passing Through: Ulnar artery (Guyon's), palmar arches

Biomechanical Role: Maintains flexor mechanical advantage

Force Transmission: Grip force transmission through carpal arch

Mobility: Low — tendons glide in tunnel

Sliding Characteristics: Tendon gliding beneath TCL

Clinical Importance: Carpal tunnel syndrome primary structure

Palpation: Tinel over carpal tunnel

Ultrasound Appearance: High-resolution US: Flexor Retinaculum hyperechoic fibrillar layer; assess thickness, adhesion, and inter-layer sliding.

MRI Appearance: MRI: Flexor Retinaculum low T1/T2 signal; STIR hyperintensity with inflammation; defines compartment boundaries.

Fibrosis: Post-release scarring; pillar pain

Restrictions: Night paresthesia, thenar weakness

Common Injuries: CTS from repetitive grip

Manual Therapy: Median nerve gliding; TCL soft tissue

Stretching: Wrist flexor stretching; prayer stretch

Mobility Exercises: Tendon and nerve gliding programs

Rehabilitation: CTS post-surgical nerve glide progression

Evidence: AAOS CTS clinical practice guideline

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Flexor Retinaculum]

RECORD: Palmar Fascia

ID: FAS-UE-015

Name: Palmar Fascia

Region: Hand — palm

Layer: Palmar aponeurosis

Thickness: Thick central triangle

Continuity: Continuous with flexor retinaculum and digital fascia

Attachments: Thenar/hypothenar fascia, skin, MCP joints

Compartments: Central palm and digital rays

Structures Enclosed: Flexor tendons, lumbricals, neurovascular bundles

Muscles Covered: Thenar/hypothenar, lumbricals

Nerves Passing Through: Digital nerves and arteries

Vessels Passing Through: Palmar arches

Biomechanical Role: Grip force distribution; skin anchoring

Force Transmission: Longitudinal grip force to forearm

Mobility: Moderate with grip

Sliding Characteristics: Skin-fascia glide for power grip

Clinical Importance: Dupuytren contracture

Palpation: Nodules and cords palpable in palm

Ultrasound Appearance: High-resolution US: Palmar Fascia hyperechoic fibrillar layer; assess thickness, adhesion, and inter-layer sliding.

MRI Appearance: MRI: Palmar Fascia low T1/T2 signal; STIR hyperintensity with inflammation; defines compartment boundaries.

Fibrosis: Dupuytren cord thickening

Restrictions: MCP/PIP flexion contracture

Common Injuries: Dupuytren disease

Manual Therapy: Skin and fascial mobilization pre/post surgery

Stretching: Digit extension stretching

Mobility Exercises: Intrinsic strengthening

Rehabilitation: Dupuytren post-fasciectomy therapy

Evidence: Dupuytren society rehab guidelines

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Palmar Fascia]

RECORD: Thenar Fascia

ID: FAS-UE-016

Name: Thenar Fascia

Region: Hand — thumb

Layer: Fascia over thenar eminence

Thickness: Moderate

Continuity: From flexor retinaculum to thumb MCP

Attachments: First metacarpal, thumb rays

Compartments: Thenar compartment

Structures Enclosed: Thenar muscles, thumb tendons

Muscles Covered: Abductor pollicis brevis, opponens, FPB

Nerves Passing Through: Median nerve motor branch

Vessels Passing Through: Superficial palmar branch radial artery

Biomechanical Role: Thumb opposition and grip stability

Force Transmission: Pinch force transmission

Mobility: High with thumb motion

Sliding Characteristics: Thenar glide during opposition

Clinical Importance: Thenar weakness in CTS and CMC arthritis

Palpation: Thenar eminence palpation with APB test

Ultrasound Appearance: High-resolution US: Thenar Fascia hyperechoic fibrillar layer; assess thickness, adhesion, and inter-layer sliding.

MRI Appearance: MRI: Thenar Fascia low T1/T2 signal; STIR hyperintensity with inflammation; defines compartment boundaries.

Fibrosis: Post-op scarring after CMC surgery

Restrictions: Opposition loss

Common Injuries: Skier's thumb; CMC sprain

Manual Therapy: Thenar soft tissue mobilization

Stretching: Thumb opposition stretches

Mobility Exercises: Opposition and abduction strengthening

Rehabilitation: CMC arthritis thumb stabilization program

Evidence: Eaton CMC arthritis staging and rehab

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Thenar Fascia]

RECORD: Hypothenar Fascia

ID: FAS-UE-017

Name: Hypothenar Fascia

Region: Hand — ulnar

Layer: Fascia over hypothenar eminence

Thickness: Moderate

Continuity: From flexor retinaculum/pisiform to 5th ray

Attachments: Pisiform, hook hamate, 5th metacarpal

Compartments: Hypothenar compartment

Structures Enclosed: Hypothenar muscles, ulnar nerve/artery

Muscles Covered: Abductor digiti minimi, flexor digiti minimi, opponens digiti minimi

Nerves Passing Through: Ulnar nerve deep branch nearby

Vessels Passing Through: Ulnar artery in Guyon's canal

Biomechanical Role: Power grip ulnar border stability

Force Transmission: Ulnar grip force transmission

Mobility: Moderate

Sliding Characteristics: Hypothenar glide with grip

Clinical Importance: Guyon's canal syndrome; hook hamate fracture

Palpation: Tenderness over pisiform/hook hamate

Ultrasound Appearance: High-resolution US: Hypothenar Fascia hyperechoic fibrillar layer; assess thickness, adhesion, and inter-layer sliding.

MRI Appearance: MRI: Hypothenar Fascia low T1/T2 signal; STIR hyperintensity with inflammation; defines compartment boundaries.

Fibrosis: Post-fracture fibrosis

Restrictions: Ulnar-sided wrist pain with grip

Common Injuries: Hypothenar hammer syndrome

Manual Therapy: Pisiform mobilization when appropriate

Stretching: Wrist ulnar deviation stretch

Mobility Exercises: Hypothenar strengthening

Rehabilitation: Guyon's canal decompression rehab

Evidence: Clinical Guyon's canal management consensus

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Hypothenar Fascia]

RECORD: Digital Fascia

ID: FAS-UE-018

Name: Digital Fascia

Region: Fingers

Layer: Fibrous flexor sheaths and lateral digital fascia

Thickness: Thin over dorsum; thick volar sheath

Continuity: Continuous with palmar aponeurosis and extensor hood

Attachments: MCP, PIP, DIP pulleys (A1–A5, C1–C3)

Compartments: Digital compartments

Structures Enclosed: Flexor tendons, neurovascular bundles, extensor mechanism

Muscles Covered: Lumbricals, interossei, flexors

Nerves Passing Through: Digital nerves

Vessels Passing Through: Proper digital arteries

Biomechanical Role: Pulley system for flexor mechanical advantage

Force Transmission: Precision and power grip force to hand

Mobility: High tendon excursion

Sliding Characteristics: Tendon gliding within pulleys (2–3 cm FDP)

Clinical Importance: Trigger finger; pulley injury in climbers

Palpation: Tenderness at A1 pulley with triggering

Ultrasound Appearance: High-resolution US: Digital Fascia hyperechoic fibrillar layer; assess thickness, adhesion, and inter-layer sliding.

MRI Appearance: MRI: Digital Fascia low T1/T2 signal; STIR hyperintensity with inflammation; defines compartment boundaries.

Fibrosis: Trigger finger nodule at A1

Restrictions: Locking digit with pain

Common Injuries: Pulley rupture (A2/A4) in climbing

Manual Therapy: Tendon gliding; pulley soft tissue care

Stretching: Tendon gliding exercises grades I–IV

Mobility Exercises: Blocking exercises for flexion

Rehabilitation: Trigger finger post-release and conservative

Evidence: Wrist/hand tendon gliding evidence (Cannon)

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Digital Fascia]

4. Biomechanics Movements

RECORD: Cervical Flexion

ID: MOV-UE-001

Movement: Cervical Flexion

Joint: Cervical spine

Body Region: Cervical

Plane: Sagittal

Axis: Mediolateral

Open Chain: Chin tuck with head moving forward in space

Closed Chain: Quadruped neck neutral with scapular load

Primary Movers: Longus colli, longus capitis, sternocleidomastoid

Secondary Movers: Scalenes, platysma

Stabilizers: Deep cervical flexors; upper cervical extensors eccentrically

Force Couples: Agonist-antagonist pairs coordinating cervical flexion with regional stabilizers

Prime Stabilizers: Deep cervical flexors

Dynamic Stabilizers: Scalenes, platysma

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Inferior facet slides posterior-superior; anterior vertebral bodies approximate

Osteokinematics: Head translates anterior; chin moves toward sternum

Joint Reaction Force: Compression increases with load during cervical flexion especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Longus colli as primary during standardized cervical flexion

Normal ROM: 0–45° total (varies by level)

Functional ROM: 20–30° for reading and device use

Compensations: Cervical protraction compensating for thoracic kyphosis

Common Dysfunctions: Deep flexor inhibition; upper crossed syndrome

Common Pain Patterns: Suboccipital and upper trapezius referral

Clinical Relevance: Neck pain with prolonged flexion posture; whiplash screening

Functional Activities: ADL and occupational tasks requiring cervical flexion within symptom-free range

Sport-Specific Applications: Swimming starts, wrestling takedown posture

Common Injuries: Whiplash, disc herniation aggravation

Assessment: Active ROM, CCFT, postural assessment

Treatment Implications: Deep neck flexor training before stretching

Exercise Progressions: Craniocervical flexion progression → seated postural re-education

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Cervical Flexion]

RECORD: Cervical Extension

ID: MOV-UE-002

Movement: Cervical Extension

Joint: Cervical spine

Body Region: Cervical

Plane: Sagittal

Axis: Mediolateral

Open Chain: Head extends in open chain from neutral

Closed Chain: Prone extension with thoracic support

Primary Movers: Semispinalis capitis/cervicis, splenius, upper trapezius

Secondary Movers: Levator scapulae, erector spinae

Stabilizers: Deep cervical extensors; anterior neck flexors eccentrically

Force Couples: Agonist-antagonist pairs coordinating cervical extension with regional stabilizers

Prime Stabilizers: Deep cervical extensors

Dynamic Stabilizers: Levator scapulae, erector spinae

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Facet joints approximate; spinous processes separate

Osteokinematics: Head moves posterior-superior

Joint Reaction Force: Compression increases with load during cervical extension especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Semispinalis capitis/cervicis as primary during standardized cervical extension

Normal ROM: 0–55° total

Functional ROM: 15–25° for overhead visual tasks

Compensations: Thoracic extension substitute for cervical extension

Common Dysfunctions: Extensor dominance; loss of flexor balance

Common Pain Patterns: Suboccipital and interscapular pain

Clinical Relevance: Extension-intolerant neck pain; stenosis caution

Functional Activities: ADL and occupational tasks requiring cervical extension within symptom-free range

Sport-Specific Applications: Gymnastics, dance, overhead sport visual tracking

Common Injuries: Facet syndrome, stenosis

Assessment: Extension ROM, Spurling if indicated

Treatment Implications: Limit end-range extension in stenosis

Exercise Progressions: Isometric extension → controlled AROM with flexor co-contraction

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Cervical Extension]

RECORD: Cervical Rotation

ID: MOV-UE-003

Movement: Cervical Rotation

Joint: Cervical spine

Body Region: Cervical

Plane: Transverse
Axis: Vertical
Open Chain: Head rotates in space
Closed Chain: Quadruped rotation with trunk stabilized
Primary Movers: Sternocleidomastoid contralateral, splenius ipsilateral
Secondary Movers: Scalenes, levator scapulae
Stabilizers: Deep rotators, upper cervical stabilizers
Force Couples: Agonist-antagonist pairs coordinating cervical rotation with regional stabilizers
Prime Stabilizers: Deep rotators, upper cervical stabilizers
Dynamic Stabilizers: Scalenes, levator scapulae
Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range
Arthrokinematics: Atlas rotates on axis at C1–C2; lower levels couple with lateral flexion
Osteokinematics: Head turns left/right
Joint Reaction Force: Compression increases with load during cervical rotation especially at end-range
Compression Forces: Articular approximation with muscle co-contraction and axial loading
Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability
Torque: Muscular torque balanced by eccentric antagonists for controlled motion
Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range
Lever Class: Third-class lever dominant in open-chain upper extremity motion
Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition
EMG Evidence: EMG confirms Sternocleidomastoid contralateral as primary during standardized cervical rotation
Normal ROM: 0–80° each side (C1–C2 ~50% of motion)
Functional ROM: 60–70° for driving mirror checks
Compensations: Thoracic rotation compensating
Common Dysfunctions: Rotation loss after whiplash
Common Pain Patterns: Occipital and trapezius pain
Clinical Relevance: Vertebrobasilar insufficiency screening with rotation
Functional Activities: ADL and occupational tasks requiring cervical rotation within symptom-free range
Sport-Specific Applications: Tennis serve prep, checking blind spots
Common Injuries: Whiplash, facet pain
Assessment: Rotation ROM, vertebral artery symptom monitoring
Treatment Implications: Gradual rotation in VBI history
Exercise Progressions: Rotation AROM in pain-free range → combined patterns
References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Cervical Rotation]

RECORD: Cervical Side Flexion

ID: MOV-UE-004
Movement: Cervical Side Flexion
Joint: Cervical spine
Body Region: Cervical
Plane: Frontal
Axis: Anteroposterior
Open Chain: Ear toward shoulder open chain
Closed Chain: Side-lying with head supported

Primary Movers: Scalenes, levator scapulae, upper trapezius

Secondary Movers: Sternocleidomastoid, splenius

Stabilizers: Deep cervical stabilizers contralateral

Force Couples: Agonist-antagonist pairs coordinating cervical side flexion with regional stabilizers

Prime Stabilizers: Deep cervical stabilizers contralateral

Dynamic Stabilizers: Sternocleidomastoid, splenius

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Uncinate processes guide lateral gliding; ipsilateral facets open

Osteokinematics: Head tilts laterally

Joint Reaction Force: Compression increases with load during cervical side flexion especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Scalenes as primary during standardized cervical side flexion

Normal ROM: 0–45° each side

Functional ROM: 30° for phone holding tasks

Compensations: Shoulder hiking substitute

Common Dysfunctions: Upper crossed posture

Common Pain Patterns: Trapezius and levator scapulae trigger points

Clinical Relevance: Radiculopathy may increase with side flexion toward painful side

Functional Activities: ADL and occupational tasks requiring cervical side flexion within symptom-free range

Sport-Specific Applications: Swimming breathing side preference

Common Injuries: Disc herniation, facet irritation

Assessment: Side flexion ROM, upper limb tension tests if indicated

Treatment Implications: Avoid forced side flexion in acute radiculopathy

Exercise Progressions: Side flexion AROM → combined with rotation patterns

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Cervical Side Flexion]

RECORD: Shoulder Flexion

ID: MOV-UE-005

Movement: Shoulder Flexion

Joint: Glenohumeral joint

Body Region: Shoulder

Plane: Sagittal

Axis: Mediolateral

Open Chain: Arm elevates forward in space

Closed Chain: Hand fixed — body moves over hand (push-up top)

Primary Movers: Anterior deltoid, pectoralis major clavicular head

Secondary Movers: Coracobrachialis, biceps long head

Stabilizers: Rotator cuff, serratus anterior, lower trapezius

Force Couples: Agonist-antagonist pairs coordinating shoulder flexion with regional stabilizers

Prime Stabilizers: Rotator cuff, serratus anterior, lower trapezius

Dynamic Stabilizers: Coracobrachialis, biceps long head

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Humeral head rolls and glides inferior-posterior on glenoid with elevation

Osteokinematics: Humerus moves anterior-superior in sagittal plane

Joint Reaction Force: Compression increases with load during shoulder flexion especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Anterior deltoid as primary during standardized shoulder flexion

Normal ROM: 0–180°

Functional ROM: 120–150° for most ADL overhead

Compensations: Lumbar extension with overhead reach

Common Dysfunctions: Scapular downward rotation with flexion

Common Pain Patterns: Lateral deltoid and subacromial pain

Clinical Relevance: Impingement with painful arc; scapular dyskinesis common

Functional Activities: ADL and occupational tasks requiring shoulder flexion within symptom-free range

Sport-Specific Applications: Swimming, volleyball, overhead throw

Common Injuries: Impingement, biceps tendinopathy

Assessment: Flexion ROM, painful arc, Neer test

Treatment Implications: Scapular stabilization before high flexion loading

Exercise Progressions: Scaption → full flexion with scapular control

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Shoulder Flexion]

RECORD: Shoulder Extension

ID: MOV-UE-006

Movement: Shoulder Extension

Joint: Glenohumeral joint

Body Region: Shoulder

Plane: Sagittal

Axis: Mediolateral

Open Chain: Arm moves posterior in space

Closed Chain: Closed chain pulling body forward

Primary Movers: Latissimus dorsi, teres major, posterior deltoid

Secondary Movers: Triceps long head, sternal pectoralis

Stabilizers: Rotator cuff posterior, rhomboids

Force Couples: Agonist-antagonist pairs coordinating shoulder extension with regional stabilizers

Prime Stabilizers: Rotator cuff posterior, rhomboids

Dynamic Stabilizers: Triceps long head, sternal pectoralis

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Posterior roll-glide of humeral head

Osteokinematics: Humerus moves posterior-inferior

Joint Reaction Force: Compression increases with load during shoulder extension especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Latissimus dorsi as primary during standardized shoulder extension

Normal ROM: 0–60°

Functional ROM: 30–45° for reaching behind back

Compensations: Scapular anterior tilt substitute

Common Dysfunctions: Posterior capsule tightness

Common Pain Patterns: Posterior shoulder and scapular pain

Clinical Relevance: Posterior shoulder tightness limits extension

Functional Activities: ADL and occupational tasks requiring shoulder extension within symptom-free range

Sport-Specific Applications: Rowing pull-through, gymnastics swing

Common Injuries: Posterior impingement in throwers

Assessment: Extension ROM, internal rotation composite

Treatment Implications: Posterior capsule mobility for extension

Exercise Progressions: Prone extension → resisted extension with scapular retraction

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Shoulder Extension]

RECORD: Shoulder Abduction

ID: MOV-UE-007

Movement: Shoulder Abduction

Joint: Glenohumeral joint

Body Region: Shoulder

Plane: Frontal

Axis: Anteroposterior

Open Chain: Arm elevates laterally

Closed Chain: Weight-bearing on fixed hand

Primary Movers: Middle deltoid, supraspinatus

Secondary Movers: Trapezius upper, serratus anterior

Stabilizers: Rotator cuff, lower trapezius, serratus

Force Couples: Agonist-antagonist pairs coordinating shoulder abduction with regional stabilizers

Prime Stabilizers: Rotator cuff, lower trapezius, serratus

Dynamic Stabilizers: Trapezius upper, serratus anterior

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Humeral head inferior glide with abduction; scapular upward rotation after ~30°

Osteokinematics: Humerus moves lateral away from trunk

Joint Reaction Force: Compression increases with load during shoulder abduction especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Middle deltoid as primary during standardized shoulder abduction

Normal ROM: 0–180° with scapulothoracic contribution

Functional ROM: 90–120° functional reaching

Compensations: Scapular elevation without upward rotation

Common Dysfunctions: Hitting painful arc; shrugging

Common Pain Patterns: Lateral shoulder pain

Clinical Relevance: Subacromial impingement at 60–120° painful arc

Functional Activities: ADL and occupational tasks requiring shoulder abduction within symptom-free range

Sport-Specific Applications: Swimming, lateral raises, volleyball

Common Injuries: Supraspinatus tendinopathy, impingement

Assessment: Abduction ROM, Hawkins/Kennedy

Treatment Implications: Rotator cuff and scapular rehab before heavy abduction

Exercise Progressions: Scaption → abduction with serratus activation

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Shoulder Abduction]

RECORD: Shoulder Adduction

ID: MOV-UE-008

Movement: Shoulder Adduction

Joint: Glenohumeral joint

Body Region: Shoulder

Plane: Frontal

Axis: Anteroposterior

Open Chain: Arm moves toward midline

Closed Chain: Closed chain adduction force in bear crawl

Primary Movers: Pectoralis major, latissimus dorsi, teres major

Secondary Movers: Coracobrachialis, short head biceps

Stabilizers: Rotator cuff, rhomboids

Force Couples: Agonist-antagonist pairs coordinating shoulder adduction with regional stabilizers

Prime Stabilizers: Rotator cuff, rhomboids

Dynamic Stabilizers: Coracobrachialis, short head biceps

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Superior roll-glide with adduction force

Osteokinematics: Humerus moves medial toward trunk

Joint Reaction Force: Compression increases with load during shoulder adduction especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Pectoralis major as primary during standardized shoulder adduction

Normal ROM: 0–30° across body

Functional ROM: Functional hugging, equipment carry

Compensations: Scapular protraction substitute

Common Dysfunctions: Pec tightness limiting ER

Common Pain Patterns: AC joint and anterior shoulder pain

Clinical Relevance: AC joint pain with cross-body adduction

Functional Activities: ADL and occupational tasks requiring shoulder adduction within symptom-free range

Sport-Specific Applications: Gymnastics iron cross prep, tackling

Common Injuries: AC sprain, pec strain

Assessment: Horizontal adduction ROM, cross-body test

Treatment Implications: AC joint protection in painful cross-body

Exercise Progressions: Isometric adduction → band pull-apart reverse

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Shoulder Adduction]

RECORD: Shoulder Horizontal Abduction

ID: MOV-UE-009

Movement: Shoulder Horizontal Abduction

Joint: Glenohumeral joint

Body Region: Shoulder

Plane: Transverse

Axis: Vertical

Open Chain: Arm moves posterior in horizontal plane

Closed Chain: Reverse fly closed chain against wall

Primary Movers: Posterior deltoid, infraspinatus, teres minor

Secondary Movers: Middle trapezius, rhomboids

Stabilizers: Rotator cuff, scapular retractors

Force Couples: Agonist-antagonist pairs coordinating shoulder horizontal abduction with regional stabilizers

Prime Stabilizers: Rotator cuff, scapular retractors

Dynamic Stabilizers: Middle trapezius, rhomboids

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Posterior glide of humeral head

Osteokinematics: Humerus moves posterior horizontal

Joint Reaction Force: Compression increases with load during shoulder horizontal abduction especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Posterior deltoid as primary during standardized shoulder horizontal abduction

Normal ROM: 0–30°

Functional ROM: 20° for reaching behind

Compensations: Trunk rotation substitute

Common Dysfunctions: Rhomboid dominance with winging
Common Pain Patterns: Posterior shoulder pain
Clinical Relevance: Posterior cuff strengthening for balance
Functional Activities: ADL and occupational tasks requiring shoulder horizontal abduction within symptom-free range
Sport-Specific Applications: Swimming backstroke, rowing finish
Common Injuries: Posterior impingement
Assessment: Horizontal abduction strength, ER at 90
Treatment Implications: Posterior cuff and scapular retraction focus
Exercise Progressions: Prone T → reverse fly progression
References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Shoulder Horizontal Abduction]

RECORD: Shoulder Horizontal Adduction

ID: MOV-UE-010

Movement: Shoulder Horizontal Adduction

Joint: Glenohumeral joint

Body Region: Shoulder

Plane: Transverse

Axis: Vertical

Open Chain: Arm crosses body horizontally

Closed Chain: Push-up and bench press closed chain

Primary Movers: Pectoralis major, anterior deltoid

Secondary Movers: Coracobrachialis, latissimus

Stabilizers: Serratus anterior, rotator cuff

Force Couples: Agonist-antagonist pairs coordinating shoulder horizontal adduction with regional stabilizers

Prime Stabilizers: Serratus anterior, rotator cuff

Dynamic Stabilizers: Coracobrachialis, latissimus

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Anterior glide with horizontal adduction

Osteokinematics: Humerus crosses midline anteriorly

Joint Reaction Force: Compression increases with load during shoulder horizontal adduction especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Pectoralis major as primary during standardized shoulder horizontal adduction

Normal ROM: 0–130°

Functional ROM: 90° for push and hug tasks

Compensations: Scapular abduction substitute

Common Dysfunctions: Rounded shoulder posture

Common Pain Patterns: Anterior shoulder and pec pain

Clinical Relevance: Pec dominance in bench press

Functional Activities: ADL and occupational tasks requiring shoulder horizontal adduction within symptom-free range

Sport-Specific Applications: Bench press, push-ups, blocking in football

Common Injuries: Pec strain, AC injury

Assessment: Horizontal adduction ROM, pec length

Treatment Implications: Balance with posterior cuff

Exercise Progressions: Push-up plus → bench press progressive loading

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Shoulder Horizontal Adduction]

RECORD: Shoulder Internal Rotation

ID: MOV-UE-011

Movement: Shoulder Internal Rotation

Joint: Glenohumeral joint

Body Region: Shoulder

Plane: Transverse

Axis: Vertical

Open Chain: Forearm rotates medially with humerus fixed or moving

Closed Chain: Closed chain IR in handstand prep

Primary Movers: Subscapularis, pectoralis major, latissimus dorsi

Secondary Movers: Teres major, anterior deltoid

Stabilizers: Rotator cuff posterior eccentrically

Force Couples: Agonist-antagonist pairs coordinating shoulder internal rotation with regional stabilizers

Prime Stabilizers: Rotator cuff posterior eccentrically

Dynamic Stabilizers: Teres major, anterior deltoid

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Posterior-inferior roll-glide with IR

Osteokinematics: Humerus rotates medially

Joint Reaction Force: Compression increases with load during shoulder internal rotation especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Subscapularis as primary during standardized shoulder internal rotation

Normal ROM: 0–70° at 0° abduction; 0–70° at 90° abduction

Functional ROM: 60° for behind-back tasks

Compensations: Scapular protraction with IR

Common Dysfunctions: Loss of IR after immobilization

Common Pain Patterns: Posterior shoulder and anterior cuff pain

Clinical Relevance: GIRD in throwers — IR tightness common

Functional Activities: ADL and occupational tasks requiring shoulder internal rotation within symptom-free range

Sport-Specific Applications: Throwing, tennis serve, swimming

Common Injuries: Subscapularis strain, GIRD

Assessment: IR ROM side-lying, total arc motion

Treatment Implications: Posterior capsule stretch in throwers carefully

Exercise Progressions: Side-lying IR → sleeper stretch protocol

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Shoulder Internal Rotation]

RECORD: Shoulder External Rotation

ID: MOV-UE-012

Movement: Shoulder External Rotation

Joint: Glenohumeral joint

Body Region: Shoulder

Plane: Transverse

Axis: Vertical

Open Chain: Forearm rotates laterally

Closed Chain: Closed chain ER in quadruped

Primary Movers: Infraspinatus, teres minor

Secondary Movers: Posterior deltoid, middle trapezius

Stabilizers: Subscapularis eccentrically; scapular stabilizers

Force Couples: Agonist-antagonist pairs coordinating shoulder external rotation with regional stabilizers

Prime Stabilizers: Subscapularis eccentrically

Dynamic Stabilizers: Posterior deltoid, middle trapezius

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Anterior-superior roll-glide with ER

Osteokinematics: Humerus rotates laterally

Joint Reaction Force: Compression increases with load during shoulder external rotation especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Infraspinatus as primary during standardized shoulder external rotation

Normal ROM: 0–90° at 0° abduction; 0–90° at 90° abduction

Functional ROM: 70° at 90° abduction for throwers

Compensations: Abduction substitute for ER

Common Dysfunctions: ER weakness with scapular winging

Common Pain Patterns: Lateral and posterior shoulder pain

Clinical Relevance: ER weakness linked to impingement and instability

Functional Activities: ADL and occupational tasks requiring shoulder external rotation within symptom-free range

Sport-Specific Applications: Throwing, racquet sports, swimming

Common Injuries: Rotator cuff tear, impingement

Assessment: ER strength at 0° and 90° abduction

Treatment Implications: Rotator cuff ER progressive loading

Exercise Progressions: Side-lying ER → standing ER at 90° abduction

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Shoulder External Rotation]

RECORD: Scaption

ID: MOV-UE-013

Movement: Scaption

Joint: Glenohumeral joint

Body Region: Shoulder

Plane: Scapular plane

Axis: Oblique

Open Chain: Arm elevates 30° anterior to frontal plane

Closed Chain: Hand-supported scapular plane loading

Primary Movers: Supraspinatus, deltoid middle/anterior

Secondary Movers: Trapezius, serratus anterior

Stabilizers: Rotator cuff, lower trapezius

Force Couples: Agonist-antagonist pairs coordinating scaption with regional stabilizers

Prime Stabilizers: Rotator cuff, lower trapezius

Dynamic Stabilizers: Trapezius, serratus anterior

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Optimal supraspinatus line of pull in scapular plane

Osteokinematics: Humerus elevates in plane of scapula

Joint Reaction Force: Compression increases with load during scaption especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Supraspinatus as primary during standardized scaption

Normal ROM: 0–180° in scapular plane (~30° anterior to frontal)

Functional ROM: 90–120° functional

Compensations: Pure frontal plane substitute

Common Dysfunctions: Shrugging during scaption

Common Pain Patterns: Subacromial pain

Clinical Relevance: Preferred plane for impingement-minimized elevation

Functional Activities: ADL and occupational tasks requiring scaption within symptom-free range

Sport-Specific Applications: Overhead sport, lifting

Common Injuries: Supraspinatus tendinopathy

Assessment: Scaption strength and ROM

Treatment Implications: Rehab standard for cuff-friendly elevation

Exercise Progressions: Light dumbbell scaption → progressive load

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Scaption]

RECORD: Scapular Elevation

ID: MOV-UE-014

Movement: Scapular Elevation

Joint: Scapulothoracic joint

Body Region: Scapula

Plane: Frontal

Axis: Anteroposterior

Open Chain: Scapula moves superior on thorax

Closed Chain: Handstand closed chain elevation

Primary Movers: Upper trapezius, levator scapulae

Secondary Movers: Rhomboids (minor elevation component)

Stabilizers: Serratus anterior lower fibers eccentrically

Force Couples: Agonist-antagonist pairs coordinating scapular elevation with regional stabilizers

Prime Stabilizers: Serratus anterior lower fibers eccentrically

Dynamic Stabilizers: Rhomboids (minor elevation component)

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Scapula rotates upward slightly with elevation

Osteokinematics: Scapula translates superiorly

Joint Reaction Force: Compression increases with load during scapular elevation especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Upper trapezius as primary during standardized scapular elevation

Normal ROM: 0–15°

Functional ROM: 10° for shrugging objects

Compensations: Cervical side flexion substitute

Common Dysfunctions: Elevated resting scapula posture

Common Pain Patterns: Neck and upper trapezius pain

Clinical Relevance: Upper trapezius dominance in neck pain

Functional Activities: ADL and occupational tasks requiring scapular elevation within symptom-free range

Sport-Specific Applications: Weightlifting shrugs, climbing

Common Injuries: Upper trapezius strain

Assessment: Scapular elevation observation, SHR test

Treatment Implications: Lower trapezius/serratus balance

Exercise Progressions: Scapular setting → shrug with lower trap activation

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Scapular Elevation]

RECORD: Scapular Depression

ID: MOV-UE-015

Movement: Scapular Depression

Joint: Scapulothoracic joint

Body Region: Scapula

Plane: Frontal

Axis: Anteroposterior

Open Chain: Scapula moves inferior on thorax

Closed Chain: Supported dip position

Primary Movers: Lower trapezius, pectoralis minor (eccentric)

Secondary Movers: Latissimus dorsi, serratus anterior

Stabilizers: Rhomboids; upper trapezius eccentrically

Force Couples: Agonist-antagonist pairs coordinating scapular depression with regional stabilizers

Prime Stabilizers: Rhomboids

Dynamic Stabilizers: Latissimus dorsi, serratus anterior

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Inferior glide of scapula on thoracic wall

Osteokinematics: Scapula translates inferiorly

Joint Reaction Force: Compression increases with load during scapular depression especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Lower trapezius as primary during standardized scapular depression

Normal ROM: 0–10°

Functional ROM: 5° for reaching down

Compensations: Shoulder elevation substitute

Common Dysfunctions: Scapular winging with depression weakness

Common Pain Patterns: Lower trapezius region pain

Clinical Relevance: Depression important for upward rotation rhythm

Functional Activities: ADL and occupational tasks requiring scapular depression within symptom-free range

Sport-Specific Applications: Dips, swimming, pull-ups

Common Injuries: Long thoracic nerve injury

Assessment: Scapular depression strength

Treatment Implications: Serratus and lower trap co-activation

Exercise Progressions: Scapular depression in wall slide → dip progression

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Scapular Depression]

RECORD: Scapular Upward Rotation

ID: MOV-UE-016

Movement: Scapular Upward Rotation

Joint: Scapulothoracic joint

Body Region: Scapula

Plane: Frontal

Axis: Anteroposterior

Open Chain: Scapula rotates glenoid superiorly

Closed Chain: Push-up plus, handstand

Primary Movers: Serratus anterior, upper and lower trapezius

Secondary Movers: Pectoralis minor (stabilizer)

Stabilizers: Rotator cuff; force couple UT/LT/SA

Force Couples: Agonist-antagonist pairs coordinating scapular upward rotation with regional stabilizers

Prime Stabilizers: Rotator cuff

Dynamic Stabilizers: Pectoralis minor (stabilizer)

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Posterior tilt often couples with upward rotation

Osteokinematics: Inferior angle moves lateral and superior

Joint Reaction Force: Compression increases with load during scapular upward rotation especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Serratus anterior as primary during standardized scapular upward rotation

Normal ROM: 0–60° upward rotation with arm elevation

Functional ROM: 45° for overhead reach

Compensations: GH elevation without scapular rotation

Common Dysfunctions: Winging; premature shrugging

Common Pain Patterns: Periscapular pain

Clinical Relevance: Serratus weakness causes winging and impingement

Functional Activities: ADL and occupational tasks requiring scapular upward rotation within symptom-free range

Sport-Specific Applications: Overhead throwing, swimming

Common Injuries: Scapular dyskinesis, impingement

Assessment: Scapular dyskinesis assessment, wall slide

Treatment Implications: Serratus punch, push-up plus

Exercise Progressions: Serratus punch → Y raise → overhead loading

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Scapular Upward Rotation]

RECORD: Scapular Downward Rotation

ID: MOV-UE-017

Movement: Scapular Downward Rotation

Joint: Scapulothoracic joint

Body Region: Scapula

Plane: Frontal

Axis: Anteroposterior

Open Chain: Scapula rotates glenoid inferiorly

Closed Chain: Controlled lowering in pull-up eccentric

Primary Movers: Rhomboids, levator scapulae, pectoralis minor

Secondary Movers: Latissimus dorsi

Stabilizers: Middle trapezius

Force Couples: Agonist-antagonist pairs coordinating scapular downward rotation with regional stabilizers

Prime Stabilizers: Middle trapezius

Dynamic Stabilizers: Latissimus dorsi

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Anterior tilt may couple

Osteokinematics: Superior scapula moves medial-inferior

Joint Reaction Force: Compression increases with load during scapular downward rotation especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Rhomboids as primary during standardized scapular downward rotation

Normal ROM: Returns from upward rotation

Functional ROM: Functional lowering from overhead

Compensations: Excessive cervical extension

Common Dysfunctions: Downward rotation dominance

Common Pain Patterns: Interscapular and neck pain

Clinical Relevance: Overactive downward rotators in neck pain

Functional Activities: ADL and occupational tasks requiring scapular downward rotation within symptom-free range

Sport-Specific Applications: Pull-ups, lat pulldown

Common Injuries: Rhomboid spasm

Assessment: Observation during lowering phase

Treatment Implications: Balance with upward rotators

Exercise Progressions: Eccentric serratus/low trap → rhomboid flexibility

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Scapular Downward Rotation]

RECORD: Scapular Protraction

ID: MOV-UE-018

Movement: Scapular Protraction

Joint: Scapulothoracic joint

Body Region: Scapula

Plane: Transverse

Axis: Vertical

Open Chain: Scapula moves anterior around thorax

Closed Chain: Push-up protraction phase

Primary Movers: Serratus anterior

Secondary Movers: Pectoralis major/minor

Stabilizers: Lower trapezius; rhomboids eccentrically

Force Couples: Agonist-antagonist pairs coordinating scapular protraction with regional stabilizers

Prime Stabilizers: Lower trapezius

Dynamic Stabilizers: Pectoralis major/minor

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Scapula wraps anterior on curved thorax

Osteokinematics: Scapula translates anteriorly

Joint Reaction Force: Compression increases with load during scapular protraction especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Serratus anterior as primary during standardized scapular protraction

Normal ROM: 0–25°

Functional ROM: 15° for reaching forward

Compensations: Shoulder rolling forward substitute

Common Dysfunctions: Winging with protraction weakness

Common Pain Patterns: Periscapular and anterior chest pain

Clinical Relevance: Serratus weakness in long thoracic nerve palsy

Functional Activities: ADL and occupational tasks requiring scapular protraction within symptom-free range

Sport-Specific Applications: Punching, push-ups, swimming

Common Injuries: Scapular winging

Assessment: Push-up plus, serratus wall slide

Treatment Implications: Serratus anterior strengthening

Exercise Progressions: Serratus punch → push-up plus → bear crawl

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Scapular Protraction]

RECORD: Scapular Retraction

ID: MOV-UE-019

Movement: Scapular Retraction

Joint: Scapulothoracic joint

Body Region: Scapula

Plane: Transverse

Axis: Vertical

Open Chain: Scapula moves posterior toward spine

Closed Chain: Rowing closed chain retraction

Primary Movers: Rhomboids, middle trapezius

Secondary Movers: Latissimus dorsi, posterior deltoid

Stabilizers: Serratus anterior eccentrically

Force Couples: Agonist-antagonist pairs coordinating scapular retraction with regional stabilizers

Prime Stabilizers: Serratus anterior eccentrically

Dynamic Stabilizers: Latissimus dorsi, posterior deltoid

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Scapula adducts on thorax

Osteokinematics: Scapula translates posteriorly

Joint Reaction Force: Compression increases with load during scapular retraction especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Rhomboids as primary during standardized scapular retraction

Normal ROM: 0–25°

Functional ROM: 15° for rowing finish

Compensations: Cervical extension with retraction

Common Dysfunctions: Rounded shoulders with weak retraction

Common Pain Patterns: Interscapular pain

Clinical Relevance: Over-retraction in military posture

Functional Activities: ADL and occupational tasks requiring scapular retraction within symptom-free range

Sport-Specific Applications: Rowing, climbing, swimming backstroke

Common Injuries: Rhomboid strain

Assessment: Row test, scapular retraction endurance

Treatment Implications: Balance retraction with serratus

Exercise Progressions: Band row → face pull → heavy row

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Scapular Retraction]

RECORD: Scapular Anterior Tilt

ID: MOV-UE-020

Movement: Scapular Anterior Tilt

Joint: Scapulothoracic joint

Body Region: Scapula

Plane: Sagittal

Axis: Mediolateral

Open Chain: Coracoid tilts inferior-anterior

Closed Chain: Scapula tips anterior

Primary Movers: Pectoralis minor, serratus anterior

Secondary Movers: Latissimus dorsi

Stabilizers: Lower trapezius eccentrically

Force Couples: Agonist-antagonist pairs coordinating scapular anterior tilt with regional stabilizers

Prime Stabilizers: Lower trapezius eccentrically

Dynamic Stabilizers: Latissimus dorsi

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Anterior tilt increases subacromial space in some positions

Osteokinematics: Observation in wall angel

Joint Reaction Force: Compression increases with load during scapular anterior tilt especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Pectoralis minor as primary during standardized scapular anterior tilt

Normal ROM: Variable coupling with elevation

Functional ROM: Functional in push and reach

Compensations: Not specified for this grip pattern

Common Dysfunctions: Pec minor stretch → serratus activation in protraction

Common Pain Patterns: Thoracic flexion substitute

Clinical Relevance: Pec minor length, serratus control

Functional Activities: ADL and occupational tasks requiring scapular anterior tilt within symptom-free range

Sport-Specific Applications: Anterior shoulder pain

Common Injuries: Excessive anterior tilt in desk posture

Assessment: Push-ups, bench press

Treatment Implications: Pec minor syndrome

Exercise Progressions: Not specified for this grip pattern

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Scapular Anterior Tilt]

RECORD: Scapular Posterior Tilt

ID: MOV-UE-021

Movement: Scapular Posterior Tilt

Joint: Scapulothoracic joint

Body Region: Scapula

Plane: Sagittal

Axis: Mediolateral

Open Chain: Coracoid moves superior-posterior

Closed Chain: Scapula tips posterior

Primary Movers: Lower trapezius, serratus anterior

Secondary Movers: Rhomboids

Stabilizers: Upper trapezius

Force Couples: Agonist-antagonist pairs coordinating scapular posterior tilt with regional stabilizers

Prime Stabilizers: Upper trapezius

Dynamic Stabilizers: Rhomboids

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Posterior tilt reduces impingement in elevation

Osteokinematics: Wall slide with posterior tilt cue

Joint Reaction Force: Compression increases with load during scapular posterior tilt especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Lower trapezius as primary during standardized scapular posterior tilt

Normal ROM: Couples with upward rotation

Functional ROM: Overhead sport mechanics

Compensations: Not specified for this grip pattern

Common Dysfunctions: Wall slides → Y raises with posterior tilt emphasis

Common Pain Patterns: Lumbar hyperextension substitute

Clinical Relevance: Lower trap and serratus training

Functional Activities: ADL and occupational tasks requiring scapular posterior tilt within symptom-free range

Sport-Specific Applications: Subacromial pain

Common Injuries: Loss of posterior tilt in elevation

Assessment: Overhead throwing, serving

Treatment Implications: Impingement

Exercise Progressions: Not specified for this grip pattern

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Scapular Posterior Tilt]

RECORD: Scapular Internal Rotation

ID: MOV-UE-022

Movement: Scapular Internal Rotation

Joint: Scapulothoracic joint

Body Region: Scapula

Plane: Transverse

Axis: Vertical

Open Chain: Glenoid rotates medial

Closed Chain: Scapula rotates around vertical axis medially

Primary Movers: Pectoralis minor, rhomboids

Secondary Movers: Levator scapulae

Stabilizers: Serratus anterior

Force Couples: Agonist-antagonist pairs coordinating scapular internal rotation with regional stabilizers

Prime Stabilizers: Serratus anterior

Dynamic Stabilizers: Levator scapulae

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Associated with rounded shoulder posture

Osteokinematics: Resting scapular posture

Joint Reaction Force: Compression increases with load during scapular internal rotation especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Pectoralis minor as primary during standardized scapular internal rotation

Normal ROM: Medial border prominence variable

Functional ROM: Postural assessment

Compensations: Not specified for this grip pattern

Common Dysfunctions: Postural re-education → scapular control drills

Common Pain Patterns: Shoulder protraction substitute

Clinical Relevance: Pec minor and rhomboid balance

Functional Activities: ADL and occupational tasks requiring scapular internal rotation within symptom-free range

Sport-Specific Applications: Anterior chest and interscapular pain

Common Injuries: Medial border prominence

Assessment: Desk work posture

Treatment Implications: Scapular dyskinesis

Exercise Progressions: Not specified for this grip pattern

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Scapular Internal Rotation]

RECORD: Scapular External Rotation

ID: MOV-UE-023

Movement: Scapular External Rotation

Joint: Scapulothoracic joint

Body Region: Scapula

Plane: Transverse

Axis: Vertical

Open Chain: Glenoid rotates lateral

Closed Chain: Scapula rotates around vertical axis laterally

Primary Movers: Serratus anterior, lower trapezius

Secondary Movers: Middle trapezius

Stabilizers: Rhomboids eccentrically

Force Couples: Agonist-antagonist pairs coordinating scapular external rotation with regional stabilizers

Prime Stabilizers: Rhomboids eccentrically

Dynamic Stabilizers: Middle trapezius

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Important for overhead athlete scapular mechanics

Osteokinematics: Scapular control assessment

Joint Reaction Force: Compression increases with load during scapular external rotation especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Serratus anterior as primary during standardized scapular external rotation

Normal ROM: Couples with upward rotation

Functional ROM: Overhead athletes

Compensations: Not specified for this grip pattern

Common Dysfunctions: Serratus wall slides → overhead holds

Common Pain Patterns: Compensatory GH elevation

Clinical Relevance: Serratus/low trap emphasis

Functional Activities: ADL and occupational tasks requiring scapular external rotation within symptom-free range

Sport-Specific Applications: Periscapular fatigue

Common Injuries: Difficulty maintaining ER in overhead hold

Assessment: Swimming, volleyball

Treatment Implications: Scapular dyskinesis

Exercise Progressions: Not specified for this grip pattern

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Scapular External Rotation]

RECORD: Elbow Flexion

ID: MOV-UE-024

Movement: Elbow Flexion

Joint: Elbow joint

Body Region: Elbow

Plane: Sagittal

Axis: Mediolateral

Open Chain: Hand moves toward shoulder

Closed Chain: Pull-up, chin-up closed chain

Primary Movers: Biceps brachii, brachialis, brachioradialis

Secondary Movers: Pronator teres (weak flexor)

Stabilizers: Anconeus eccentrically; capsule

Force Couples: Agonist-antagonist pairs coordinating elbow flexion with regional stabilizers

Prime Stabilizers: Anconeus eccentrically

Dynamic Stabilizers: Pronator teres (weak flexor)

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Olecranon moves away from humerus; anterior capsule distends

Osteokinematics: Forearm approximates humerus anteriorly

Joint Reaction Force: Compression increases with load during elbow flexion especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Biceps brachii as primary during standardized elbow flexion

Normal ROM: 0–140° (individual variation)

Functional ROM: 120–130° for feeding and grooming

Compensations: Shoulder elevation substitute

Common Dysfunctions: Flexion loss post immobilization

Common Pain Patterns: Anterior elbow and biceps pain

Clinical Relevance: Flexion contracture after fracture/immobilization

Functional Activities: ADL and occupational tasks requiring elbow flexion within symptom-free range

Sport-Specific Applications: Weightlifting curls, climbing

Common Injuries: Distal biceps rupture, flexion contracture

Assessment: Elbow flexion ROM goniometry

Treatment Implications: Progressive flexion after fracture protocol

Exercise Progressions: Active flexion → resisted flexion progression

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Elbow Flexion]

RECORD: Elbow Extension

ID: MOV-UE-025

Movement: Elbow Extension

Joint: Elbow joint

Body Region: Elbow

Plane: Sagittal

Axis: Mediolateral

Open Chain: Forearm moves away from humerus

Closed Chain: Push-up, bench press lockout

Primary Movers: Triceps brachii

Secondary Movers: Anconeus

Stabilizers: Biceps eccentrically; capsule

Force Couples: Agonist-antagonist pairs coordinating elbow extension with regional stabilizers

Prime Stabilizers: Biceps eccentrically

Dynamic Stabilizers: Anconeus

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Olecranon locks into olecranon fossa

Osteokinematics: Forearm moves posterior away from humerus

Joint Reaction Force: Compression increases with load during elbow extension especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Triceps brachii as primary during standardized elbow extension

Normal ROM: 0° to 5° hyperextension common

Functional ROM: 0° for push-off tasks

Compensations: Shoulder internal rotation substitute

Common Dysfunctions: Extension lag after triceps injury

Common Pain Patterns: Posterior elbow pain

Clinical Relevance: Olecranon bursitis with repetitive extension

Functional Activities: ADL and occupational tasks requiring elbow extension within symptom-free range

Sport-Specific Applications: Throwing, push-ups, pressing

Common Injuries: Triceps tendinopathy, olecranon bursitis

Assessment: Extension ROM, triceps strength

Treatment Implications: Triceps progressive loading

Exercise Progressions: Triceps isometrics → extension against gravity

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Elbow Extension]

RECORD: Forearm Pronation

ID: MOV-UE-026

Movement: Forearm Pronation

Joint: Proximal/distal radioulnar joints

Body Region: Forearm

Plane: Transverse

Axis: Longitudinal

Open Chain: Forearm rotates palm down open chain

Closed Chain: Weight-bearing pronation in push-up

Primary Movers: Pronator teres, pronator quadratus

Secondary Movers: Flexor carpi radialis, palmaris longus

Stabilizers: Interosseous membrane; joint capsules

Force Couples: Agonist-antagonist pairs coordinating forearm pronation with regional stabilizers

Prime Stabilizers: Interosseous membrane

Dynamic Stabilizers: Flexor carpi radialis, palmaris longus

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Radius crosses over ulna at proximal and distal RU joints

Osteokinematics: Palm rotates posterior/inferior

Joint Reaction Force: Compression increases with load during forearm pronation especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Pronator teres as primary during standardized forearm pronation

Normal ROM: 0–80°

Functional ROM: 70° for typing and tool use

Compensations: Shoulder internal rotation substitute

Common Dysfunctions: Pronation stiffness post cast

Common Pain Patterns: Volar forearm pain

Clinical Relevance: Pronation loss after fracture (Essex-Lopresti)

Functional Activities: ADL and occupational tasks requiring forearm pronation within symptom-free range

Sport-Specific Applications: Tennis forehand, screwdriver use

Common Injuries: Pronator teres syndrome, fracture

Assessment: Pronation/supination ROM

Treatment Implications: Pronator strengthening after fracture

Exercise Progressions: Pronation AROM → resisted pronation

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Forearm Pronation]

RECORD: Forearm Supination

ID: MOV-UE-027

Movement: Forearm Supination

Joint: Proximal/distal radioulnar joints

Body Region: Forearm

Plane: Transverse

Axis: Longitudinal

Open Chain: Forearm rotates palm up

Closed Chain: Closed chain supination in chin-up

Primary Movers: Biceps brachii, supinator

Secondary Movers: Brachioradialis (from pronated)

Stabilizers: Interosseous membrane

Force Couples: Agonist-antagonist pairs coordinating forearm supination with regional stabilizers

Prime Stabilizers: Interosseous membrane

Dynamic Stabilizers: Brachioradialis (from pronated)

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Radius parallels ulna; supinator and biceps rotate radius

Osteokinematics: Palm rotates anterior/superior

Joint Reaction Force: Compression increases with load during forearm supination especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Biceps brachii as primary during standardized forearm supination

Normal ROM: 0–80°

Functional ROM: 70° for supinated carry (palms up)

Compensations: External rotation substitute

Common Dysfunctions: Supination loss post immobilization

Common Pain Patterns: Lateral elbow and forearm pain

Clinical Relevance: Supination weakness after radial head injury

Functional Activities: ADL and occupational tasks requiring forearm supination within symptom-free range

Sport-Specific Applications: Supinated curl, racquet backhand

Common Injuries: Radial head fracture, PIN palsy

Assessment: Pronation/supination ROM, supination strength

Treatment Implications: Supinator and biceps rehab

Exercise Progressions: Supination AROM → hammer rotation exercises

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Forearm Supination]

RECORD: Wrist Flexion

ID: MOV-UE-028

Movement: Wrist Flexion

Joint: Wrist joint

Body Region: Wrist

Plane: Sagittal

Axis: Mediolateral

Open Chain: Hand moves toward volar forearm

Closed Chain: Weight-bearing wrist flexion in gymnastics

Primary Movers: Flexor carpi radialis, flexor carpi ulnaris, flexor digitorum superficialis

Secondary Movers: Palmaris longus

Stabilizers: Extensors eccentrically; wrist capsule

Force Couples: Agonist-antagonist pairs coordinating wrist flexion with regional stabilizers

Prime Stabilizers: Extensors eccentrically

Dynamic Stabilizers: Palmaris longus

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Proximal carpal row glides posterior on radius/ulna

Osteokinematics: Hand moves palmar toward forearm

Joint Reaction Force: Compression increases with load during wrist flexion especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Flexor carpi radialis as primary during standardized wrist flexion

Normal ROM: 0–80°

Functional ROM: 60° for grip and typing

Compensations: Finger flexion without wrist flexion

Common Dysfunctions: Flexion loss after fracture

Common Pain Patterns: Volar wrist and forearm pain

Clinical Relevance: Flexor tendinopathy in climbers and typists

Functional Activities: ADL and occupational tasks requiring wrist flexion within symptom-free range

Sport-Specific Applications: Golf, climbing, keyboard use

Common Injuries: Carpal tunnel, flexor tendinopathy

Assessment: Wrist flexion ROM, Phalen if indicated

Treatment Implications: Eccentric flexor loading for tendinopathy

Exercise Progressions: Wrist flexion AROM → eccentric flexor exercises

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Wrist Flexion]

RECORD: Wrist Extension

ID: MOV-UE-029

Movement: Wrist Extension

Joint: Wrist joint

Body Region: Wrist

Plane: Sagittal

Axis: Mediolateral

Open Chain: Hand moves toward dorsal forearm

Closed Chain: Push-up wrist extension load

Primary Movers: Extensor carpi radialis longus/brevis, extensor carpi ulnaris

Secondary Movers: Extensor digitorum

Stabilizers: Flexors eccentrically; wrist capsule

Force Couples: Agonist-antagonist pairs coordinating wrist extension with regional stabilizers

Prime Stabilizers: Flexors eccentrically

Dynamic Stabilizers: Extensor digitorum

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Proximal row glides anterior on radius

Osteokinematics: Hand moves dorsal toward forearm

Joint Reaction Force: Compression increases with load during wrist extension especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Extensor carpi radialis longus/brevis as primary during standardized wrist extension

Normal ROM: 0–70°

Functional ROM: 50° for push-off and typing

Compensations: Metacarpophalangeal extension substitute

Common Dysfunctions: Extension weakness post fracture

Common Pain Patterns: Dorsal wrist and lateral elbow pain

Clinical Relevance: Extensor tendinopathy in racquet sports

Functional Activities: ADL and occupational tasks requiring wrist extension within symptom-free range

Sport-Specific Applications: Tennis, typing, push-ups

Common Injuries: Intersection syndrome, lateral epicondylalgia

Assessment: Extension ROM, resisted extension test

Treatment Implications: Eccentric wrist extension for lateral epicondyle

Exercise Progressions: Wrist extension AROM → Tyler twist progression

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Wrist Extension]

RECORD: Wrist Radial Deviation

ID: MOV-UE-030

Movement: Wrist Radial Deviation

Joint: Wrist joint

Body Region: Wrist

Plane: Frontal

Axis: Anteroposterior

Open Chain: Hand moves toward thumb/radius

Closed Chain: Closed chain radial bias in punch

Primary Movers: Extensor carpi radialis longus/brevis, flexor carpi radialis

Secondary Movers: Abductor pollicis longus

Stabilizers: FCU eccentrically

Force Couples: Agonist-antagonist pairs coordinating wrist radial deviation with regional stabilizers

Prime Stabilizers: FCU eccentrically

Dynamic Stabilizers: Abductor pollicis longus

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Scaphoid translates on radius

Osteokinematics: Hand moves toward radius

Joint Reaction Force: Compression increases with load during wrist radial deviation especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Extensor carpi radialis longus/brevis as primary during standardized wrist radial deviation

Normal ROM: 0–20°

Functional ROM: 15° for hammering radially

Compensations: Shoulder abduction substitute

Common Dysfunctions: Radial deviation pain in CTS

Common Pain Patterns: Radial wrist and thumb base pain

Clinical Relevance: De Quervain aggravated by radial deviation

Functional Activities: ADL and occupational tasks requiring wrist radial deviation within symptom-free range

Sport-Specific Applications: Hammering, racquet sports

Common Injuries: De Quervain tenosynovitis

Assessment: Radial/ulnar deviation ROM

Treatment Implications: FCR/ECRL balance training

Exercise Progressions: Radial deviation AROM → resisted deviation

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Wrist Radial Deviation]

RECORD: Wrist Ulnar Deviation

ID: MOV-UE-031

Movement: Wrist Ulnar Deviation

Joint: Wrist joint

Body Region: Wrist

Plane: Frontal

Axis: Anteroposterior

Open Chain: Hand moves toward ulna/pinky side

Closed Chain: Golf swing ulnar deviation load

Primary Movers: Extensor carpi ulnaris, flexor carpi ulnaris

Secondary Movers: Flexor digitorum

Stabilizers: ECRL eccentrically

Force Couples: Agonist-antagonist pairs coordinating wrist ulnar deviation with regional stabilizers

Prime Stabilizers: ECRL eccentrically

Dynamic Stabilizers: Flexor digitorum

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Triquetrum moves on ulna

Osteokinematics: Hand moves toward ulna

Joint Reaction Force: Compression increases with load during wrist ulnar deviation especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Extensor carpi ulnaris as primary during standardized wrist ulnar deviation

Normal ROM: 0–30°

Functional ROM: 25° for power grip ulnar bias

Compensations: Elbow valgus substitute

Common Dysfunctions: Ulnar deviation weakness

Common Pain Patterns: Ulnar wrist pain

Clinical Relevance: ECU tendinopathy in golfers and tennis players

Functional Activities: ADL and occupational tasks requiring wrist ulnar deviation within symptom-free range

Sport-Specific Applications: Golf, tennis, lifting

Common Injuries: ECU subsheath tear, TFCC injury

Assessment: Ulnar deviation ROM, ECU testing

Treatment Implications: ECU stabilization for ulnar-sided wrist pain

Exercise Progressions: Ulnar deviation AROM → ECU isometrics

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Wrist Ulnar Deviation]

RECORD: Wrist Circumduction

ID: MOV-UE-032

Movement: Wrist Circumduction

Joint: Wrist joint

Body Region: Wrist

Plane: Multiple

Axis: Multiple

Open Chain: Wrist traces cone shape

Closed Chain: Closed chain circular loading rare

Primary Movers: Combined wrist flexors and extensors

Secondary Movers: Intrinsic stabilizers

Stabilizers: Forearm pronators/supinators

Force Couples: Agonist-antagonist pairs coordinating wrist circumduction with regional stabilizers

Prime Stabilizers: Forearm pronators/supinators

Dynamic Stabilizers: Intrinsic stabilizers

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Sequential carpal glides in multiple planes

Osteokinematics: Hand traces circle at wrist

Joint Reaction Force: Compression increases with load during wrist circumduction especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Combined wrist flexors and extensors as primary during standardized wrist circumduction

Normal ROM: Composite motion

Functional ROM: Functional circular wrist motion

Compensations: Elbow circumduction substitute

Common Dysfunctions: Compensatory forearm rotation

Common Pain Patterns: Diffuse wrist pain

Clinical Relevance: Assesses global wrist mobility

Functional Activities: ADL and occupational tasks requiring wrist circumduction within symptom-free range

Sport-Specific Applications: Dance, gymnastics wrist circles

Common Injuries: Global wrist stiffness post cast

Assessment: Composite wrist ROM observation

Treatment Implications: Multi-planar wrist mobility drills

Exercise Progressions: Slow circles → resisted multi-planar ball control

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Wrist Circumduction]

RECORD: Dart Thrower's Motion

ID: MOV-UE-033

Movement: Dart Thrower's Motion

Joint: Wrist/midcarpal

Body Region: Wrist

Plane: Oblique

Axis: Oblique

Open Chain: Wrist moves extension-radial to flexion-ulnar

Closed Chain: Functional closed chain in throwing

Primary Movers: ECRB, FCR, midcarpal stabilizers

Secondary Movers: Extrinsic flexors/extensors

Stabilizers: Intrinsic ligaments

Force Couples: Agonist-antagonist pairs coordinating dart thrower's motion with regional stabilizers

Prime Stabilizers: Intrinsic ligaments

Dynamic Stabilizers: Extrinsic flexors/extensors

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Midcarpal joint primary driver (~60% of motion)

Osteokinematics: Hand follows oblique dart-throw arc

Joint Reaction Force: Compression increases with load during dart thrower's motion especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms ECRB as primary during standardized dart thrower's motion

Normal ROM: Extension-radial to flexion-ulnar arc ~30° each component

Functional ROM: Throwing and hammering arc

Compensations: Planar flexion/extension only

Common Dysfunctions: Loss of smooth dart arc

Common Pain Patterns: Dorsal radial wrist pain

Clinical Relevance: Critical for functional wrist athletic motion

Functional Activities: ADL and occupational tasks requiring dart thrower's motion within symptom-free range

Sport-Specific Applications: Baseball throw, hammering

Common Injuries: Scapholunate injury, midcarpal instability

Assessment: Dart thrower ER test for SL ligament

Treatment Implications: Midcarpal stability rehab

Exercise Progressions: Dart thrower motion AROM → weighted ball drills

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Dart Thrower's Motion]

RECORD: MCP Flexion

ID: MOV-UE-034

Movement: MCP Flexion

Joint: Metacarpophalangeal joints

Body Region: Fingers

Plane: Sagittal

Axis: Mediolateral

Open Chain: Fingers flex at knuckles open chain

Closed Chain: Hook grip closed chain

Primary Movers: Flexor digitorum superficialis/profundus, lumbricals, interossei

Secondary Movers: Palmar interossei

Stabilizers: Extensor hood eccentrically; collateral ligaments

Force Couples: Agonist-antagonist pairs coordinating mcp flexion with regional stabilizers

Prime Stabilizers: Extensor hood eccentrically

Dynamic Stabilizers: Palmar interossei

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Metacarpal head rolls and glides palmarly

Osteokinematics: Proximal phalanx flexes toward palm

Joint Reaction Force: Compression increases with load during mcp flexion especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Flexor digitorum superficialis/profundus as primary during standardized mcp flexion

Normal ROM: 0–90°

Functional ROM: 70° for power grip

Compensations: IP flexion without MCP flexion

Common Dysfunctions: MCP extension contracture (claw hand)

Common Pain Patterns: Palmar MCP pain

Clinical Relevance: MCP flexion essential for grip strength

Functional Activities: ADL and occupational tasks requiring mcp flexion within symptom-free range

Sport-Specific Applications: Rock climbing, weightlifting

Common Injuries: MCP sprain, capsulitis

Assessment: MCP ROM, grip strength

Treatment Implications: Intrinsic plus/minus exercises

Exercise Progressions: MCP flexion AROM → grip strengthening

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [MCP Flexion]

RECORD: MCP Extension

ID: MOV-UE-035

Movement: MCP Extension

Joint: Metacarpophalangeal joints

Body Region: Fingers

Plane: Sagittal

Axis: Mediolateral

Open Chain: Fingers extend at knuckles

Closed Chain: Open hand push-up MCP extension load

Primary Movers: Extensor digitorum, lumbricals, interossei via extensor hood

Secondary Movers: Extensor indicis/propius

Stabilizers: Flexors eccentrically; sagittal bands

Force Couples: Agonist-antagonist pairs coordinating mcp extension with regional stabilizers

Prime Stabilizers: Flexors eccentrically

Dynamic Stabilizers: Extensor indicis/propius

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Metacarpal head glides dorsally

Osteokinematics: Proximal phalanx extends

Joint Reaction Force: Compression increases with load during mcp extension especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Extensor digitorum as primary during standardized mcp extension

Normal ROM: 0–30° hyperextension variable

Functional ROM: 20° for hand flat on table

Compensations: Wrist extension substitute

Common Dysfunctions: Intrinsic weakness

Common Pain Patterns: Dorsal MCP pain

Clinical Relevance: MCP extension lag in claw deformity

Functional Activities: ADL and occupational tasks requiring mcp extension within symptom-free range

Sport-Specific Applications: Handstand, finger locks climbing

Common Injuries: Sagittal band rupture, MCP dislocation

Assessment: MCP extension observation, intrinsic tests

Treatment Implications: Intrinsic muscle rehab

Exercise Progressions: MCP extension AROM → rubber band extensions

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [MCP Extension]

RECORD: PIP Flexion

ID: MOV-UE-036

Movement: PIP Flexion

Joint: Proximal interphalangeal joints

Body Region: Fingers

Plane: Sagittal

Axis: Mediolateral

Open Chain: Middle joint flexes open chain

Closed Chain: Crimp grip climbing

Primary Movers: Flexor digitorum superficialis (FDS)

Secondary Movers: Flexor digitorum profundus (limited at PIP)

Stabilizers: Lateral bands; collateral ligaments

Force Couples: Agonist-antagonist pairs coordinating pip flexion with regional stabilizers

Prime Stabilizers: Lateral bands

Dynamic Stabilizers: Flexor digitorum profundus (limited at PIP)

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Condylar roll-glide palmar

Osteokinematics: Middle phalanx flexes

Joint Reaction Force: Compression increases with load during pip flexion especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Flexor digitorum superficialis (FDS) as primary during standardized pip flexion

Normal ROM: 0–110°

Functional ROM: 90° for fine grip

Compensations: DIP flexion substitute in isolation

Common Dysfunctions: PIP flexion contracture

Common Pain Patterns: Lateral PIP pain

Clinical Relevance: PIP stiffness common after fracture

Functional Activities: ADL and occupational tasks requiring pip flexion within symptom-free range

Sport-Specific Applications: Climbing crimp, guitar

Common Injuries: PIP fracture, boutonniere

Assessment: PIP ROM, tenodesis effect

Treatment Implications: Tendon gliding for PIP stiffness

Exercise Progressions: Tendon gliding → blocking exercises PIP

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [PIP Flexion]

RECORD: PIP Extension

ID: MOV-UE-037

Movement: PIP Extension

Joint: Proximal interphalangeal joints

Body Region: Fingers

Plane: Sagittal

Axis: Mediolateral

Open Chain: Middle joint extends

Closed Chain: Open chain extension against gravity

Primary Movers: Central slip of extensor mechanism

Secondary Movers: Lumbricals/interossei via lateral bands

Stabilizers: FDS eccentrically; collateral ligaments

Force Couples: Agonist-antagonist pairs coordinating pip extension with regional stabilizers

Prime Stabilizers: FDS eccentrically

Dynamic Stabilizers: Lumbricals/interossei via lateral bands

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Dorsal glide of middle phalanx

Osteokinematics: Middle phalanx extends

Joint Reaction Force: Compression increases with load during pip extension especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Central slip of extensor mechanism as primary during standardized pip extension

Normal ROM: 0° to 5° hyperextension

Functional ROM: 0° for finger straight

Compensations: MCP hyperextension substitute (swan neck)

Common Dysfunctions: Extension lag

Common Pain Patterns: Dorsal PIP pain

Clinical Relevance: Boutonniere deformity — central slip injury

Functional Activities: ADL and occupational tasks requiring pip extension within symptom-free range

Sport-Specific Applications: Basketball finger, volleyball

Common Injuries: Boutonniere, PIP dislocation

Assessment: PIP extension lag, Elson test

Treatment Implications: Central slip protection and rehab

Exercise Progressions: Static extension splint → active extension

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [PIP Extension]

RECORD: DIP Flexion

ID: MOV-UE-038

Movement: DIP Flexion

Joint: Distal interphalangeal joints

Body Region: Fingers

Plane: Sagittal

Axis: Mediolateral

Open Chain: Distal joint flexes

Closed Chain: Tip pinch closed chain

Primary Movers: Flexor digitorum profundus (FDP)

Secondary Movers: None significant

Stabilizers: Oblique retinacular ligament

Force Couples: Agonist-antagonist pairs coordinating dip flexion with regional stabilizers

Prime Stabilizers: Oblique retinacular ligament

Dynamic Stabilizers: None significant

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Distal condylar roll-glide palmar

Osteokinematics: Distal phalanx flexes

Joint Reaction Force: Compression increases with load during dip flexion especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Flexor digitorum profundus (FDP) as primary during standardized dip flexion

Normal ROM: 0–90°

Functional ROM: 60° for tip pinch grip

Compensations: Composite fist without isolated DIP

Common Dysfunctions: DIP flexion loss after jersey finger

Common Pain Patterns: Volae and DIP pain

Clinical Relevance: FDP laceration — loss DIP flexion zone II

Functional Activities: ADL and occupational tasks requiring dip flexion within symptom-free range

Sport-Specific Applications: Rock climbing, guitar plucking

Common Injuries: Mallet finger (extensor), FDP avulsion (jersey finger)

Assessment: DIP flexion isolation (hold PIP)

Treatment Implications: FDP tendon repair protocol

Exercise Progressions: Isolated DIP flexion → block exercises

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [DIP Flexion]

RECORD: DIP Extension

ID: MOV-UE-039

Movement: DIP Extension

Joint: Distal interphalangeal joints

Body Region: Fingers

Plane: Sagittal

Axis: Mediolateral

Open Chain: Distal joint extends

Closed Chain: Mallet splint extension

Primary Movers: Terminal tendon of extensor mechanism

Secondary Movers: Lumbricals/interossei indirect

Stabilizers: FDP eccentrically

Force Couples: Agonist-antagonist pairs coordinating dip extension with regional stabilizers

Prime Stabilizers: FDP eccentrically

Dynamic Stabilizers: Lumbricals/interossei indirect

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Terminal tendon extends DIP

Osteokinematics: Distal phalanx extends

Joint Reaction Force: Compression increases with load during dip extension especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Terminal tendon of extensor mechanism as primary during standardized dip extension

Normal ROM: 0° to 5° hyperextension

Functional ROM: 0° for nail bed protection

Compensations: Hyperextension at PIP substitute

Common Dysfunctions: Extension lag (mallet deformity)

Common Pain Patterns: Dorsal DIP pain

Clinical Relevance: Mallet finger — terminal extensor rupture

Functional Activities: ADL and occupational tasks requiring dip extension within symptom-free range

Sport-Specific Applications: Ball sports tip injuries

Common Injuries: Mallet finger, DIP fracture

Assessment: Extension lag at DIP, mallet test

Treatment Implications: Mallet splint 6–8 weeks full-time

Exercise Progressions: Stack splint → active extension when healed

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [DIP Extension]

RECORD: Thumb Opposition

ID: MOV-UE-040

Movement: Thumb Opposition

Joint: Carpometacarpal joint (saddle)

Body Region: Thumb

Plane: Multi-planar

Axis: Oblique

Open Chain: Thumb CMC pronates and flexes with abduction

Closed Chain: Thumb moves across palm to digit

Primary Movers: Opponens pollicis, abductor pollicis brevis, flexor pollicis brevis

Secondary Movers: Adductor pollicis

Stabilizers: First dorsal interosseous; CMC ligaments

Force Couples: Agonist-antagonist pairs coordinating thumb opposition with regional stabilizers

Prime Stabilizers: First dorsal interosseous

Dynamic Stabilizers: Adductor pollicis

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Opposition critical for precision grip

Osteokinematics: Opposition test, Kapandji score

Joint Reaction Force: Compression increases with load during thumb opposition especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Opponens pollicis as primary during standardized thumb opposition

Normal ROM: Wide arc — tip to each finger pad

Functional ROM: Tip-to-tip and tip-to-base pinch

Compensations: Not specified for this grip pattern

Common Dysfunctions: Opposition AROM → pinch progression

Common Pain Patterns: Lateral pinch substitute

Clinical Relevance: Thenar strengthening progression

Functional Activities: ADL and occupational tasks requiring thumb opposition within symptom-free range

Sport-Specific Applications: Thenar and thumb base pain

Common Injuries: Opposition loss in CTS/advanced CMC

Assessment: Opposition for writing, pinching

Treatment Implications: CMC arthritis, median nerve palsy

Exercise Progressions: Not specified for this grip pattern

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Thumb Opposition]

RECORD: Thumb Reposition

ID: MOV-UE-041

Movement: Thumb Reposition

Joint: Carpometacarpal joint

Body Region: Thumb

Plane: Multi-planar

Axis: Oblique

Open Chain: Reverse CMC motion from opposition

Closed Chain: Thumb returns from palm to radial abduction

Primary Movers: Extensor pollicis brevis, abductor pollicis longus

Secondary Movers: Extensor pollicis longus

Stabilizers: Thenar muscles; CMC ligaments

Force Couples: Agonist-antagonist pairs coordinating thumb reposition with regional stabilizers

Prime Stabilizers: Thenar muscles

Dynamic Stabilizers: Extensor pollicis longus

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Controlled reposition prevents CMC irritation

Osteokinematics: Observation after opposition tasks

Joint Reaction Force: Compression increases with load during thumb reposition especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Extensor pollicis brevis as primary during standardized thumb reposition

Normal ROM: Returns from opposition to anatomical position

Functional ROM: Release from pinch

Compensations: Not specified for this grip pattern

Common Dysfunctions: EPL strengthening → controlled reposition drills

Common Pain Patterns: Forced reposition pain

Clinical Relevance: EPL/EPB strengthening

Functional Activities: ADL and occupational tasks requiring thumb reposition within symptom-free range

Sport-Specific Applications: Radial wrist pain with reposition

Common Injuries: Painful reposition in De Quervain

Assessment: Releasing tool grip

Treatment Implications: De Quervain, CMC strain

Exercise Progressions: Not specified for this grip pattern

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Thumb Reposition]

RECORD: Thumb Flexion

ID: MOV-UE-042

Movement: Thumb Flexion

Joint: CMC, MCP, IP joints

Body Region: Thumb

Plane: Sagittal

Axis: Mediolateral

Open Chain: Thumb flexes across joints open chain

Closed Chain: Hook grip thumb flexion

Primary Movers: Flexor pollicis brevis, flexor pollicis longus

Secondary Movers: Opponens pollicis

Stabilizers: Extensors eccentrically

Force Couples: Agonist-antagonist pairs coordinating thumb flexion with regional stabilizers

Prime Stabilizers: Extensors eccentrically

Dynamic Stabilizers: Opponens pollicis

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Multi-joint sequential flexion

Osteokinematics: Thumb moves into palm

Joint Reaction Force: Compression increases with load during thumb flexion especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Flexor pollicis brevis as primary during standardized thumb flexion

Normal ROM: CMC ~15°, MCP ~50°, IP ~80°

Functional ROM: Composite flexion for grip

Compensations: CMC collapse substitute

Common Dysfunctions: Trigger thumb locking

Common Pain Patterns: Volar thumb and wrist pain

Clinical Relevance: FPL tenosynovitis in texting/gaming

Functional Activities: ADL and occupational tasks requiring thumb flexion within symptom-free range

Sport-Specific Applications: Climbing, gaming, gripping

Common Injuries: Trigger thumb, FPL rupture

Assessment: Thumb flexion ROM, FPL test

Treatment Implications: FPL tendon gliding

Exercise Progressions: Thumb flexion AROM → resisted flexion

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Thumb Flexion]

RECORD: Thumb Extension

ID: MOV-UE-043

Movement: Thumb Extension

Joint: CMC, MCP, IP joints

Body Region: Thumb

Plane: Sagittal

Axis: Mediolateral

Open Chain: Thumb extends open chain

Closed Chain: Open hand position

Primary Movers: Extensor pollicis longus, extensor pollicis brevis

Secondary Movers: Abductor pollicis longus

Stabilizers: Flexors eccentrically

Force Couples: Agonist-antagonist pairs coordinating thumb extension with regional stabilizers

Prime Stabilizers: Flexors eccentrically

Dynamic Stabilizers: Abductor pollicis longus

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: EPL drives IP extension strongly

Osteokinematics: Thumb moves away from palm radially

Joint Reaction Force: Compression increases with load during thumb extension especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Extensor pollicis longus as primary during standardized thumb extension

Normal ROM: CMC ~60° extension/abduction component

Functional ROM: Extension for release and typing

Compensations: MCP hyperextension substitute

Common Dysfunctions: Extension weakness

Common Pain Patterns: Dorsal thumb and radial wrist pain

Clinical Relevance: EPL rupture in RA — loss active extension

Functional Activities: ADL and occupational tasks requiring thumb extension within symptom-free range

Sport-Specific Applications: Typing, open hand sports

Common Injuries: EPL rupture, EPB tenosynovitis (De Quervain)

Assessment: Thumb extension lag, EPL test

Treatment Implications: EPL protection and strengthening

Exercise Progressions: Thumb extension AROM → EPL strengthening

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Thumb Extension]

RECORD: Thumb Abduction

ID: MOV-UE-044

Movement: Thumb Abduction

Joint: Carpometacarpal joint

Body Region: Thumb

Plane: Frontal

Axis: Anteroposterior

Open Chain: Thumb moves away from index metacarpal

Closed Chain: Wide grip object hold

Primary Movers: Abductor pollicis longus, abductor pollicis brevis

Secondary Movers: Extensor pollicis brevis

Stabilizers: Adductor pollicis eccentrically

Force Couples: Agonist-antagonist pairs coordinating thumb abduction with regional stabilizers

Prime Stabilizers: Adductor pollicis eccentrically

Dynamic Stabilizers: Extensor pollicis brevis

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: CMC abduction in saddle joint

Osteokinematics: Thumb moves perpendicular from palm

Joint Reaction Force: Compression increases with load during thumb abduction especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Abductor pollicis longus as primary during standardized thumb abduction

Normal ROM: 0–50° palmar abduction; 45° radial abduction

Functional ROM: 40° for wide grip

Compensations: CMC flexion substitute

Common Dysfunctions: Adduction contracture (thumb-in-palm)

Common Pain Patterns: First web space pain

Clinical Relevance: First web space contracture limits abduction

Functional Activities: ADL and occupational tasks requiring thumb abduction within symptom-free range

Sport-Specific Applications: Opening jars, ball grip

Common Injuries: First web space contracture, CMC arthritis

Assessment: First web space ROM

Treatment Implications: Web space stretching and APB/APL strength

Exercise Progressions: Web space stretch → abduction strengthening

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Thumb Abduction]

RECORD: Thumb Adduction

ID: MOV-UE-045

Movement: Thumb Adduction

Joint: Carpometacarpal joint

Body Region: Thumb

Plane: Frontal

Axis: Anteroposterior

Open Chain: CMC adduction with flexion component

Closed Chain: Thumb moves toward index

Primary Movers: Adductor pollicis

Secondary Movers: First dorsal interosseous

Stabilizers: Abductors eccentrically

Force Couples: Agonist-antagonist pairs coordinating thumb adduction with regional stabilizers

Prime Stabilizers: Abductors eccentrically

Dynamic Stabilizers: First dorsal interosseous

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Adductor strength for key pinch

Osteokinematics: Key pinch strength test

Joint Reaction Force: Compression increases with load during thumb adduction especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Adductor pollicis as primary during standardized thumb adduction

Normal ROM: Returns thumb to index side

Functional ROM: Key pinch force

Compensations: Not specified for this grip pattern

Common Dysfunctions: Key pinch isometrics → dynamic pinch

Common Pain Patterns: Flexor substitute pinch

Clinical Relevance: Adductor pollicis strengthening

Functional Activities: ADL and occupational tasks requiring thumb adduction within symptom-free range

Sport-Specific Applications: Thenar eminence pain

Common Injuries: Weak key pinch in ulnar palsy

Assessment: Key pinch, tool grip

Treatment Implications: Adductor strain, CMC instability

Exercise Progressions: Not specified for this grip pattern

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Thumb Adduction]

RECORD: Power Grip

ID: MOV-UE-046

Movement: Power Grip

Joint: Hand composite

Body Region: Hand/grip

Plane: Multi-planar

Axis: Multiple

Open Chain: All digits flex around object with wrist extension

Closed Chain: Isometric grip on object

Primary Movers: Flexor digitorum profundus/superficialis, flexor pollicis longus

Secondary Movers: Interossei, lumbricals, thenar/hypothenar

Stabilizers: Wrist extensors stabilize; intrinsic stabilize arches

Force Couples: Agonist-antagonist pairs coordinating power grip with regional stabilizers

Prime Stabilizers: Wrist extensors stabilize

Dynamic Stabilizers: Interossei, lumbricals, thenar/hypothenar

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Carpometacarpal and MCP joints stabilize against flexion force

Osteokinematics: Hand closes maximally on object

Joint Reaction Force: Compression increases with load during power grip especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Flexor digitorum profundus/superficialis as primary during standardized power grip

Normal ROM: Full composite finger flexion with wrist ~30° extension

Functional ROM: Max force grip tasks

Compensations: Excessive wrist flexion during grip

Common Dysfunctions: Weak grip post fracture

Common Pain Patterns: Forearm and hand fatigue pain

Clinical Relevance: Grip strength assessment in hand pathology

Functional Activities: ADL and occupational tasks requiring power grip within symptom-free range

Sport-Specific Applications: Weightlifting, climbing, manual labor

Common Injuries: Flexor tendinopathy, epicondylalgia

Assessment: Dynamometer grip test

Treatment Implications: Progressive grip strengthening

Exercise Progressions: Isometric grip → dynamometer progressive loading

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Power Grip]

RECORD: Hook Grip

ID: MOV-UE-047

Movement: Hook Grip

Joint: Hand composite

Body Region: Hand/grip

Plane: Sagittal

Axis: Mediolateral

Open Chain: Fingers flex at IP joints with extended MCPs

Closed Chain: Pull-up bar hook grip

Primary Movers: Flexor digitorum profundus, flexor digitorum superficialis

Secondary Movers: Lumbricals (MCP extended position)

Stabilizers: Extensors maintain MCP extension

Force Couples: Agonist-antagonist pairs coordinating hook grip with regional stabilizers

Prime Stabilizers: Extensors maintain MCP extension

Dynamic Stabilizers: Lumbricals (MCP extended position)

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: MCP extended — collateral ligaments tight

Osteokinematics: Fingers form hook without thumb opposition

Joint Reaction Force: Compression increases with load during hook grip especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Flexor digitorum profundus as primary during standardized hook grip

Normal ROM: PIP/DIP flexion with MCP extended ~0–30°

Functional ROM: Carrying bags, pull-up hook

Compensations: Full fist grip substitute

Common Dysfunctions: Inability to maintain hook in climbers

Common Pain Patterns: Volar finger and pulley pain

Clinical Relevance: Climbing and lifting carry technique

Functional Activities: ADL and occupational tasks requiring hook grip within symptom-free range

Sport-Specific Applications: Climbing, deadlift hook grip

Common Injuries: Pulley injury climbing, flexor strain

Assessment: Hook grip strength observation

Treatment Implications: FDP strengthening in MCP extension

Exercise Progressions: Hook grip hangs → progressive load

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Hook Grip]

RECORD: Cylindrical Grip

ID: MOV-UE-048

Movement: Cylindrical Grip

Joint: Hand composite

Body Region: Hand/grip

Plane: Multi-planar

Axis: Multiple

Open Chain: Digits wrap around cylinder

Closed Chain: Carrying cylindrical object

Primary Movers: Flexor digitorum, FPL, interossei

Secondary Movers: Thenar/hypothenar

Stabilizers: Wrist extensors

Force Couples: Agonist-antagonist pairs coordinating cylindrical grip with regional stabilizers

Prime Stabilizers: Wrist extensors

Dynamic Stabilizers: Thenar/hypothenar

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: MCP and IP flexion adapt to cylinder diameter

Osteokinematics: Hand encircles cylinder

Joint Reaction Force: Compression increases with load during cylindrical grip especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Flexor digitorum as primary during standardized cylindrical grip

Normal ROM: Composite around cylinder diameter

Functional ROM: Holding bottles, hammer handles

Compensations: Tip pinch substitute for small objects

Common Dysfunctions: Grip size intolerance post injury

Common Pain Patterns: Forearm and hand pain

Clinical Relevance: Common functional grip pattern

Functional Activities: ADL and occupational tasks requiring cylindrical grip within symptom-free range

Sport-Specific Applications: Hammer, tennis racquet, bottles

Common Injuries: Epicondylalgia with repetitive cylindrical grip

Assessment: Cylinder grip test various diameters

Treatment Implications: Grip training multiple diameters

Exercise Progressions: Progressive cylinder diameters → functional carry

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Cylindrical Grip]

RECORD: Spherical Grip

ID: MOV-UE-049

Movement: Spherical Grip

Joint: Hand composite

Body Region: Hand/grip

Plane: Multi-planar

Axis: Multiple

Open Chain: Digits conform to spherical shape

Closed Chain: Ball catch and hold

Primary Movers: Flexor digitorum, thenar/hypothenar, interossei

Secondary Movers: Lumbricals

Stabilizers: Wrist stabilizers

Force Couples: Agonist-antagonist pairs coordinating spherical grip with regional stabilizers

Prime Stabilizers: Wrist stabilizers

Dynamic Stabilizers: Lumbricals

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Multi-joint adaptive flexion

Osteokinematics: Hand molds to sphere

Joint Reaction Force: Compression increases with load during spherical grip especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Flexor digitorum as primary during standardized spherical grip

Normal ROM: Adaptive to sphere size

Functional ROM: Holding balls, doorknobs

Compensations: Flat palm substitute

Common Dysfunctions: Fear of gripping post fracture

Common Pain Patterns: Palm and finger pain

Clinical Relevance: Ball sports and rehabilitation standard

Functional Activities: ADL and occupational tasks requiring spherical grip within symptom-free range

Sport-Specific Applications: Basketball, therapy ball exercises

Common Injuries: Metacarpal fracture, finger sprain

Assessment: Spherical object grip test

Treatment Implications: Ball grip progressive sizes

Exercise Progressions: Soft ball grip → sport-specific ball drills

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Spherical Grip]

RECORD: Precision Grip

ID: MOV-UE-050

Movement: Precision Grip

Joint: Hand composite

Body Region: Hand/grip

Plane: Multi-planar

Axis: Multiple

Open Chain: Thumb opposes fingertip with IP flexion

Closed Chain: Precision pinch on small object

Primary Movers: Flexor pollicis longus, flexor digitorum profundus index

Secondary Movers: Opponens pollicis, APB

Stabilizers: Interossei stabilize

Force Couples: Agonist-antagonist pairs coordinating precision grip with regional stabilizers

Prime Stabilizers: Interossei stabilize

Dynamic Stabilizers: Opponens pollicis, APB

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Fine motor control at fingertips

Osteokinematics: Thumb and finger tips oppose

Joint Reaction Force: Compression increases with load during precision grip especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Flexor pollicis longus as primary during standardized precision grip

Normal ROM: Tip-to-tip precision

Functional ROM: Writing, tweezers, small object manipulation

Compensations: Lateral pinch substitute

Common Dysfunctions: Weak tip pinch in nerve palsy

Common Pain Patterns: Thenar and fingertip pain

Clinical Relevance: Essential for fine motor ADL

Functional Activities: ADL and occupational tasks requiring precision grip within symptom-free range

Sport-Specific Applications: Surgery, electronics, writing

Common Injuries: Median/ulnar nerve palsy

Assessment: Tip pinch strength, Jebsen-Taylor test

Treatment Implications: Precision pinch training

Exercise Progressions: Pegboard → tip pinch loading

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Precision Grip]

RECORD: Lateral Pinch

ID: MOV-UE-051

Movement: Lateral Pinch

Joint: Hand composite

Body Region: Hand/grip

Plane: Frontal

Axis: Anteroposterior

Open Chain: Thumb adducts against radial side index

Closed Chain: Lateral key pinch on object

Primary Movers: Adductor pollicis, flexor pollicis longus

Secondary Movers: First dorsal interosseous

Stabilizers: Thenar stabilizers

Force Couples: Agonist-antagonist pairs coordinating lateral pinch with regional stabilizers

Prime Stabilizers: Thenar stabilizers

Dynamic Stabilizers: First dorsal interosseous

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Stable key pinch for tool use

Osteokinematics: Key pinch force measurement

Joint Reaction Force: Compression increases with load during lateral pinch especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms Adductor pollicis as primary during standardized lateral pinch

Normal ROM: Key pinch between thumb pad and lateral index middle phalanx

Functional ROM: Key and tool lateral pinch

Compensations: Not specified for this grip pattern

Common Dysfunctions: Key pinch trainer → functional tools

Common Pain Patterns: Tip pinch substitute

Clinical Relevance: Key pinch isometrics → functional tool use

Functional Activities: ADL and occupational tasks requiring lateral pinch within symptom-free range

Sport-Specific Applications: Thumb base and index radial pain

Common Injuries: Weak lateral pinch in ulnar palsy

Assessment: Key use, scissors, pliers

Treatment Implications: CMC arthritis, adductor strain

Exercise Progressions: Not specified for this grip pattern

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Lateral Pinch]

RECORD: Tip Pinch

ID: MOV-UE-052

Movement: Tip Pinch

Joint: Hand composite

Body Region: Hand/grip

Plane: Multi-planar

Axis: Multiple

Open Chain: Opposition with IP flexion both digits

Closed Chain: Tip pinch on bead or coin

Primary Movers: FPL, FDP index, opponens pollicis

Secondary Movers: APB, interossei

Stabilizers: MCP stabilizers

Force Couples: Agonist-antagonist pairs coordinating tip pinch with regional stabilizers

Prime Stabilizers: MCP stabilizers

Dynamic Stabilizers: APB, interossei

Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range

Arthrokinematics: Highest precision demand

Osteokinematics: Tip pinch dynamometry

Joint Reaction Force: Compression increases with load during tip pinch especially at end-range

Compression Forces: Articular approximation with muscle co-contraction and axial loading

Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability

Torque: Muscular torque balanced by eccentric antagonists for controlled motion

Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range

Lever Class: Third-class lever dominant in open-chain upper extremity motion

Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition

EMG Evidence: EMG confirms FPL as primary during standardized tip pinch
Normal ROM: Thumb tip to fingertip
Functional ROM: Pick up small objects
Compensations: Not specified for this grip pattern
Common Dysfunctions: Smallest object pinch → graded resistance
Common Pain Patterns: Tripod substitute
Clinical Relevance: Fine motor pinch progression
Functional Activities: ADL and occupational tasks requiring tip pinch within symptom-free range
Sport-Specific Applications: Fingertip pain
Common Injuries: Cannot oppose in advanced CTS
Assessment: Coin picking, sewing
Treatment Implications: Median nerve palsy
Exercise Progressions: Not specified for this grip pattern
References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Tip Pinch]

RECORD: Tripod Pinch

ID: MOV-UE-053
Movement: Tripod Pinch
Joint: Hand composite
Body Region: Hand/grip
Plane: Multi-planar
Axis: Multiple
Open Chain: Three-point precision grip
Closed Chain: Pen hold, chopstick control
Primary Movers: FPL, FDP index/middle, opponens pollicis
Secondary Movers: APB, lumbricals
Stabilizers: Interossei
Force Couples: Agonist-antagonist pairs coordinating tripod pinch with regional stabilizers
Prime Stabilizers: Interossei
Dynamic Stabilizers: APB, lumbricals
Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range
Arthrokinematics: Standard pen grip pattern
Osteokinematics: Tripod pinch observation
Joint Reaction Force: Compression increases with load during tripod pinch especially at end-range
Compression Forces: Articular approximation with muscle co-contraction and axial loading
Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability
Torque: Muscular torque balanced by eccentric antagonists for controlled motion
Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range
Lever Class: Third-class lever dominant in open-chain upper extremity motion
Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition
EMG Evidence: EMG confirms FPL as primary during standardized tripod pinch
Normal ROM: Thumb opposes index and middle tips simultaneously
Functional ROM: Writing tripod, chopstick grip
Compensations: Not specified for this grip pattern

Common Dysfunctions: Tripod pinch beads → writing practice
Common Pain Patterns: Lateral pinch writing substitute
Clinical Relevance: Pen grip retraining
Functional Activities: ADL and occupational tasks requiring tripod pinch within symptom-free range
Sport-Specific Applications: Thenar and index pain with writing
Common Injuries: Poor tripod in pediatric motor delay
Assessment: Writing, chopsticks, tweezers
Treatment Implications: Writer's cramp, thumb CMC pain
Exercise Progressions: Not specified for this grip pattern
References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Tripod Pinch]

RECORD: Lumbrical Grip

ID: MOV-UE-054
Movement: Lumbrical Grip
Joint: Hand composite
Body Region: Hand/grip
Plane: Multi-planar
Axis: Multiple
Open Chain: Lumbricals flex MCP extensors action at IP via lateral bands
Closed Chain: Flat hand on plate/table
Primary Movers: Lumbricals (extend MCP, flex IP)
Secondary Movers: Interossei, flexor digitorum
Stabilizers: Intrinsic arch stabilizers
Force Couples: Agonist-antagonist pairs coordinating lumbrical grip with regional stabilizers
Prime Stabilizers: Intrinsic arch stabilizers
Dynamic Stabilizers: Interossei, flexor digitorum
Static Stabilizers: Capsuloligamentous restraints and bony architecture at end-range
Arthrokinematics: Intrinsic stability for flat hand tasks
Osteokinematics: Intrinsic plus/minus tests
Joint Reaction Force: Compression increases with load during lumbrical grip especially at end-range
Compression Forces: Articular approximation with muscle co-contraction and axial loading
Shear Forces: Shear minimized by capsular tension; excessive shear linked to instability
Torque: Muscular torque balanced by eccentric antagonists for controlled motion
Moment Arm: Moment arms vary with joint angle; typically optimal near mid-range
Lever Class: Third-class lever dominant in open-chain upper extremity motion
Length-Tension Relationship: Mid-range often optimizes force; end-range weakness common with inhibition
EMG Evidence: EMG confirms Lumbricals (extend MCP as primary during standardized lumbrical grip)
Normal ROM: MCP extension with IP flexion — intrinsic plus position
Functional ROM: Plate grip, flat hand carry
Compensations: Not specified for this grip pattern
Common Dysfunctions: Intrinsic plus exercises → flat hand weight bearing
Common Pain Patterns: Hook grip substitute
Clinical Relevance: Lumbrical blocking exercises
Functional Activities: ADL and occupational tasks requiring lumbrical grip within symptom-free range

Sport-Specific Applications: Palm and MCP pain

Common Injuries: Claw deformity without lumbricals

Assessment: Carrying plate, keyboard hand flat

Treatment Implications: Intrinsic muscle paralysis

Exercise Progressions: Not specified for this grip pattern

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Lumbrical Grip]

5. EMG Exercises

RECORD: Face Pull

ID: EMG-UE-001

Exercise: Face Pull

Primary Muscle: Middle trapezius

Secondary Muscles: Posterior deltoid, infraspinatus, rhomboids

EMG %: 60–80% MVIC upper trapezius/middle trapezius ratio favoring mid trap

Study: Cools et al. scapular muscle EMG review

Population: Healthy adults

Resistance: Band or cable at face height

Equipment: Cable machine, resistance band

Movement Speed: Controlled 2-1-2 tempo

Reliability: Moderate ICC for surface EMG

Evidence Level: Level B

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Cools scapular rehabilitation EMG]

RECORD: Push-Up Plus

ID: EMG-UE-002

Exercise: Push-Up Plus

Primary Muscle: Serratus anterior

Secondary Muscles: Pectoralis major, lower trapezius

EMG %: 80–100% MVIC serratus at top plus phase

Study: Decker et al. push-up plus EMG

Population: Healthy adults

Resistance: Bodyweight to weighted

Equipment: Floor, parallettes

Movement Speed: Controlled plus at top

Reliability: Good reliability surface EMG

Evidence Level: Level B

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Decker serratus push-up plus]

RECORD: External Rotation at 0°

ID: EMG-UE-003

Exercise: External Rotation at 0°

Primary Muscle: Infraspinatus

Secondary Muscles: Teres minor, posterior deltoid

EMG %: 40–60% MVIC infraspinatus dominant

Study: Reinold et al. rotator cuff EMG

Population: Healthy adults

Resistance: Band or dumbbell at side

Equipment: Resistance band, dumbbell

Movement Speed: Slow controlled

Reliability: Moderate reliability

Evidence Level: Level B

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Reinold cuff EMG study]

RECORD: External Rotation at 90°

ID: EMG-UE-004

Exercise: External Rotation at 90°

Primary Muscle: Infrapinatus

Secondary Muscles: Teres minor, supraspinatus

EMG %: 70–90% MVIC infrapinatus/teres minor

Study: Reinold et al. side-lying ER 90 abduction

Population: Healthy adults

Resistance: Light dumbbell or band

Equipment: Bench, dumbbell

Movement Speed: Controlled arc

Reliability: Moderate reliability

Evidence Level: Level B

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Reinold ER 90 abduction]

RECORD: Serratus Punch

ID: EMG-UE-005

Exercise: Serratus Punch

Primary Muscle: Serratus anterior

Secondary Muscles: Pectoralis major, lower trapezius

EMG %: 75–95% MVIC serratus

Study: Hardwick et al. serratus exercises

Population: Healthy adults

Resistance: Band punch forward

Equipment: Resistance band

Movement Speed: Punch and protract

Reliability: Good reliability

Evidence Level: Level B

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Hardwick serratus punch]

RECORD: Scaption

ID: EMG-UE-006

Exercise: Scaption

Primary Muscle: Supraspinatus

Secondary Muscles: Middle deltoid, upper trapezius

EMG %: 50–70% MVIC supraspinatus in scapular plane

Study: Reinold scapular plane elevation

Population: Healthy adults

Resistance: Light dumbbells 30° plane

Equipment: Dumbbells

Movement Speed: Controlled elevation

Reliability: Moderate reliability

Evidence Level: Level B

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Reinold scaption EMG]

RECORD: Y Raise

ID: EMG-UE-007

Exercise: Y Raise

Primary Muscle: Lower trapezius

Secondary Muscles: Serratus anterior, middle trapezius

EMG %: 70–90% MVIC lower trapezius

Study: Escamilla et al. shoulder EMG

Population: Healthy adults

Resistance: Light plate or dumbbells

Equipment: Incline bench optional

Movement Speed: Thumbs up Y pattern

Reliability: Moderate reliability

Evidence Level: Level B

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Escamilla lower trap exercises]

RECORD: T Raise

ID: EMG-UE-008

Exercise: T Raise

Primary Muscle: Middle trapezius

Secondary Muscles: Rhomboids, posterior deltoid

EMG %: 60–80% MVIC middle trapezius

Study: Escamilla shoulder EMG

Population: Healthy adults

Resistance: Light dumbbells prone or bent

Equipment: Bench, dumbbells

Movement Speed: Controlled T raise

Reliability: Moderate reliability

Evidence Level: Level B

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Escamilla T raise EMG]

RECORD: Row

ID: EMG-UE-009

Exercise: Row

Primary Muscle: Middle trapezius

Secondary Muscles: Rhomboids, latissimus dorsi, posterior deltoid

EMG %: 60–80% MVIC mid trap/rhomboid

Study: Lehman et al. rowing EMG

Population: Healthy adults

Resistance: Band, cable, barbell

Equipment: Cable, barbell, band

Movement Speed: Controlled pull

Reliability: Good reliability

Evidence Level: Level B

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Lehman rowing muscle activation]

RECORD: Bench Press

ID: EMG-UE-010

Exercise: Bench Press

Primary Muscle: Pectoralis major

Secondary Muscles: Anterior deltoid, triceps

EMG %: 80–100% MVIC pec major at moderate loads

Study: Barnett EMG bench press

Population: Trained adults

Resistance: Moderate to heavy load

Equipment: Barbell, dumbbell

Movement Speed: Controlled press

Reliability: Moderate reliability

Evidence Level: Level B

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Barnett bench press EMG]

RECORD: Pull-Up

ID: EMG-UE-011

Exercise: Pull-Up

Primary Muscle: Latissimus dorsi

Secondary Muscles: Biceps, middle trapezius, rhomboids

EMG %: 80–100% MVIC lats

Study: Snarr pull-up EMG analysis

Population: Trained adults

Resistance: Bodyweight

Equipment: Pull-up bar

Movement Speed: Controlled pull

Reliability: Moderate reliability

Evidence Level: Level B

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Snarr pull-up EMG]

RECORD: Dead Hang

ID: EMG-UE-012

Exercise: Dead Hang

Primary Muscle: Flexor digitorum profundus/superficialis

Secondary Muscles: Brachioradialis, latissimus, shoulder stabilizers

EMG %: High forearm flexor activity; moderate cuff

Study: Climbing hang EMG studies

Population: Climbers and healthy adults

Resistance: Bodyweight

Equipment: Pull-up bar

Movement Speed: Static hang

Reliability: Variable reliability

Evidence Level: Level C

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Climbing hang forearm EMG]

6. Force Couples

RECORD: Scapular Upward Rotation Force Couple

ID: FC-UE-001

Movement: Scapular upward rotation / arm elevation

Muscle 1: Upper trapezius

Muscle 2: Lower trapezius

Muscle 3: Serratus anterior

Biomechanical Role: UT rotates upward; LT and SA rotate glenoid superiorly and protract — balanced upward rotation without excessive elevation

Clinical Importance: Essential for overhead function and impingement prevention

Common Dysfunction: Serratus weakness or upper trap dominance → winging and impingement

Rehabilitation: Push-up plus, serratus punch, Y raise, lower/upper trap balance drills

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Cools scapular force couple literature]

RECORD: Glenohumeral Compression Force Couple

ID: FC-UE-002

Movement: Glenohumeral joint centration

Muscle 1: Subscapularis

Muscle 2: Infraspinatus

Muscle 3: Teres minor

Biomechanical Role: Rotator cuff compresses humeral head into glenoid while antagonist pair balances rotation — dynamic stability

Clinical Importance: Critical for shoulder stability in athletes and post-dislocation rehab

Common Dysfunction: Cuff imbalance → superior migration and impingement

Rehabilitation: Rotator cuff co-contraction drills, ER/IR at multiple abduction angles, rhythmic stabilization

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Neer cuff force couple; Reinold cuff EMG]

RECORD: Wrist Flexor/Extensor Co-contraction Gripping

ID: FC-UE-003

Movement: Power grip and wrist stability

Muscle 1: Wrist flexors (FCR, FCU, FDS)

Muscle 2: Wrist extensors (ECRL, ECRB, ECU)

Muscle 3: None — bilateral co-contraction

Biomechanical Role: Co-contraction stabilizes wrist in neutral during grip — prevents excessive flexion/extension under load

Clinical Importance: Epicondylalgia often involves grip imbalance; rehab targets balanced co-contraction

Common Dysfunction: Overactive flexors or extensors → medial/lateral epicondylalgia

Rehabilitation: Tyler twist, wrist neutral grip training, balanced flexor/extensor eccentric loading

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Wrist extensor/flexor co-contraction grip studies]

RECORD: Finger Extension Force Couple

ID: FC-UE-004

Movement: Finger extension / intrinsic stabilization

Muscle 1: Dorsal interossei

Muscle 2: Lumbricals

Muscle 3: Long extensors (EDC)

Biomechanical Role: Lumbricals extend IP via lateral bands while interossei extend MCP and adduct/abduct — coordinated with EDC

Clinical Importance: Intrinsic weakness → claw deformity and grip dysfunction

Common Dysfunction: Ulnar palsy → loss interossei/lumbrical coordination

Rehabilitation: Intrinsic plus/minus exercises, rubber band extensions, pegboard, tendon gliding

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Ovalle intrinsic muscle biomechanics]

7. Kinetic Chain

RECORD: Baseball Throw

ID: KIN-UE-001

Movement: Baseball Throw

Sport: Baseball/softball

Starting Position: Wind-up to ball release

Joint Sequence: Ankle → knee → hip → trunk rotation → scapular rotation → GH IR → elbow extension → wrist snap

Primary Drivers: Lower body and trunk

Energy Transfer: Kinetic chain transfers ~50% energy from lower extremity; scapular and GH sequential timing critical

Common Compensation: Early trunk opening; excessive elbow valgus; scapular dyskinesis

Injury Risk: UCL injury, labral tears, rotator cuff strain

Clinical Assessment: Throwing analysis, shoulder/elbow exam, total arc motion

Corrective Exercise: Serratus/ cuff / hip rotation drills

Progression: Interval throwing program

Evidence: ASMI interval throwing; Wilk thrower's ten

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Baseball Throw]

RECORD: Tennis Serve

ID: KIN-UE-002

Movement: Tennis Serve

Sport: Tennis

Starting Position: Platform or pinpoint stance to racket contact

Joint Sequence: Leg drive → trunk hyperextension-rotation → scapular upward rotation → GH abduction/ER → elbow extension → wrist snap

Primary Drivers: Leg drive and trunk rotation

Energy Transfer: Ground reaction force through kinetic chain to racket head speed

Common Compensation: Shoulder-only serve; insufficient knee bend; scapular lag

Injury Risk: Rotator cuff tendinopathy, impingement, tennis elbow

Clinical Assessment: Serve video analysis, cuff strength, scapular tests

Corrective Exercise: Leg drive drills, scapular strengthening, eccentric wrist extensors

Progression: Progressive serve volume

Evidence: Elliott tennis serve biomechanics

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Tennis Serve]

RECORD: Volleyball Spike

ID: KIN-UE-003

Movement: Volleyball Spike

Sport: Volleyball

Starting Position: Approach to ball contact overhead

Joint Sequence: Approach steps → penultimate step → arm swing → trunk rotation → GH elevation → elbow whip → wrist snap

Primary Drivers: Approach and trunk

Energy Transfer: Vertical jump and trunk rotation transfer to hitting arm speed
Common Compensation: Single-arm dominance without trunk rotation; poor scapular timing
Injury Risk: Rotator cuff strain, finger sprains
Clinical Assessment: Hitting mechanics review, cuff/scapular assessment
Corrective Exercise: Approach footwork, trunk rotation med ball throws
Progression: Progressive hitting reps
Evidence: Coleman volleyball spike biomechanics
References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Volleyball Spike]

RECORD: Swimming Freestyle

ID: KIN-UE-004
Movement: Swimming Freestyle
Sport: Swimming
Starting Position: Streamline push-off to pull-through
Joint Sequence: Kick → trunk rotation → catch → pull → recovery with scapular upward rotation
Primary Drivers: Trunk rotation and lat pull
Energy Transfer: Continuous chain linking kick, rotation, and pull; high repetition stress
Common Compensation: Dropped elbow catch; excessive internal rotation; inadequate body roll
Injury Risk: Swimmer's shoulder, impingement
Clinical Assessment: Stroke analysis, cuff endurance, scapular dyskinesis
Corrective Exercise: Band rotator work, scapular drills, technique correction
Progression: Progressive yardage
Evidence: Tovin swimmer's shoulder rehab
References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Swimming Freestyle]

RECORD: Push-Up

ID: KIN-UE-005
Movement: Push-Up
Sport: Calisthenics/strength
Starting Position: Plank position to bottom and return
Joint Sequence: Wrist → elbow → GH horizontal adduction → scapular protraction at top
Primary Drivers: Pectoralis and serratus
Energy Transfer: Closed chain force from hands through trunk; serratus protraction at plus phase
Common Compensation: Scapular winging; excessive cervical extension; elbow flare
Injury Risk: Wrist pain, anterior shoulder strain
Clinical Assessment: Push-up form, scapular control, wrist mobility
Corrective Exercise: Incline push-up plus progression
Progression: Floor to weighted push-up
Evidence: Calatayud push-up EMG progression
References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Push-Up]

RECORD: Pull-Up

ID: KIN-UE-006
Movement: Pull-Up

Sport: Calisthenics/strength

Starting Position: Dead hang to chin over bar

Joint Sequence: Shoulder extension/adduction → elbow flexion → scapular depression/retraction

Primary Drivers: Latissimus dorsi

Energy Transfer: Closed chain vertical pull; scapular initiation before elbow flexion

Common Compensation: Shrugging without scapular depression; kipping compensation

Injury Risk: Elbow tendinopathy, shoulder impingement

Clinical Assessment: Scapular pull, hang tolerance, grip strength

Corrective Exercise: Scapular pulls, eccentric lowering

Progression: Band-assisted to full pull-up

Evidence: Snarr pull-up activation

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Pull-Up]

RECORD: Bench Press

ID: KIN-UE-007

Movement: Bench Press

Sport: Strength training

Starting Position: Supine on bench, bar lower to chest, press

Joint Sequence: GH horizontal adduction → elbow extension; scapular retraction maintained

Primary Drivers: Pectoralis major

Energy Transfer: Bar path over shoulders; leg drive stabilizes trunk

Common Compensation: Excessive shoulder protraction loss; bench without scapular retraction

Injury Risk: Pec strain, anterior shoulder pain

Clinical Assessment: Bar path, scapular position, cuff balance

Corrective Exercise: Retracted scapular bench technique drills

Progression: Progressive load

Evidence: Green bench press scapular position

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Bench Press]

RECORD: Overhead Press

ID: KIN-UE-008

Movement: Overhead Press

Sport: Strength training

Starting Position: Rack to overhead lockout

Joint Sequence: Legs/trunk stability → scapular upward rotation → GH flexion/abduction → elbow extension

Primary Drivers: Deltoid and triceps

Energy Transfer: Vertical force through stable core; full scapular upward rotation required

Common Compensation: Lumbar hyperextension; incomplete scapular rotation

Injury Risk: Impingement, lower back strain

Clinical Assessment: Overhead mobility, scapular assessment, core stability

Corrective Exercise: Landmine press progression to full overhead

Progression: Barbell/dumbbell progressive load

Evidence: Saeterbakken overhead press EMG

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Overhead Press]

RECORD: Handstand

ID: KIN-UE-009

Movement: Handstand

Sport: Gymnastics/calisthenics

Starting Position: Kick-up or press to inverted support

Joint Sequence: Wrist → elbow → GH flexion → scapular elevation/upward rotation → core

Primary Drivers: Wrist extensors and shoulder stabilizers

Energy Transfer: Closed chain vertical load through arms to trunk; cuff co-contraction critical

Common Compensation: Excessive arch; inadequate shoulder flexion; wrist collapse

Injury Risk: Wrist pain, shoulder impingement

Clinical Assessment: Wall handstand hold, wrist prep, cuff endurance

Corrective Exercise: Wall walks, shoulder taps

Progression: Freestanding progression

Evidence: Gymnastics handstand conditioning

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Handstand]

RECORD: Rock Climbing

ID: KIN-UE-010

Movement: Rock Climbing

Sport: Climbing

Starting Position: Route start to sequence of reaches and pulls

Joint Sequence: Fingers → wrist → elbow → GH → scapular → trunk → lower body foot drive

Primary Drivers: Finger flexors and lats

Energy Transfer: Alternating closed chain upper and lower body; high finger demand

Common Compensation: Arm-only climbing; elbow hyperextension; shoulder elevation

Injury Risk: Pulley injury, A2/A4 rupture, shoulder impingement

Clinical Assessment: Climbing movement screen, finger strength, scapular control

Corrective Exercise: Footwork drills, scapular pulls, fingerboard progressive

Progression: Grade progression

Evidence: Michailov climbing injury prevention

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Rock Climbing]

RECORD: Rowing

ID: KIN-UE-011

Movement: Rowing

Sport: Rowing sport

Starting Position: Catch to drive to finish to recovery

Joint Sequence: Leg drive → trunk swing → arm pull with scapular retraction

Primary Drivers: Quadriceps then lats and scapular retractors

Energy Transfer: Leg-trunk-arm sequence; arms only finish

Common Compensation: Excessive arm pull early; rounded upper back

Injury Risk: Low back pain, rib stress, wrist extensor strain

Clinical Assessment: Erg technique analysis, hip hinge, scapular endurance

Corrective Exercise: Leg drive drills, band rows

Progression: Stroke rate/volume progression

Evidence: Hosea rowing biomechanics

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Rowing]

RECORD: Golf Swing

ID: KIN-UE-012

Movement: Golf Swing

Sport: Golf

Starting Position: Address to backswing to downswing to follow-through

Joint Sequence: Hip rotation → trunk rotation → GH adduction/IR downswing → wrist release

Primary Drivers: Hip and trunk rotators

Energy Transfer: Ground reaction torque through chain to club head speed

Common Compensation: Early upper body; reverse spine angle; lead arm excessive across body

Injury Risk: Golfer's elbow, lead wrist injury, low back pain

Clinical Assessment: Swing analysis, hip mobility, scapular control

Corrective Exercise: Hip rotation med ball, wrist neutral grip

Progression: Progressive driving range volume

Evidence: Myers golf kinetic chain research

References: Moore Clinically Oriented Anatomy; Netter Atlas; see PubMed [Golf Swing]